The LaTeX report

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1 Setup

1.1 Command history

```
ma5>
ma5>
ma5>import samples/ttbar_sl_1.lhe.gz as testset
ma5>set main.normalize = none
ma5>set main.lumi = 1.0
ma5>define e = e+ e-
ma5>define mu = mu+ mu-
ma5>
ma5>
ma5>
ma5>plot SQRTS 50 0 5000
ma5>plot SCALE 50 0 2000
ma5>plot ALPHA_QCD 50 0 0.3
ma5>plot ALPHA_QED 50 0 0.02
ma5>plot TET 50 0 5000
ma5>plot THT 50 0 5000
ma5>plot MET 50 0 1500
ma5>plot MHT 50 0 1500
ma5>plot NPID
ma5>plot NAPID
ma5>plot WEIGHTS
ma5>
ma5>
ma5>
ma5>plot PT(e) 50 0 1000 [PTordering]
ma5>plot ETA(e) 50 -8 8
ma5>plot ABSETA(e) 50 0 8
ma5>plot PHI(e) 32 -3.2 3.2
ma5>plot E(e) 50 0 2000
ma5>plot ET(e) 50 0 2000
ma5>plot M(e) 50 0 20
ma5>plot MT(e) 50 0 1000
ma5>plot MT_MET(e) 50 0 1000
ma5>plot P(e) 50 0 2000
ma5>plot PX(e) 50 -1000 1000
ma5>plot PY(e) 50 -1000 1000
ma5>plot PZ(e) 50 -2000 2000
ma5>plot R(e) 50 0 10
ma5>plot Y(e) 50 -8 8
ma5>plot BETA(e) 50 0 1.2
ma5>plot GAMMA(e) 50 1 100
ma5>
ma5>plot PT(e[1]) 50 0 1000 [PTordering]
ma5>plot ETA(e[1]) 50 -8 8 [PTordering]
ma5>plot PHI(e[1]) 32 -3.2 3.2 [PTordering]
ma5>plot E(e[1]) 50 0 2000 [PTordering]
```

```
ma5>plot MT_MET(e[1]) 50 0 1000 [PTordering]
ma5>plot PT(e[2]) 50 0 1000 [PTordering]
ma5>plot ETA(e[2]) 50 -8 8 [PTordering]
ma5>plot PHI(e[2]) 32 -3.2 3.2 [PTordering]
ma5>plot E(e[2]) 50 0 2000 [PTordering]
ma5>plot MT_MET(e[2]) 50 0 1000 [PTordering]
ma5>plot M(e[1] e[2]) 50 0 1000 [PTordering]
ma5>plot PT(e[1] e[2]) 50 0 1500 [PTordering]
ma5>plot DELTAR(e[1],e[2]) 50 0 10 [PTordering]
ma5>plot DPHI_0_PI(e[1],e[2]) 32 0 3.2 [PTordering]
ma5>plot DPHI_0_2PI(e[1],e[2]) 32 0 6.4 [PTordering]
ma5>
ma5>
ma5>
ma5>plot PT(mu) 50 0 1000 [PTordering]
ma5>plot ETA(mu) 50 -8 8
ma5>plot ABSETA(mu) 50 0 8
ma5>plot PHI(mu) 32 -3.2 3.2
ma5>plot E(mu) 50 0 2000
ma5>plot ET(mu) 50 0 2000
ma5>plot M(mu) 50 0 20
ma5>plot MT(mu) 50 0 1000
ma5>plot MT_MET(mu) 50 0 1000
ma5>plot P(mu) 50 0 2000
ma5>plot PX(mu) 50 -1000 1000
ma5>plot PY(mu) 50 -1000 1000
ma5>plot PZ(mu) 50 -2000 2000
ma5>plot R(mu) 50 0 10
ma5>plot Y(mu) 50 -8 8
ma5>plot BETA(mu) 50 0 1.2
ma5>plot GAMMA(mu) 50 1 100
ma5>
ma5>plot PT(mu[1]) 50 0 1000 [PTordering]
ma5>plot ETA(mu[1]) 50 -8 8 [PTordering]
ma5>plot PHI(mu[1]) 32 -3.2 3.2 [PTordering]
ma5>plot E(mu[1]) 50 0 2000 [PTordering]
ma5>plot MT_MET(mu[1]) 50 0 1000 [PTordering]
ma5>plot PT(mu[2]) 50 0 1000 [PTordering]
ma5>plot ETA(mu[2]) 50 -8 8 [PTordering]
ma5>plot PHI(mu[2]) 32 -3.2 3.2 [PTordering]
ma5>plot E(mu[2]) 50 0 2000 [PTordering]
ma5>plot MT_MET(mu[2]) 50 0 1000 [PTordering]
ma5>plot M(mu[1] mu[2]) 50 0 1000 [PTordering]
ma5>plot PT(mu[1] mu[2]) 50 0 1500 [PTordering]
ma5>plot DELTAR(mu[1],mu[2]) 50 0 10 [PTordering]
ma5>plot DPHI_0_PI(mu[1],mu[2]) 32 0 3.2 [PTordering]
ma5>plot DPHI_0_2PI(mu[1],mu[2]) 32 0 6.4 [PTordering]
ma5>
ma5>
```

```
ma5>
ma5>plot PT(b) 50 0 1000 [PTordering]
ma5>plot ETA(b) 50 -8 8
ma5>plot ABSETA(b) 50 0 8
ma5>plot PHI(b) 32 -3.2 3.2
ma5>plot E(b) 50 0 2000
ma5>plot ET(b) 50 0 2000
ma5>plot M(b) 50 0 50
ma5>plot MT(b) 50 0 1000
ma5>plot MT_MET(b) 50 0 1000
ma5>plot P(b) 50 0 2000
ma5>plot PX(b) 50 -1000 1000
ma5>plot PY(b) 50 -1000 1000
ma5>plot PZ(b) 50 -2000 2000
ma5>plot R(b) 50 0 10
ma5>plot Y(b) 50 -8 8
ma5>plot BETA(b) 50 0 1.2
ma5>plot GAMMA(b) 50 1 100
ma5>
ma5>plot PT(b[1]) 50 0 1000 [PTordering]
ma5>plot ETA(b[1]) 50 -8 8 [PTordering]
ma5>plot PHI(b[1]) 32 -3.2 3.2 [PTordering]
ma5>plot E(b[1]) 50 0 2000 [PTordering]
ma5>plot MT_MET(b[1]) 50 0 1000 [PTordering]
ma5>plot PT(b[2]) 50 0 1000 [PTordering]
ma5>plot ETA(b[2]) 50 -8 8 [PTordering]
ma5>plot PHI(b[2]) 32 -3.2 3.2 [PTordering]
ma5>plot E(b[2]) 50 0 2000 [PTordering]
ma5>plot MT_MET(b[2]) 50 0 1000 [PTordering]
ma5>plot M(b[1] b[2]) 50 0 2000 [PTordering]
ma5>plot PT(b[1] b[2]) 50 0 2000 [PTordering]
ma5>plot DELTAR(b[1],b[2]) 50 0 10 [PTordering]
ma5>plot DPHI_0_PI(b[1],b[2]) 32 0 3.2 [PTordering]
ma5>plot DPHI_0_2PI(b[1],b[2]) 32 0 6.4 [PTordering]
ma5>
ma5>
ma5>
ma5>plot PT(j) 50 0 1000 [PTordering]
ma5>plot ETA(j) 50 -8 8
ma5>plot ABSETA(j) 50 0 8
ma5>plot PHI(j) 32 -3.2 3.2
ma5>plot E(j) 50 0 2000
ma5>plot ET(j) 50 0 2000
ma5>plot M(j) 50 0 200
ma5>plot MT(j) 50 0 1000
ma5>plot MT_MET(j) 50 0 1000
ma5>plot P(j) 50 0 2000
ma5>plot PX(j) 50 -1000 1000
ma5>plot PY(j) 50 -1000 1000
```

```
ma5>plot PZ(j) 50 -2000 2000
ma5>plot R(j) 50 0 10
ma5>plot Y(j) 50 -8 8
ma5>plot BETA(j) 50 0 1.2
ma5>plot GAMMA(j) 50 1 100
ma5>
ma5>plot PT(j[1]) 50 0 1000 [PTordering]
ma5>plot ETA(j[1]) 50 -8 8 [PTordering]
ma5>plot PHI(j[1]) 32 -3.2 3.2 [PTordering]
ma5>plot E(j[1]) 50 0 2000 [PTordering]
ma5>plot MT_MET(j[1]) 50 0 1000 [PTordering]
ma5>plot PT(j[2]) 50 0 1000 [PTordering]
ma5>plot ETA(j[2]) 50 -8 8 [PTordering]
ma5>plot PHI(j[2]) 32 -3.2 3.2 [PTordering]
ma5>plot E(j[2]) 50 0 2000 [PTordering]
ma5>plot MT_MET(j[2]) 50 0 1000 [PTordering]
ma5>plot M(j[1] j[2]) 50 0 3000 [PTordering]
ma5>plot PT(j[1] j[2]) 50 0 3000 [PTordering]
ma5>plot DELTAR(j[1],j[2]) 50 0 10 [PTordering]
ma5>plot DPHI_0_PI(j[1],j[2]) 32 0 3.2 [PTordering]
ma5>plot DPHI_0_2PI(j[1],j[2]) 32 0 6.4 [PTordering]
ma5>
ma5>
ma5>
ma5>plot DELTAR(e,e) 50 0 10
ma5>plot DPHI_0_PI(e,e) 32 0 3.2
ma5>plot DPHI_0_2PI(e,e) 32 0 6.4
ma5>plot M(e e) 50 0 1000
ma5>plot PT(e e) 50 0 1500
ma5>plot DELTAR(mu,mu) 50 0 10
ma5>plot DPHI_0_PI(mu,mu) 32 0 3.2
ma5>plot DPHI_0_2PI(mu,mu) 32 0 6.4
ma5>plot M(mu mu) 50 0 1000
ma5>plot PT(mu mu) 50 0 1500
ma5>plot DELTAR(b,b) 50 0 10
ma5>plot DPHI_0_PI(b,b) 32 0 3.2
ma5>plot DPHI_0_2PI(b,b) 32 0 6.4
ma5>plot M(b b) 50 0 2000
ma5>plot PT(b b) 50 0 2000
ma5>plot DELTAR(j,j) 50 0 10
ma5>plot DPHI_0_PI(j,j) 32 0 3.2
ma5>plot DPHI_0_2PI(j,j) 32 0 6.4
ma5>plot M(j j) 50 0 3000
ma5>plot PT(j j) 50 0 3000
ma5>
ma5>plot DELTAR(e,mu) 50 0 10
ma5>plot DELTAR(e,j) 50 0 10
ma5>plot DELTAR(mu,j) 50 0 10
ma5>plot DELTAR(b,j) 50 0 10
```

```
ma5>plot M(e mu) 50 0 1000
ma5>plot M(e j) 50 0 2000
ma5>plot M(mu j) 50 0 2000
ma5>plot M(b j) 50 0 3000
ma5>plot PT(e mu) 50 0 1500
ma5>plot PT(b j) 50 0 3000
ma5>
ma5>plot DELTAR(e[1],mu[1]) 50 0 10 [PTordering]
ma5>plot DELTAR(e[1],j[1]) 50 0 10 [PTordering]
ma5>plot DELTAR(mu[1],j[1]) 50 0 10 [PTordering]
ma5>plot DELTAR(b[1],j[1]) 50 0 10 [PTordering]
ma5>plot M(e[1] mu[1]) 50 0 1000 [PTordering]
ma5>plot M(e[1] j[1]) 50 0 2000 [PTordering]
ma5>plot M(mu[1] j[1]) 50 0 2000 [PTordering]
ma5>plot M(b[1] j[1]) 50 0 3000 [PTordering]
ma5>plot PT(e[1] mu[1]) 50 0 1500 [PTordering]
ma5>plot PT(b[1] j[1]) 50 0 3000 [PTordering]
ma5>submit
```

1.2 Configuration

- MadAnalysis version 1.11.0 (2025/04/23).
- Histograms are not scaled.

2 Datasets

2.1 testset

- Sample consisting of: signal events.
- Generated events: 1000 events.
- Normalization to the luminosity: 15447+/- 114 events.
- Ratio (event weight): 15 - warning: please generate more events (weight larger than 1)!

Path to the event file	Nr. of events	Cross section (pb)	Negative wgts (%)
samples/ttbar_sl_1.lhe.gz	1000	15.4 @ 0.74%	0.0

3 Histos and cuts

3.1 Histogram 1

*** Plot: SQRTS**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	489.445	124.1	0.0	0.0

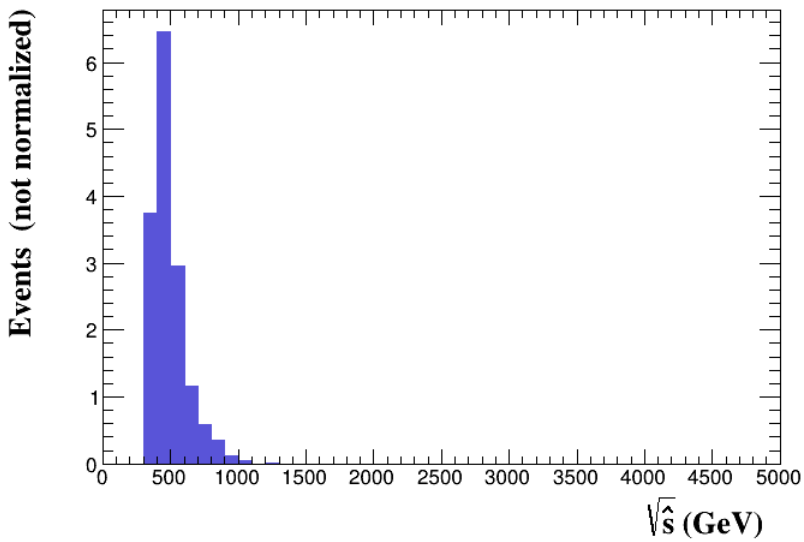

Figure 1.

3.2 Histogram 2

*** Plot: SCALE**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	214.211	45.19	0.0	0.0

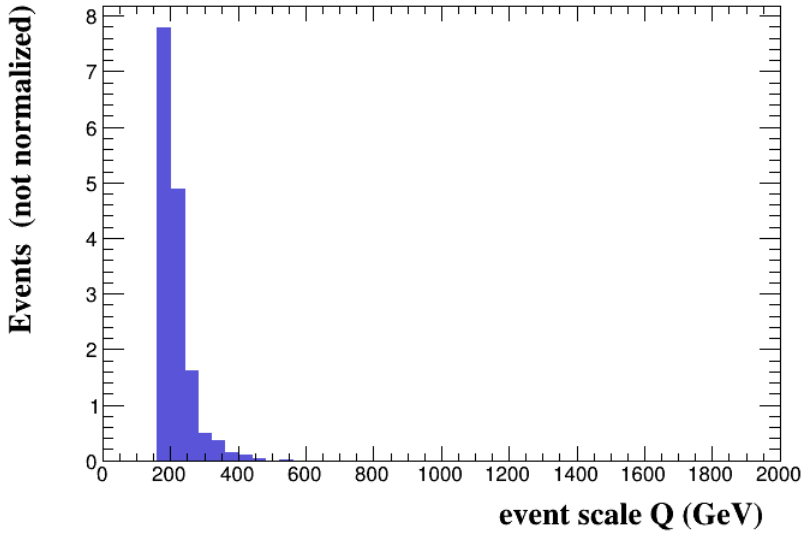

Figure 2.

3.3 Histogram 3

*** Plot: ALPHA_QCD**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	0.114843	0.002765	0.0	0.0

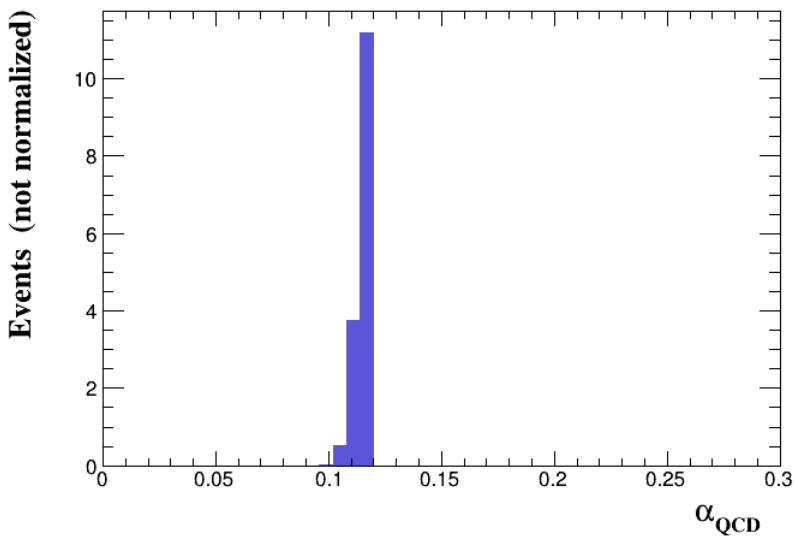

Figure 3.

3.4 Histogram 4

*** Plot: ALPHA_QED**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	0.0795775	4.683e-05	0.0	100.0

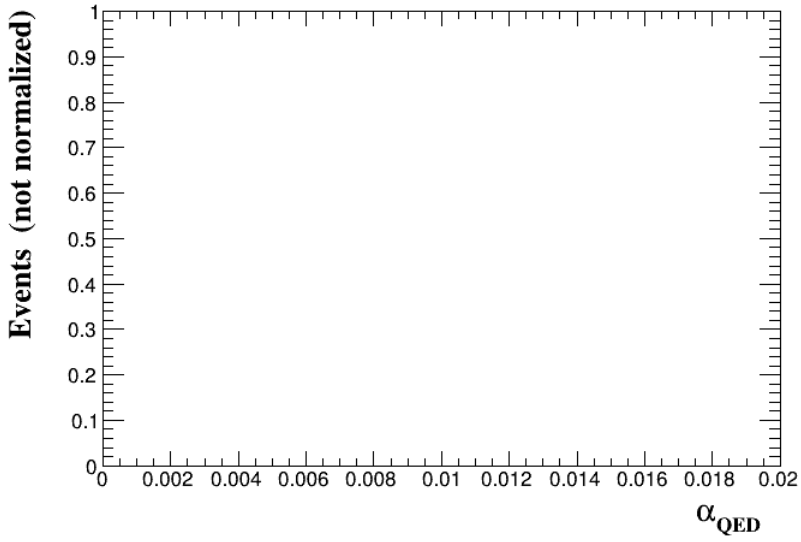

Figure 4.

3.5 Histogram 5

*** Plot: TET**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	308.078	93.57	0.0	0.0

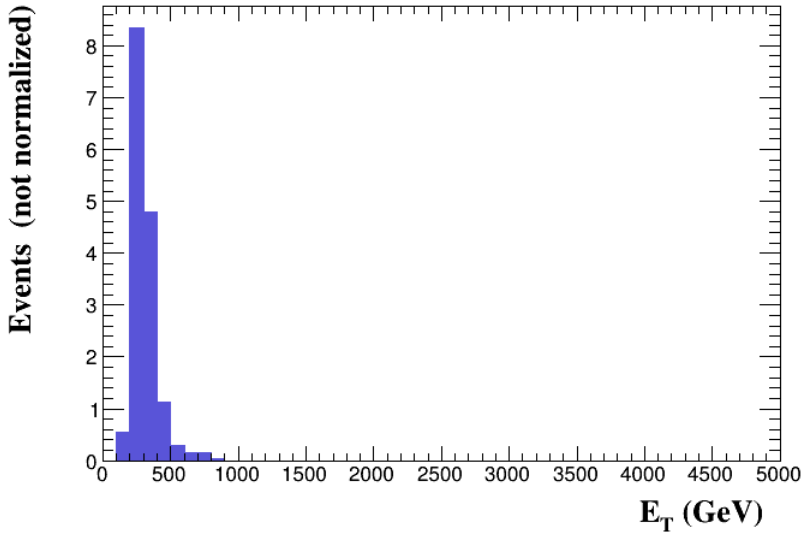

Figure 5.

3.6 Histogram 6

*** Plot: THT**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	258.83	79.71	0.0	0.0

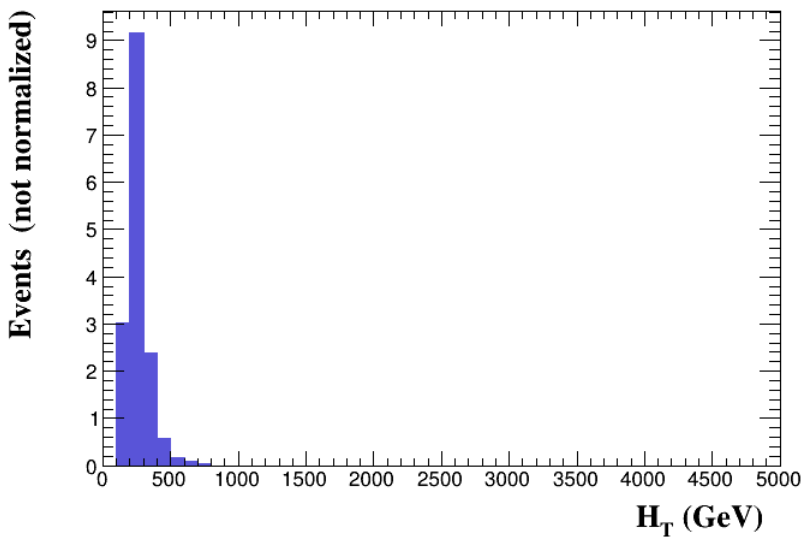

Figure 6.

3.7 Histogram 7

*** Plot: MET**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	58.7171	37.97	0.0	0.0

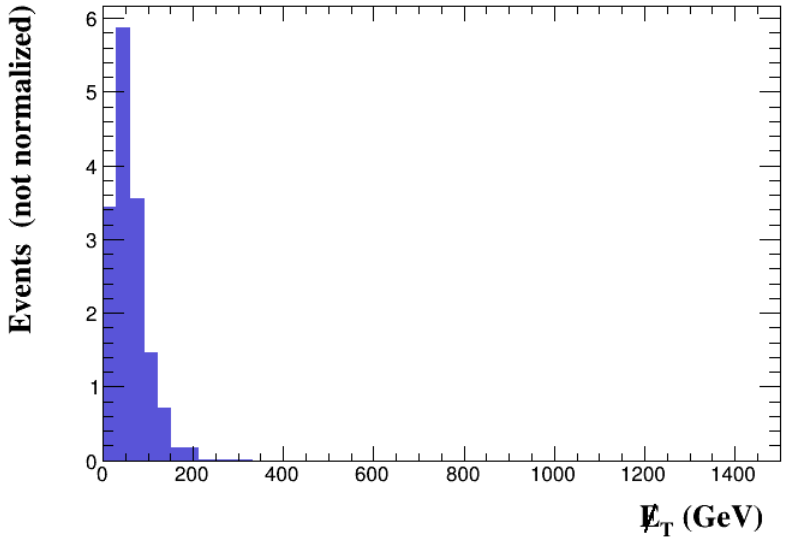

Figure 7.

3.8 Histogram 8

*** Plot: MHT**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	83.1686	52.75	0.0	0.0

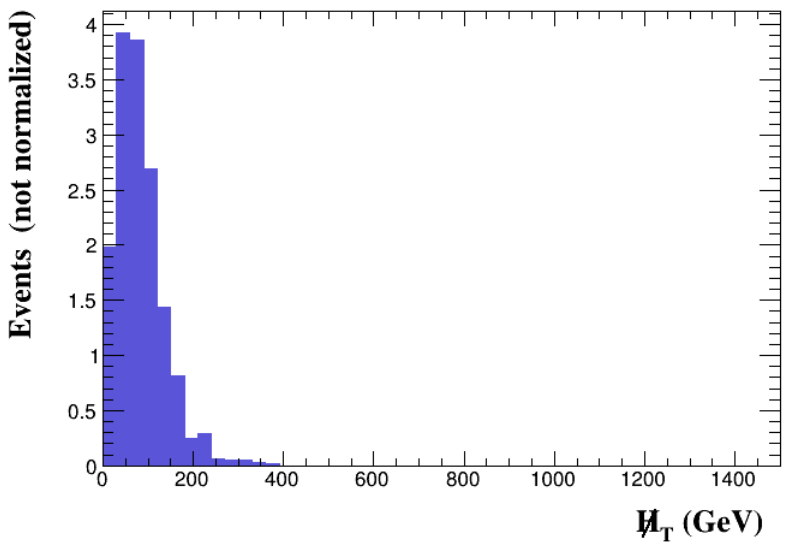

Figure 8.

3.9 Histogram 9

*** Plot: NPID**

Table 1. Statistics table

Dataset	Integral	Entries / events
testset	92.7	0.0

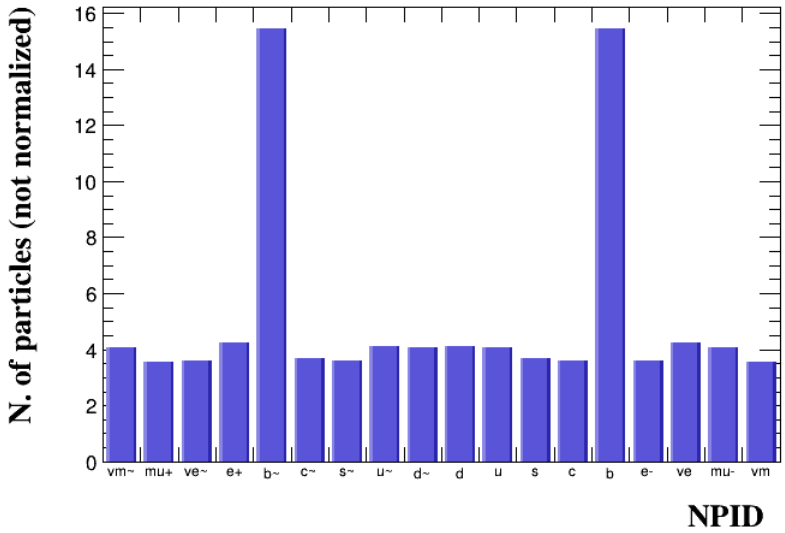

Figure 9.

3.10 Histogram 10

*** Plot: NAPID**

Table 2. Statistics table

Dataset	Integral	Entries / events
testset	92.7	0.0

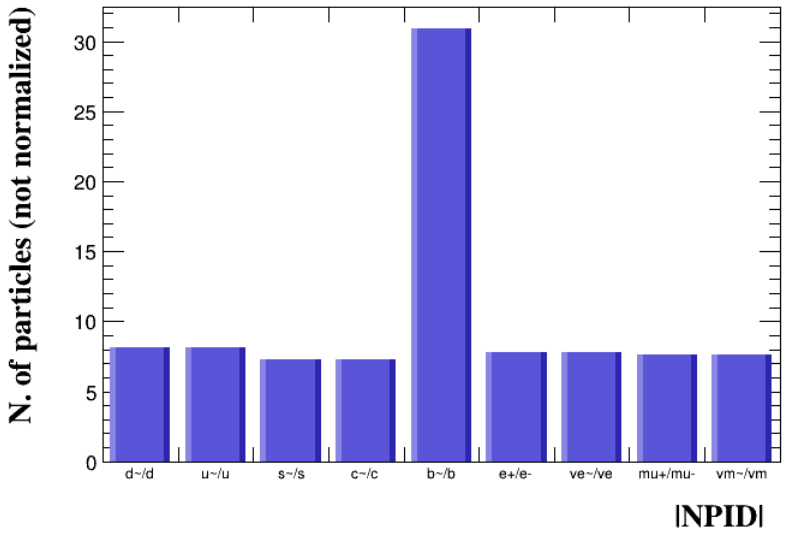

Figure 10.

3.11 Histogram 11

*** Plot: WEIGHTS**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	0.0154478	1.051e-05	0.0	0.0

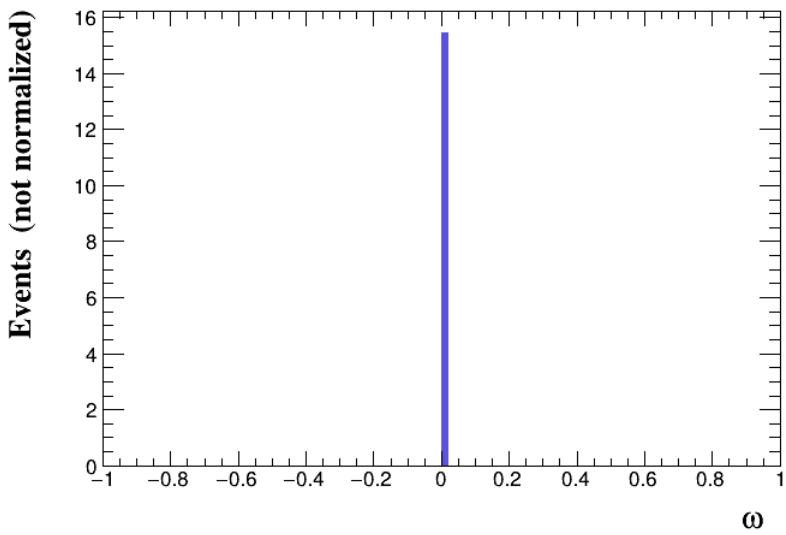

Figure 11.

3.12 Histogram 12

*** Plot: PT (e)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	49.9595	38.03	0.0	0.0

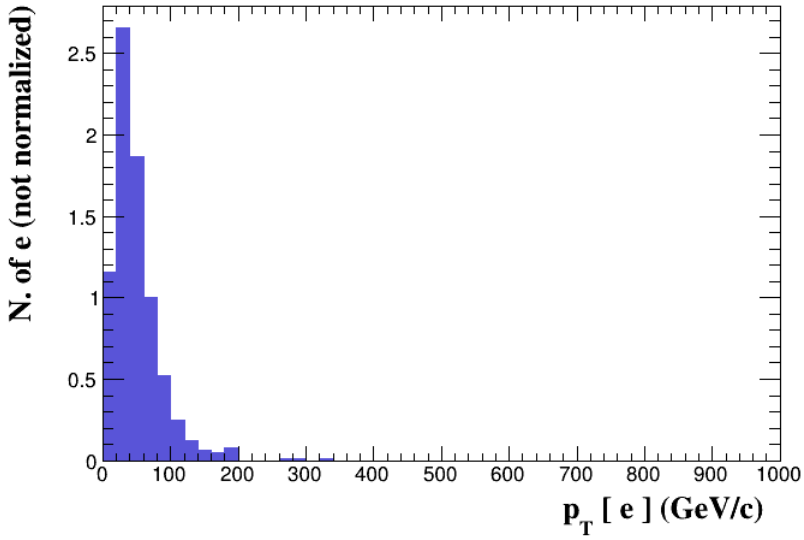

Figure 12.

3.13 Histogram 13

*** Plot: ETA (e)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	0.0509938	1.152	0.0	0.0

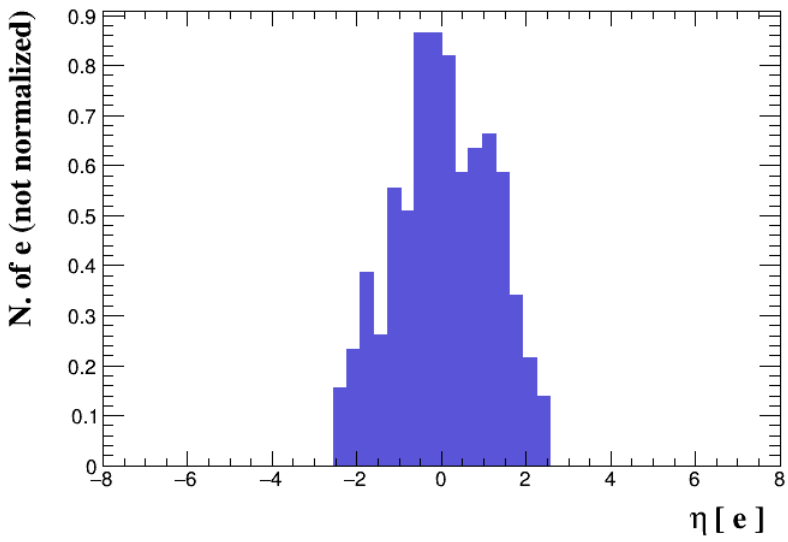

Figure 13.

3.14 Histogram 14

*** Plot: ABSETA (e)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	0.946635	0.6582	0.0	0.0

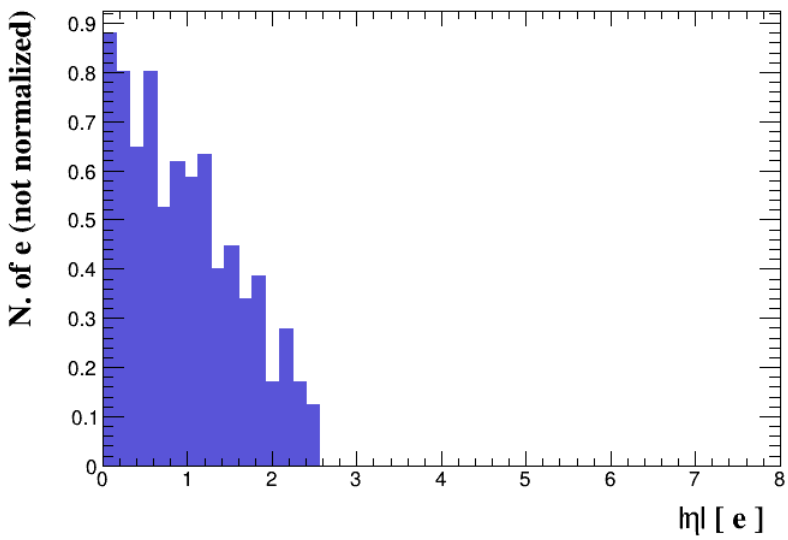

Figure 14.

3.15 Histogram 15

*** Plot: PHI (e)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	0.0207825	1.754	0.0	0.0

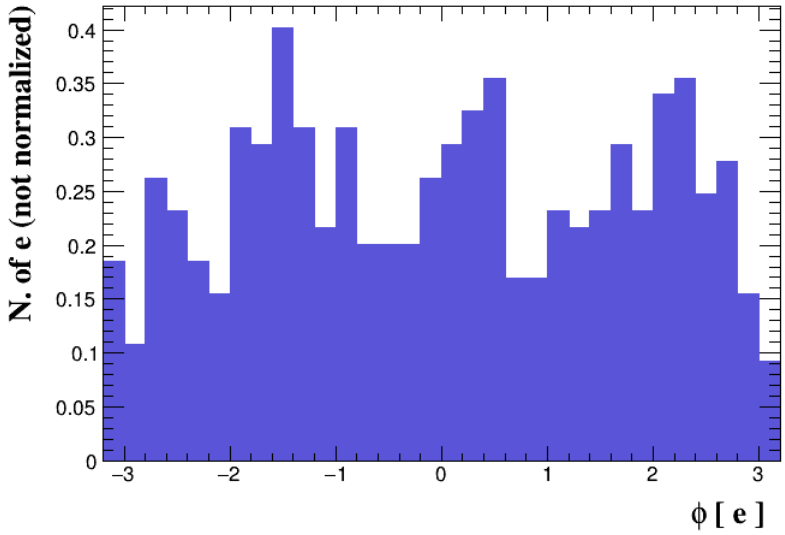

Figure 15.

3.16 Histogram 16

*** Plot: E (e)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	87.8171	77.33	0.0	0.0

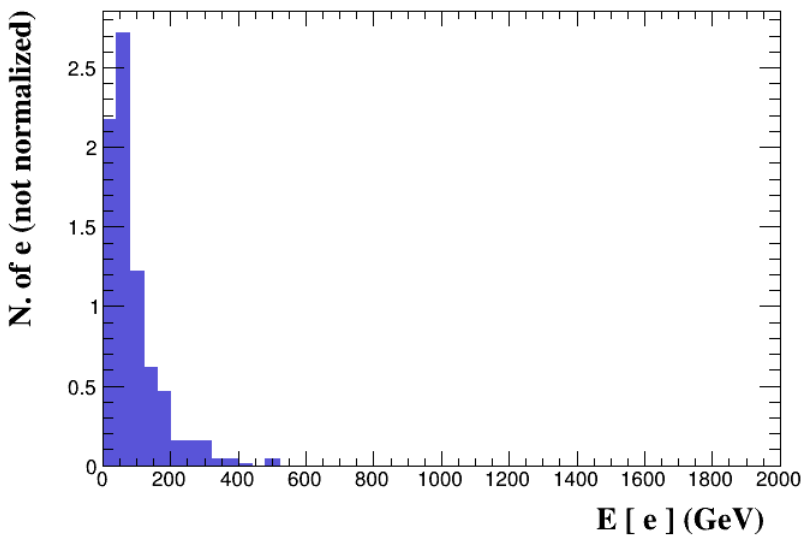

Figure 16.

3.17 Histogram 17

*** Plot: ET (e)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	49.9595	38.03	0.0	0.0

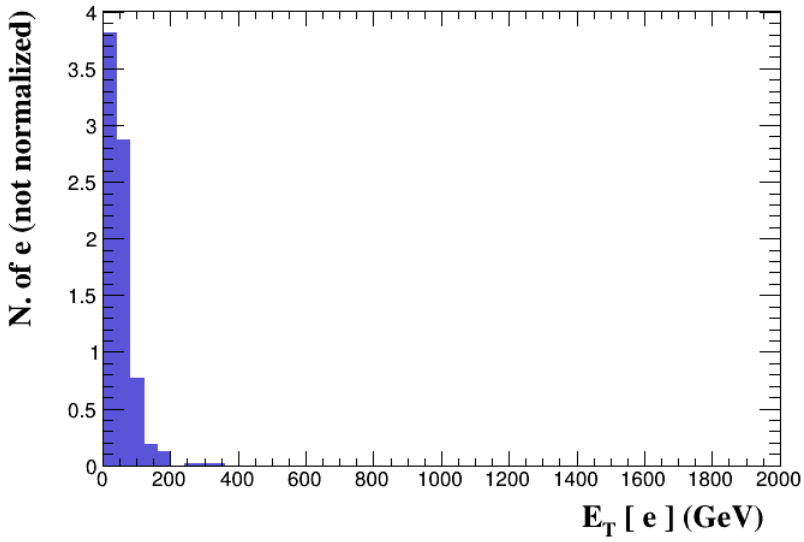

Figure 17.

3.18 Histogram 18

*** Plot: M (e)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	1.44661e-06	0.0005939	51.58	0.0

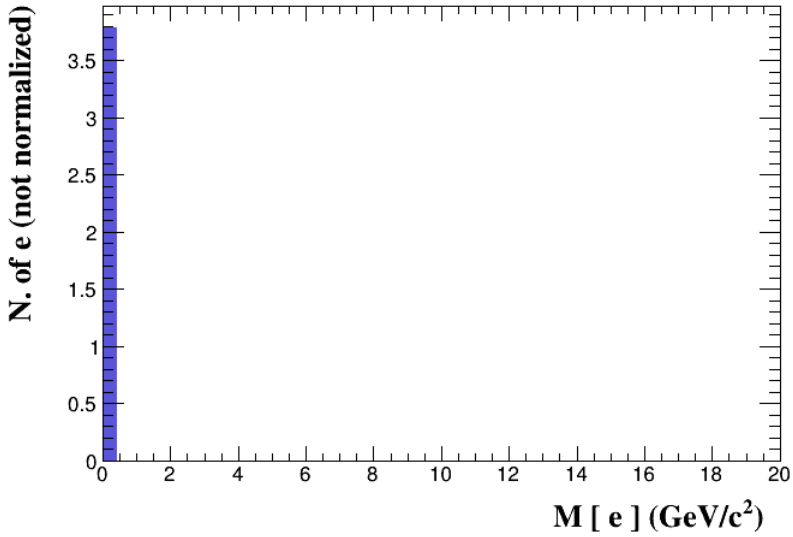

Figure 18.

3.19 Histogram 19

*** Plot: MT (e)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	0.000106287	0.0001899	0.0	0.0

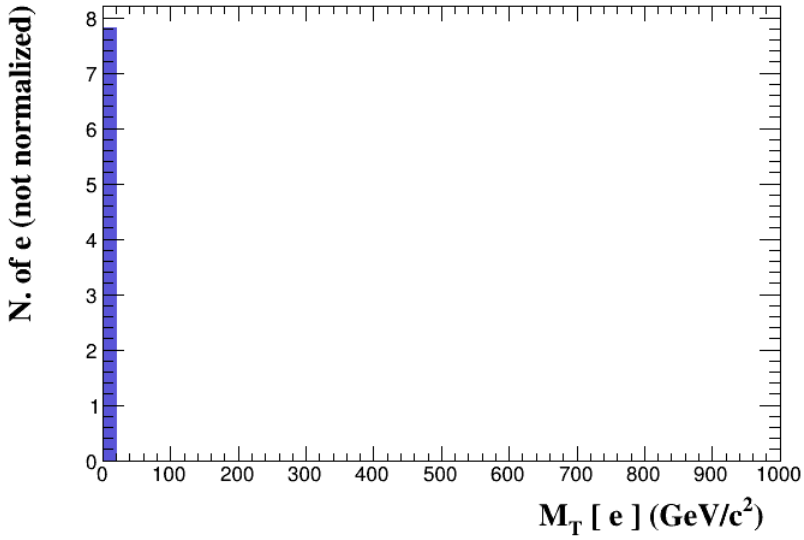

Figure 19.

3.20 Histogram 20

*** Plot: MT_MET (e)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	58.4921	22.25	0.0	0.0

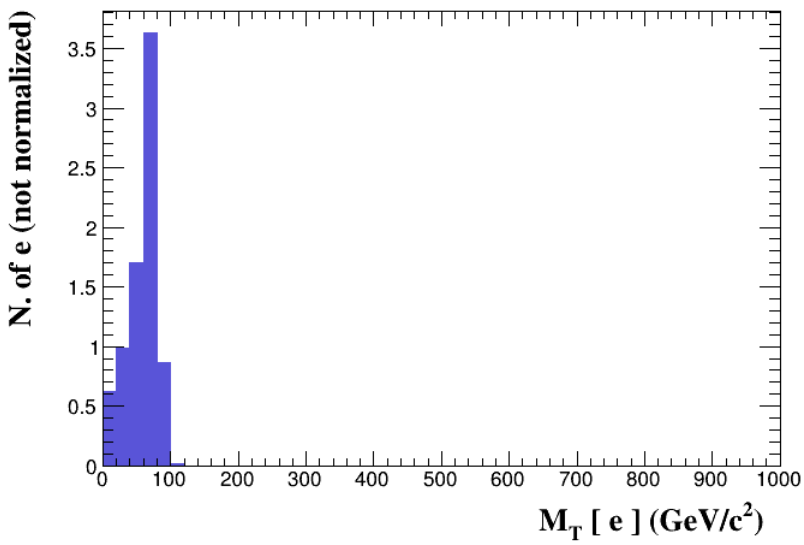

Figure 20.

3.21 Histogram 21

*** Plot: P (e)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	87.8171	77.33	0.0	0.0

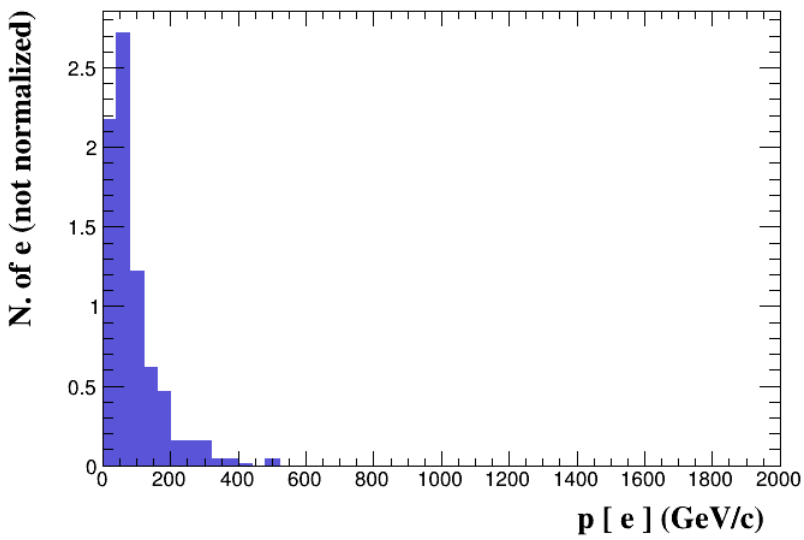

Figure 21.

3.22 Histogram 22

*** Plot: PX (e)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	2.09172	43.07	0.0	0.0

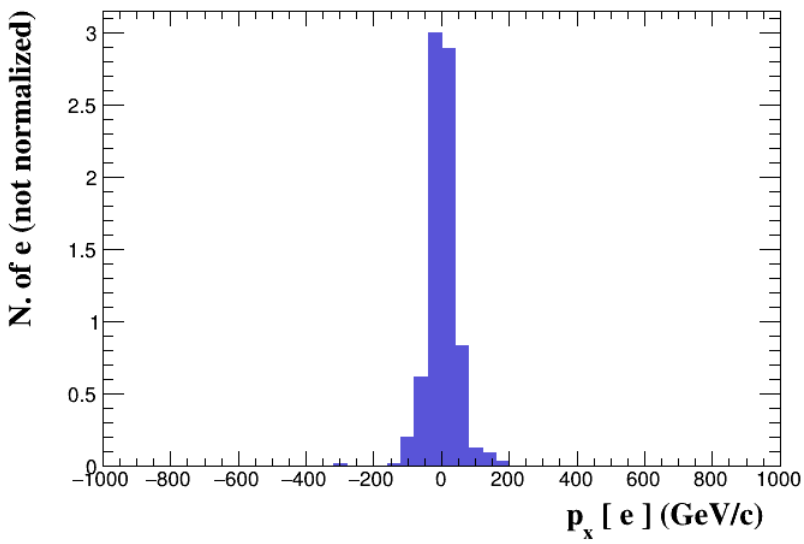

Figure 22.

3.23 Histogram 23

*** Plot: PY (e)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	0.615749	45.63	0.0	0.0

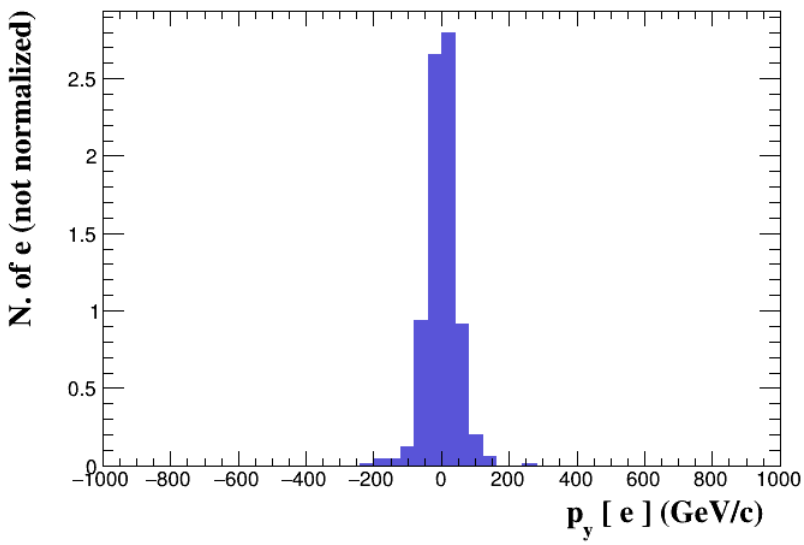

Figure 23.

3.24 Histogram 24

*** Plot: PZ (e)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	7.00982	98.49	0.0	0.0

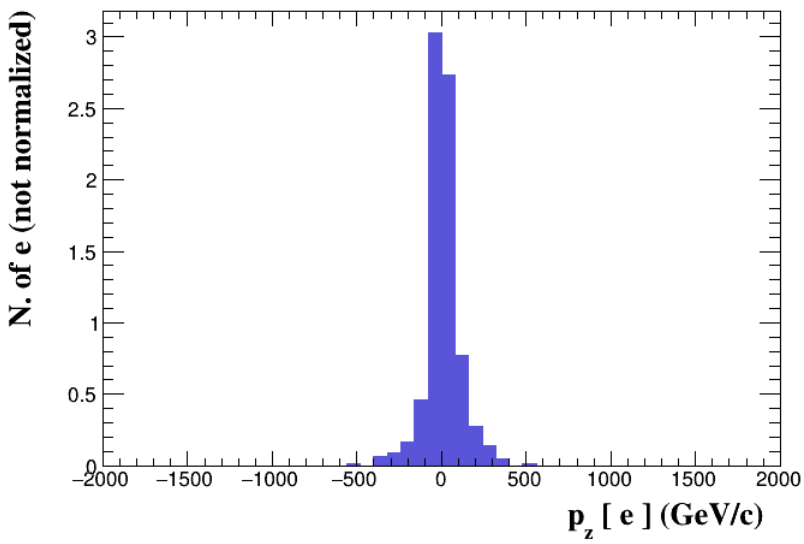

Figure 24.

3.25 Histogram 25

*** Plot: R (e)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	1.93047	0.8241	0.0	0.0

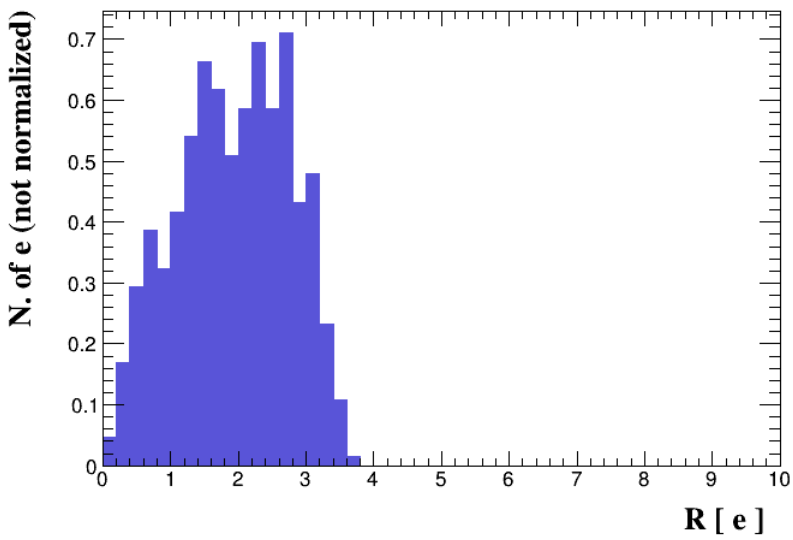

Figure 25.

3.26 Histogram 26

*** Plot: Y (e)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	0.0509938	1.152	0.0	0.0

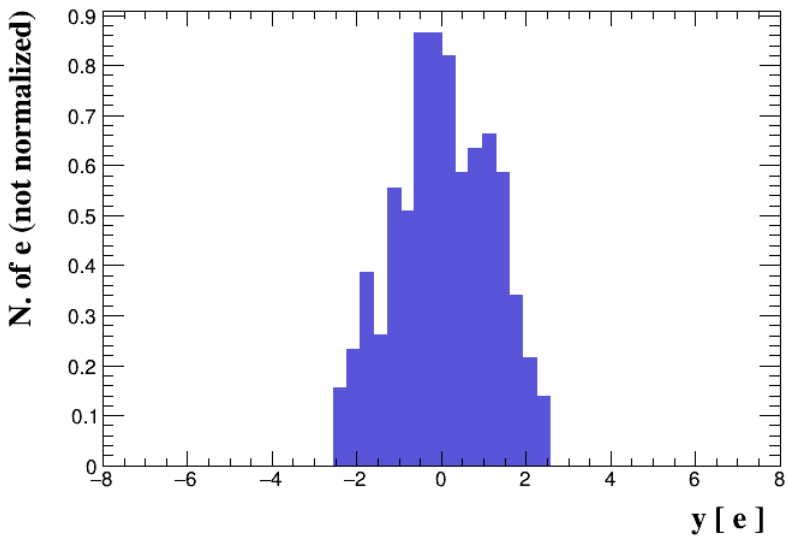

Figure 26.

3.27 Histogram 27

*** Plot: BETA (e)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	1.0	0.0	0.0	0.0

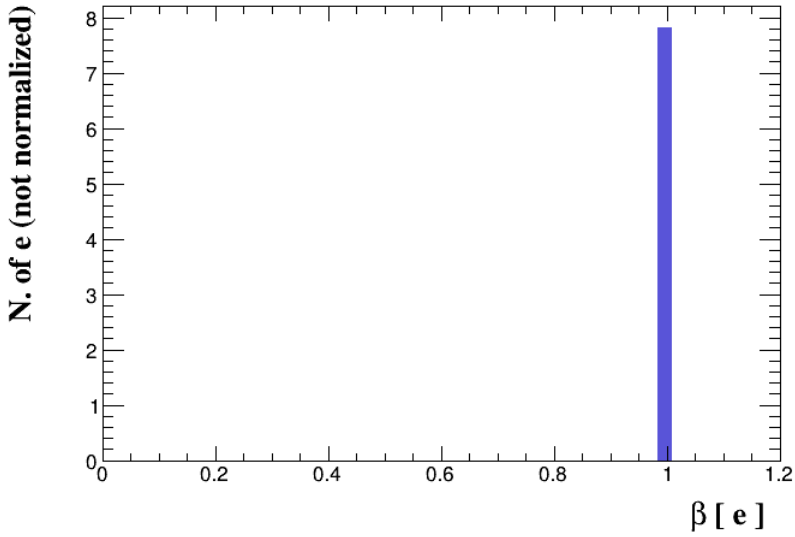

Figure 27.

3.28 Histogram 28

*** Plot: GAMMA (e)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	nan	nan	0.0	48.42

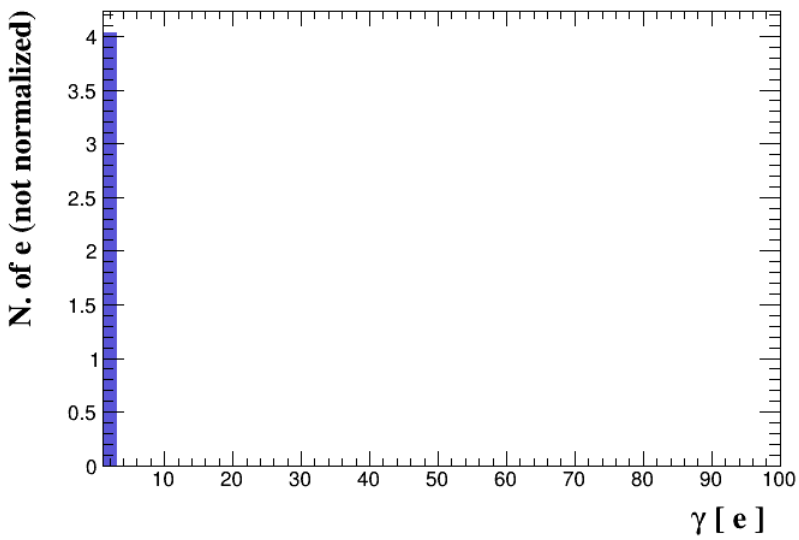

Figure 28.

3.29 Histogram 29

*** Plot: PT (e[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	49.9595	38.03	0.0	0.0

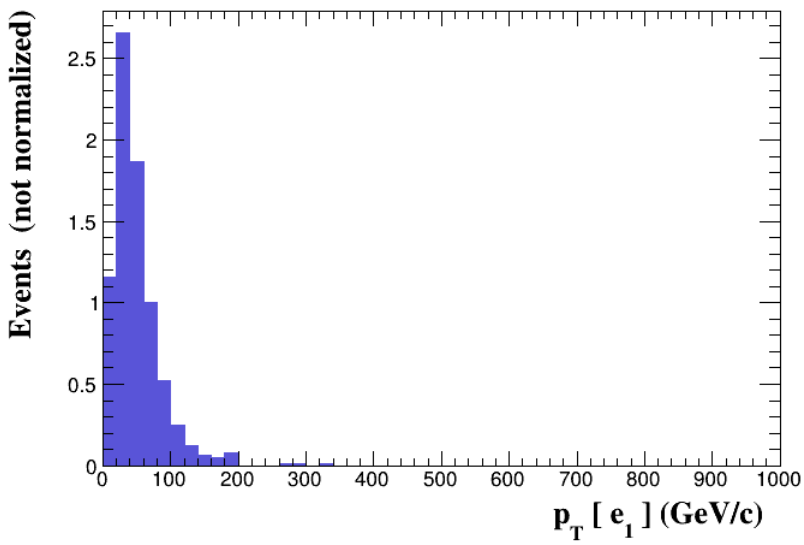

Figure 29.

3.30 Histogram 30

*** Plot: ETA (e[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	0.0509938	1.152	0.0	0.0

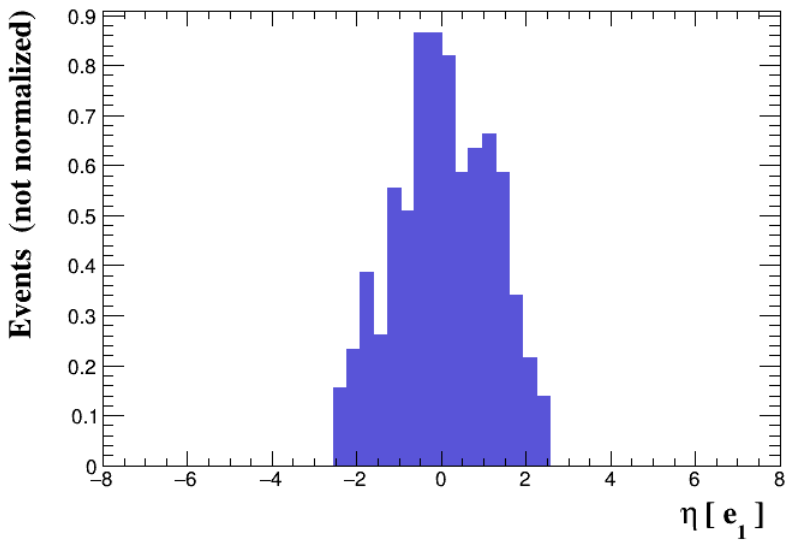

Figure 30.

3.31 Histogram 31

*** Plot: PHI (e[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	0.0207825	1.754	0.0	0.0

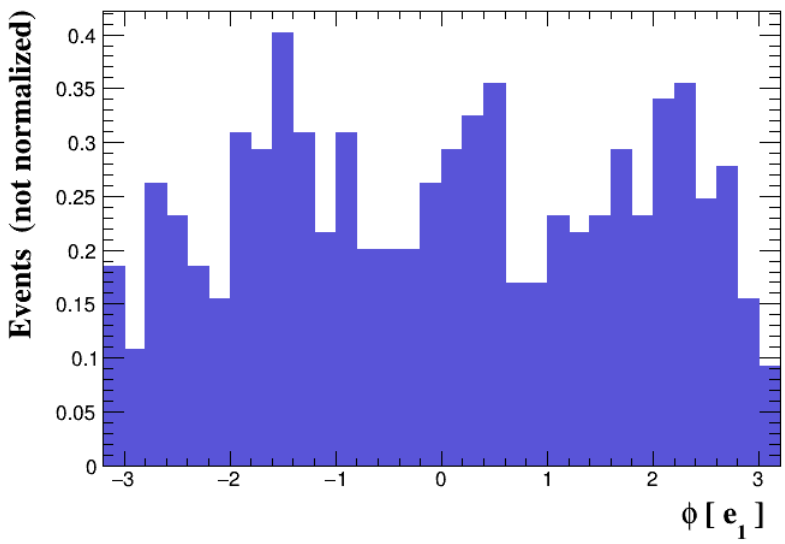

Figure 31.

3.32 Histogram 32

*** Plot: E (e[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	87.8171	77.33	0.0	0.0

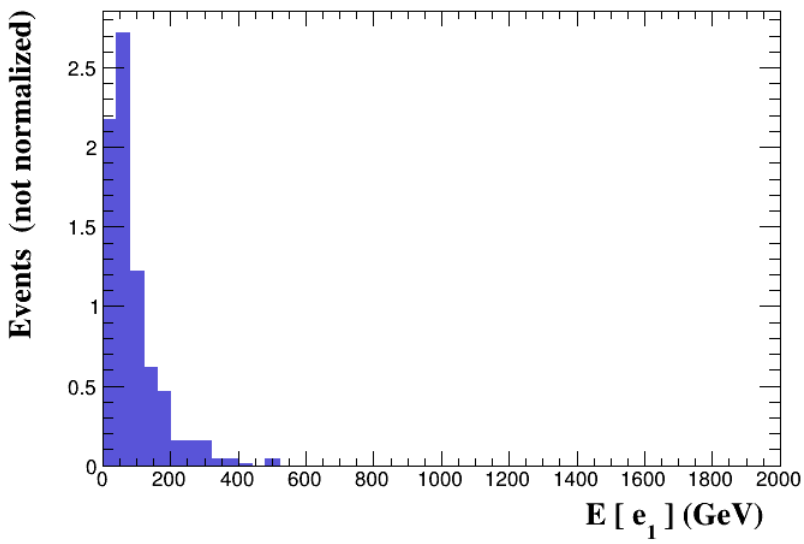

Figure 32.

3.33 Histogram 33

*** Plot: MT_MET (e[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	58.4921	22.25	0.0	0.0

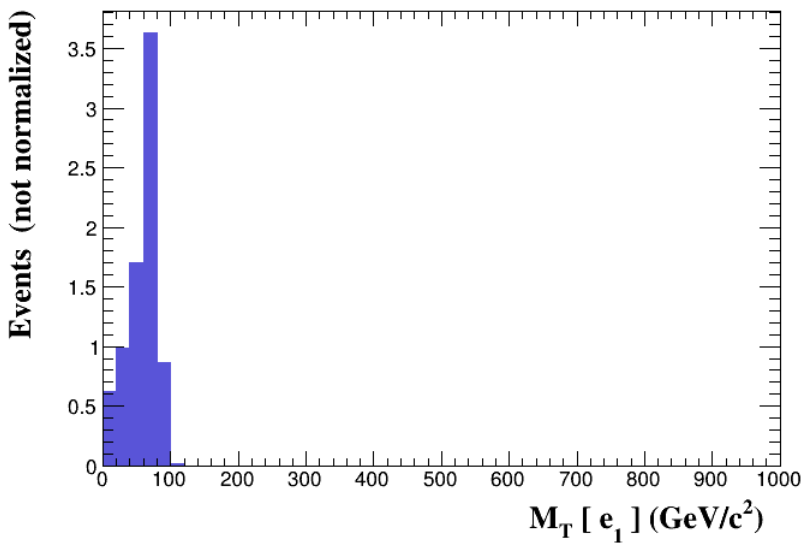

Figure 33.

3.34 Histogram 34

*** Plot: PT (e[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

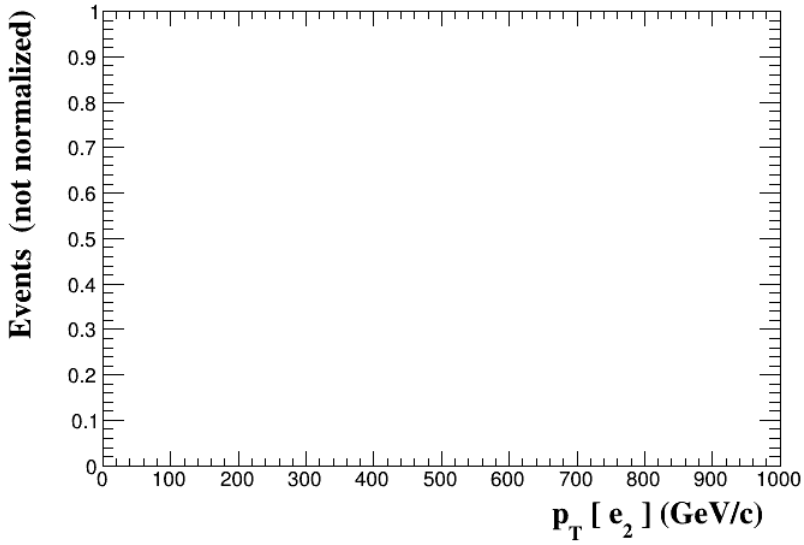

Figure 34.

3.35 Histogram 35

*** Plot: ETA (e[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

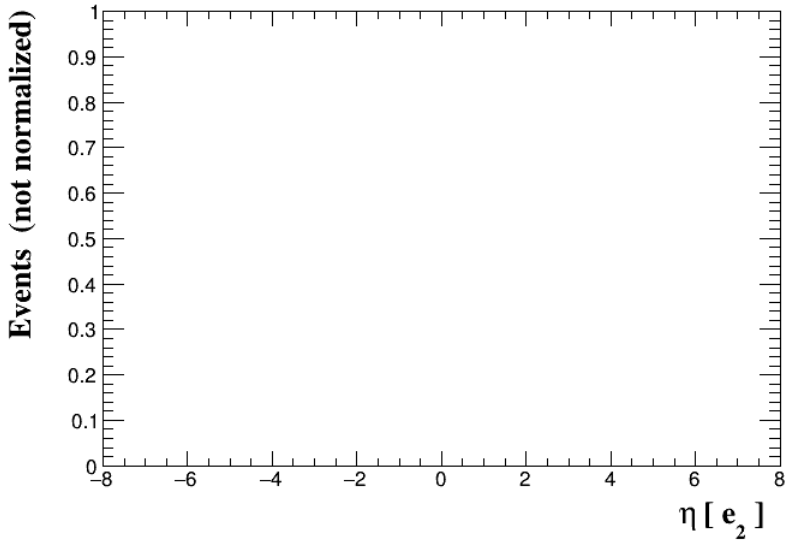

Figure 35.

3.36 Histogram 36

*** Plot: PHI (e[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

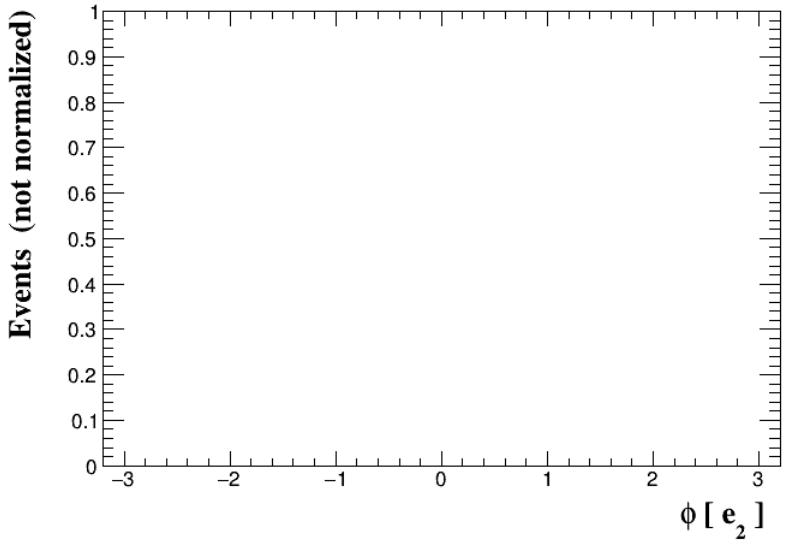

Figure 36.

3.37 Histogram 37

*** Plot: E (e[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

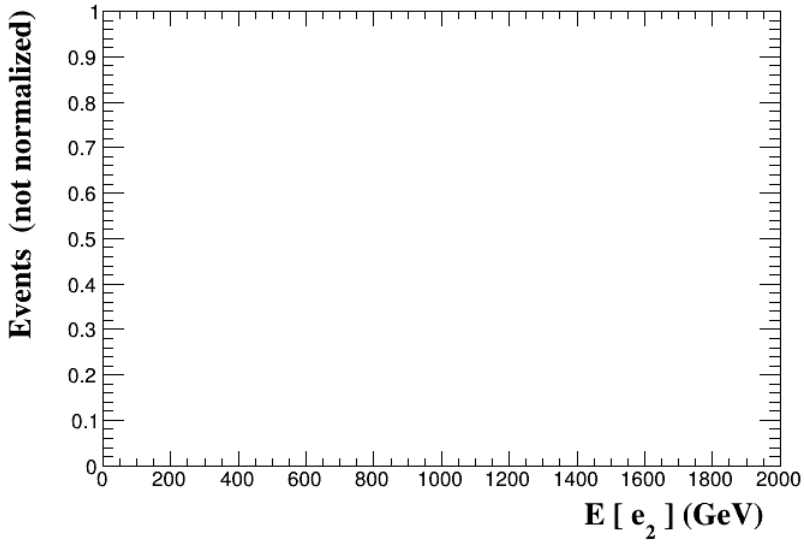

Figure 37.

3.38 Histogram 38

*** Plot: MT_MET (e[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

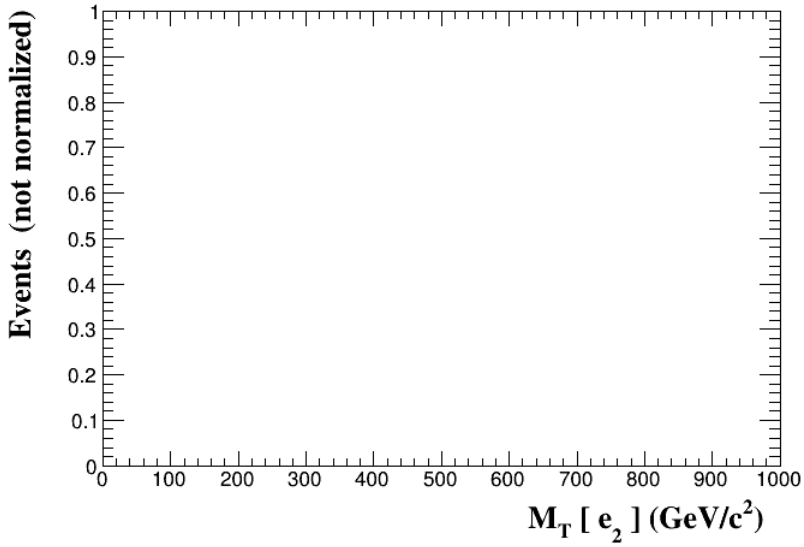

Figure 38.

3.39 Histogram 39

*** Plot: M (e[1] e[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

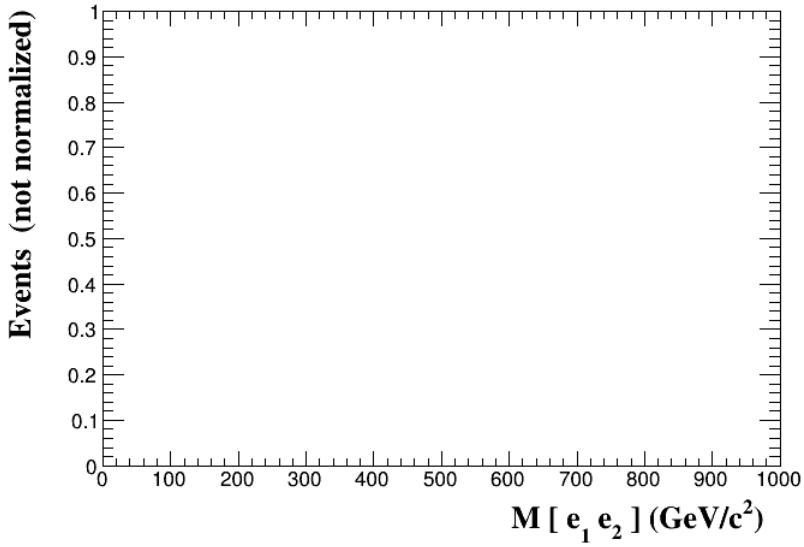

Figure 39.

3.40 Histogram 40

*** Plot: PT (e[1] e[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

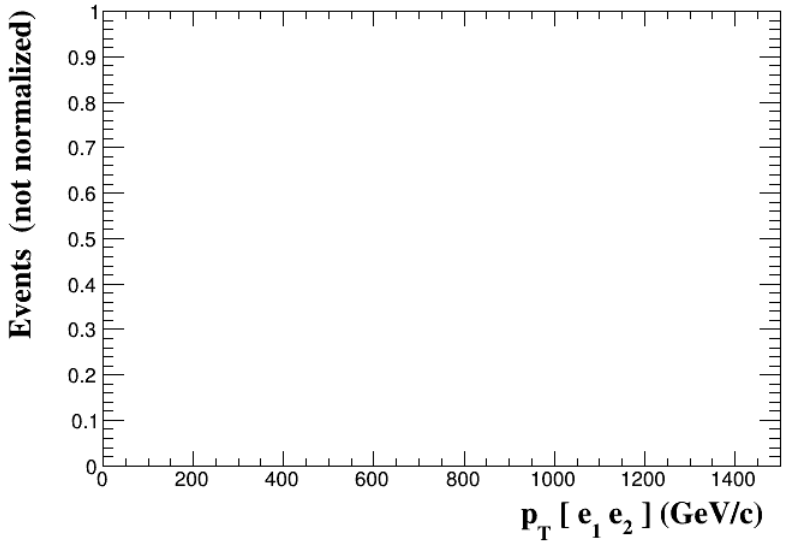

Figure 40.

3.41 Histogram 41

*** Plot: DELTAR (e[1] , e[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

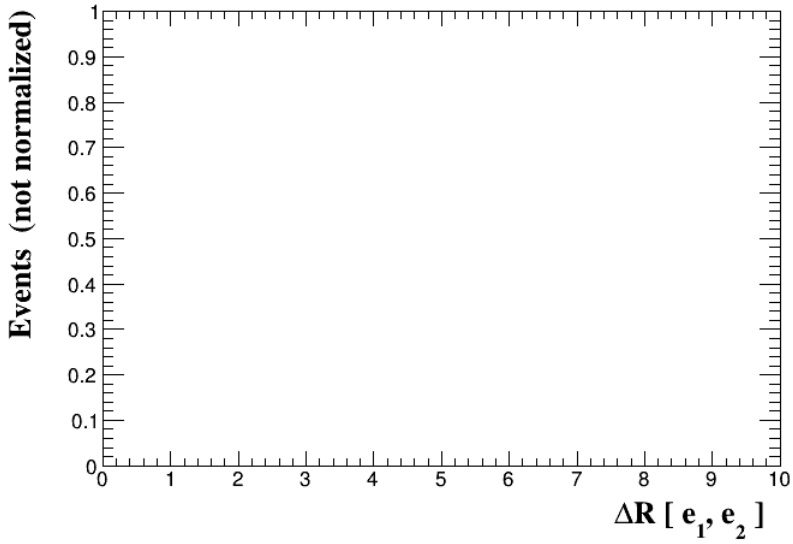

Figure 41.

3.42 Histogram 42

*** Plot: DPHI_0_PI (e[1] , e[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

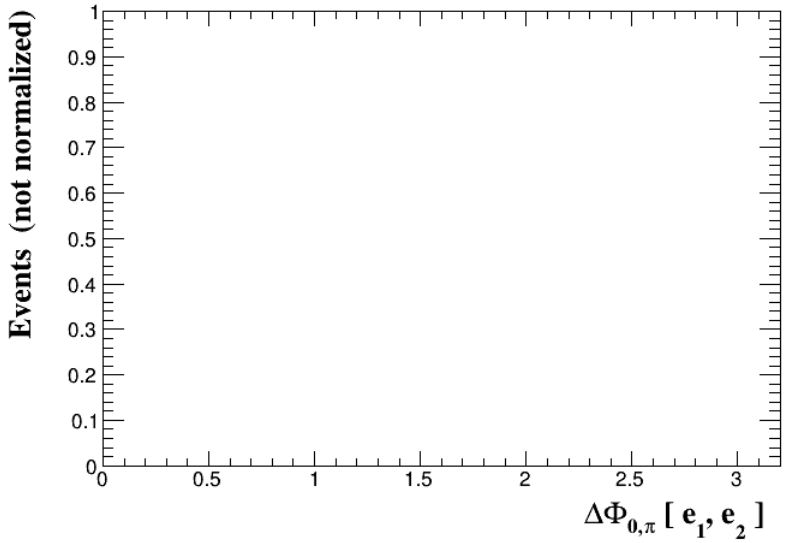

Figure 42.

3.43 Histogram 43

*** Plot: DPHI_0_2PI (e[1] , e[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

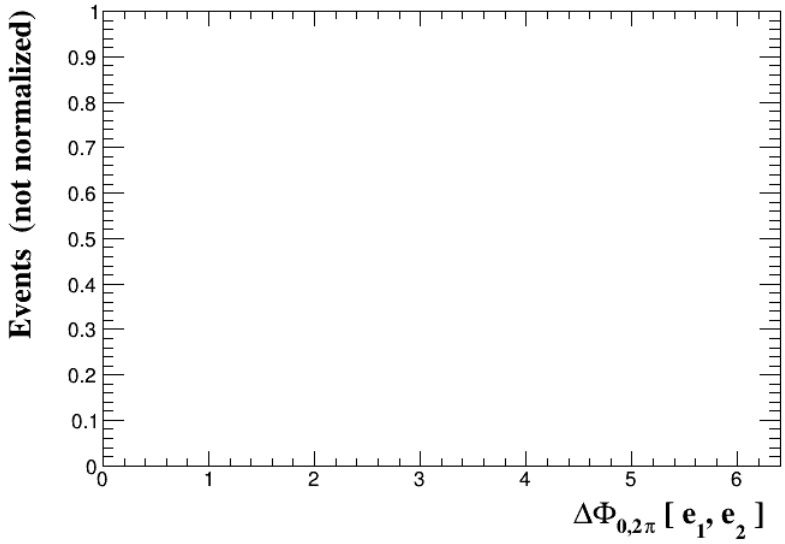

Figure 43.

3.44 Histogram 44

*** Plot: PT (mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	48.5205	33.1	0.0	0.0

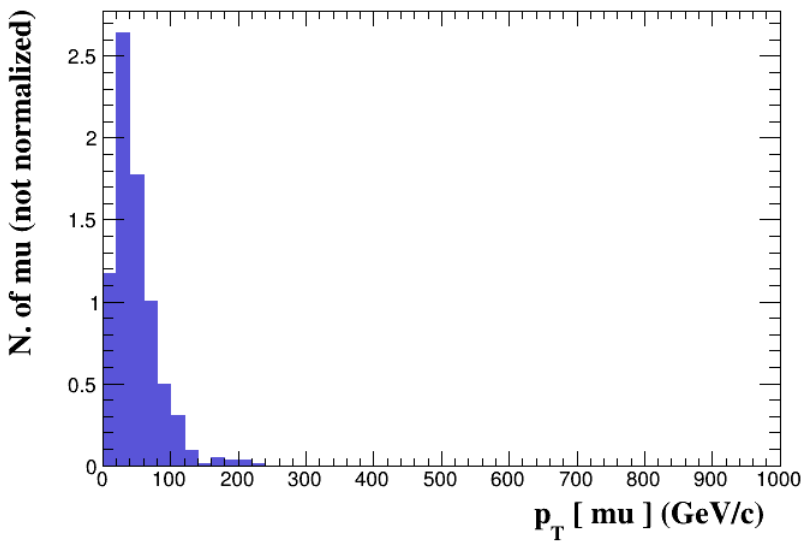

Figure 44.

3.45 Histogram 45

*** Plot: ETA (mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	0.0359047	1.13	0.0	0.0

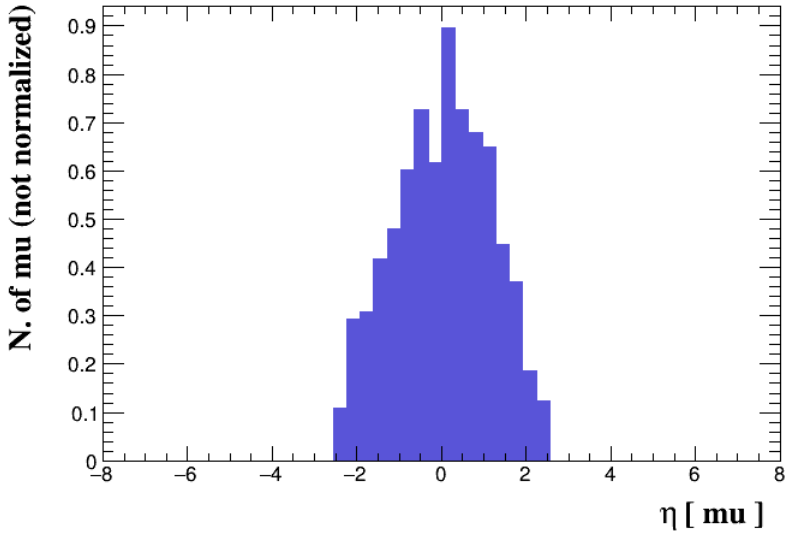

Figure 45.

3.46 Histogram 46

*** Plot: ABSETA (mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	0.938579	0.6297	0.0	0.0

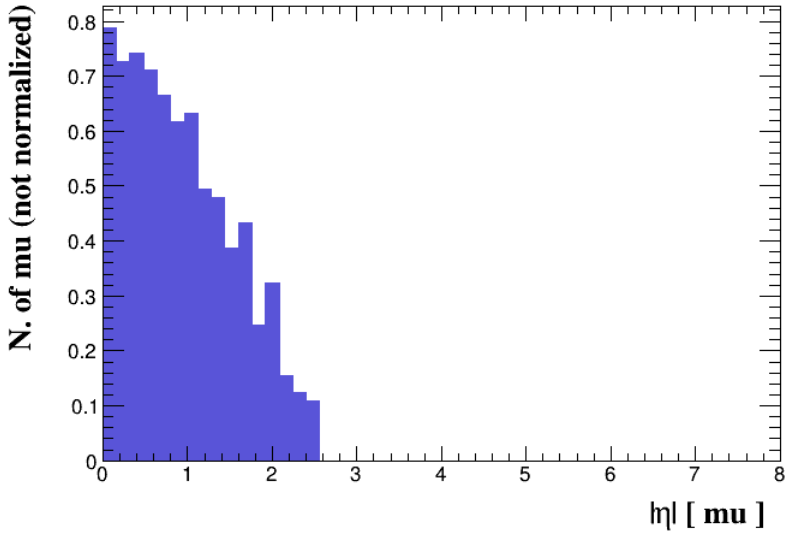

Figure 46.

3.47 Histogram 47

*** Plot: PHI (mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	-0.0970943	1.804	0.0	0.0

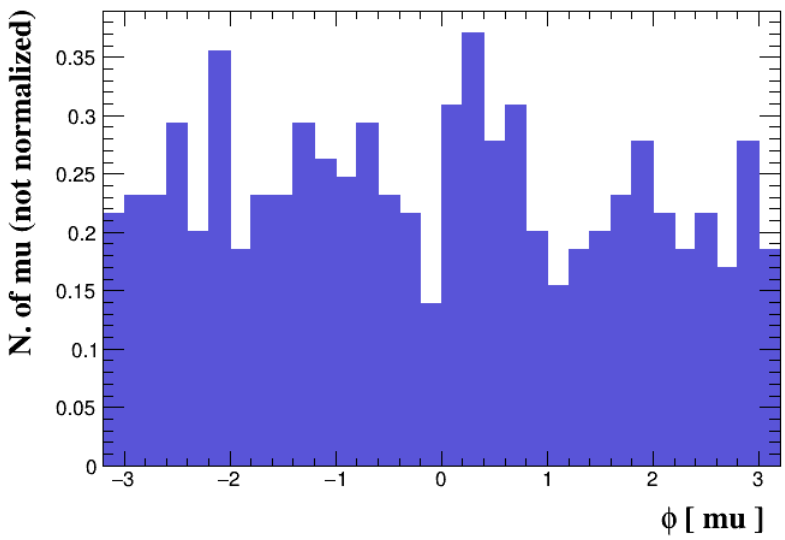

Figure 47.

3.48 Histogram 48

*** Plot: E (mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	83.6679	68.85	0.0	0.0

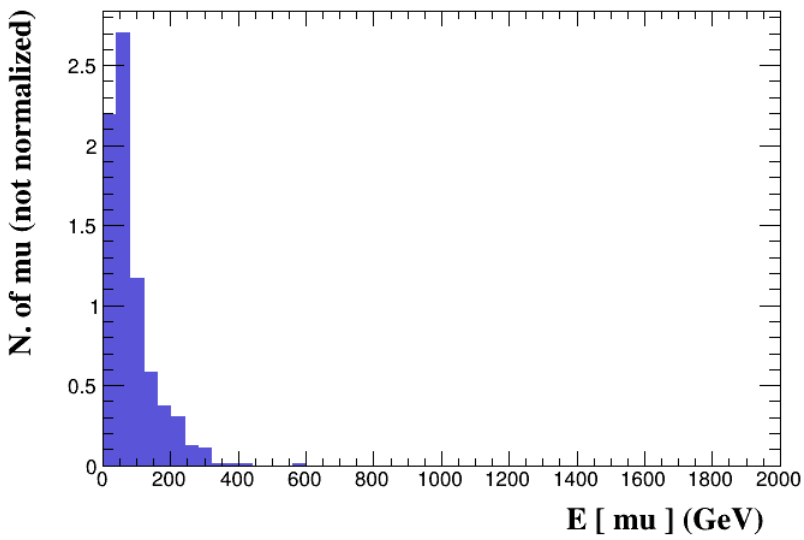

Figure 48.

3.49 Histogram 49

*** Plot: ET (mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	48.5205	33.1	0.0	0.0

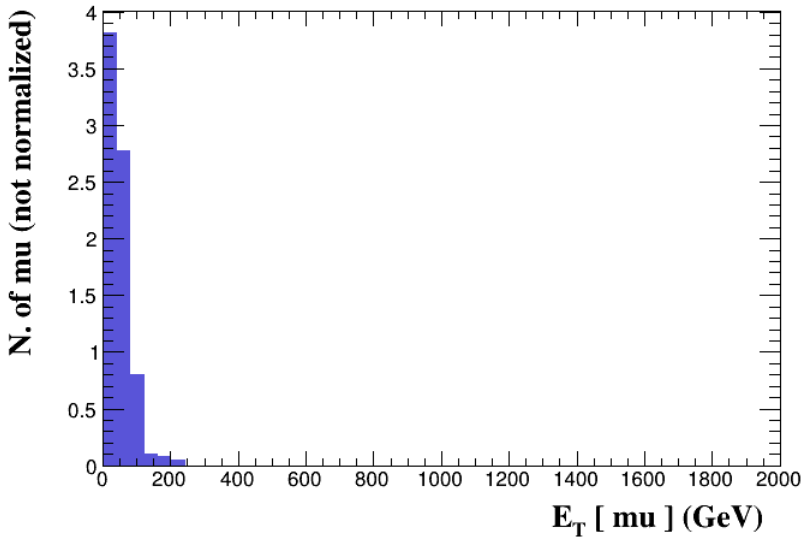

Figure 49.

3.50 Histogram 50

*** Plot: M (mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	5.6886e-06	0.0005586	47.57	0.0

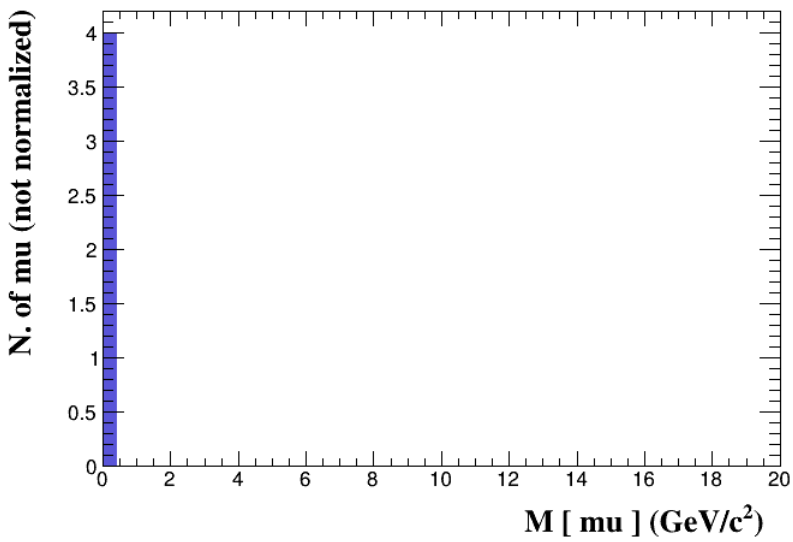

Figure 50.

3.51 Histogram 51

*** Plot: MT (mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	0.000106791	0.0001948	0.0	0.0

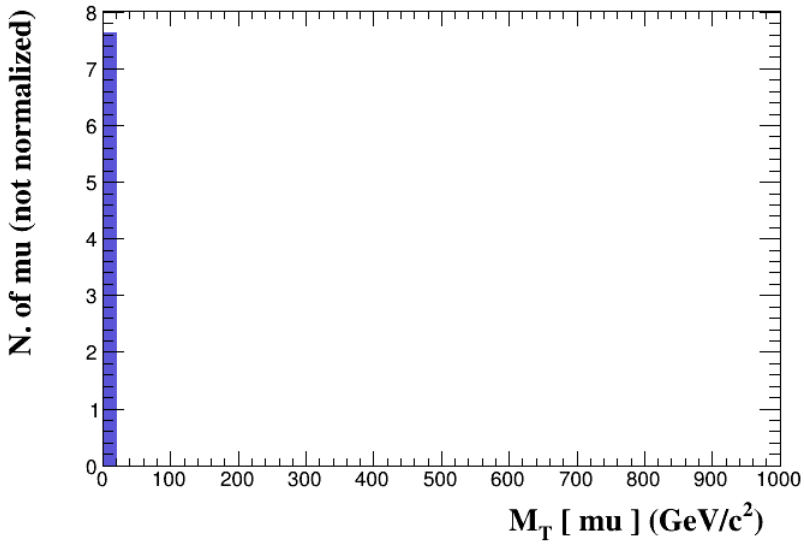

Figure 51.

3.52 Histogram 52

*** Plot: MT_MET (mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	59.2324	21.72	0.0	0.0

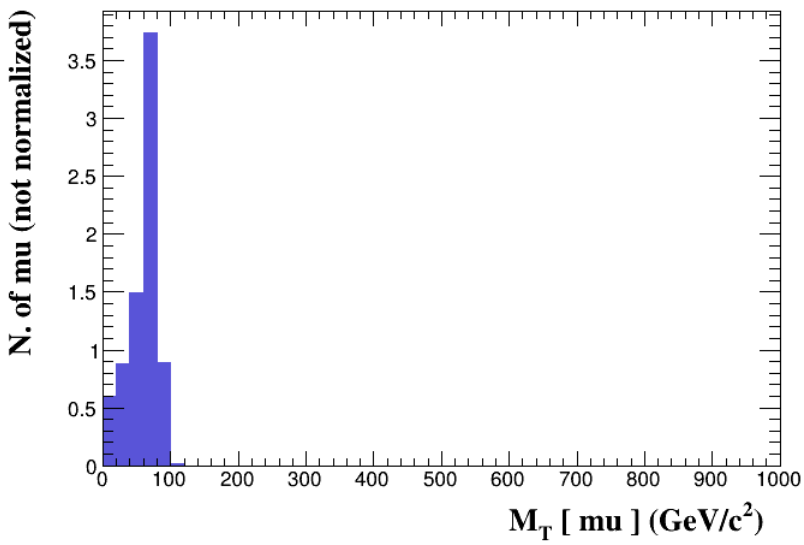

Figure 52.

3.53 Histogram 53

*** Plot: P (mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	83.6679	68.85	0.0	0.0

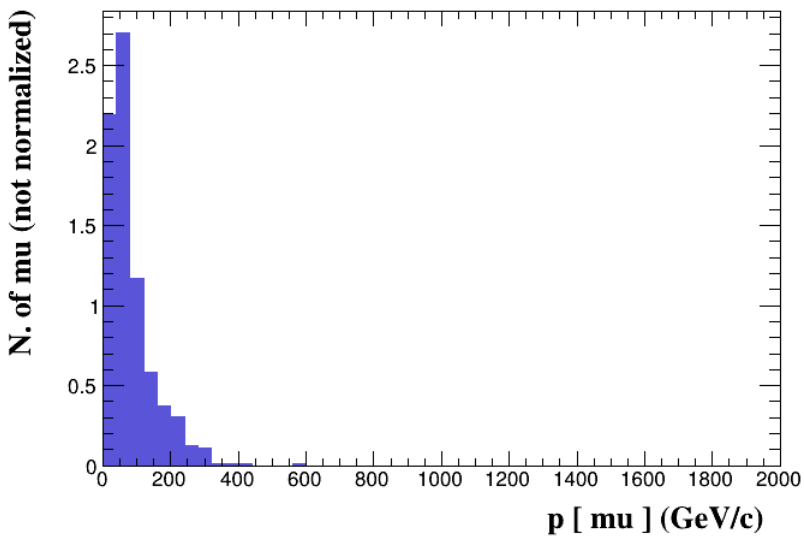

Figure 53.

3.54 Histogram 54

*** Plot: PX (mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	1.91001	42.25	0.0	0.0

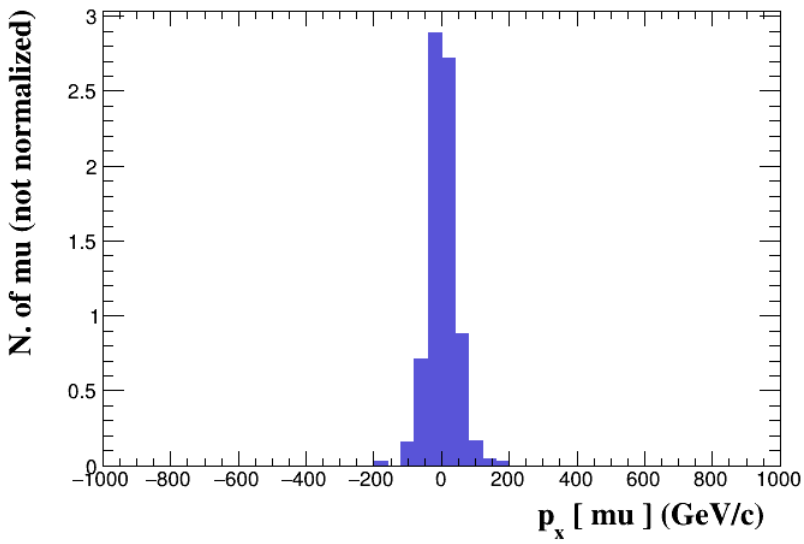

Figure 54.

3.55 Histogram 55

*** Plot: PY (mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	-0.97706	40.74	0.0	0.0

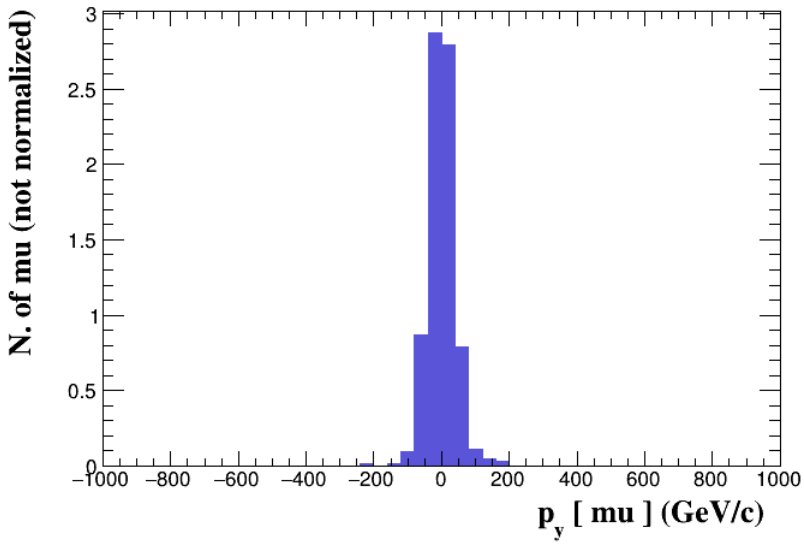

Figure 55.

3.56 Histogram 56

*** Plot: PZ (mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	-0.0695523	91.06	0.0	0.0

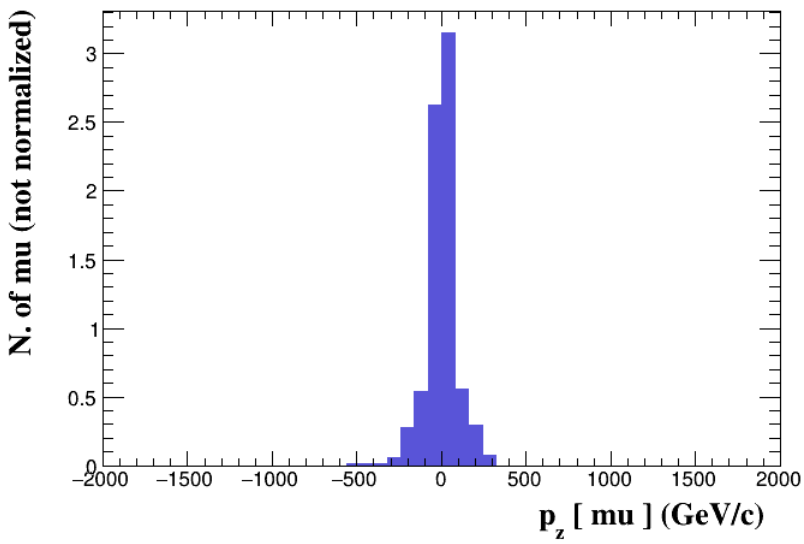

Figure 56.

3.57 Histogram 57

*** Plot: R (mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	1.9459	0.8686	0.0	0.0

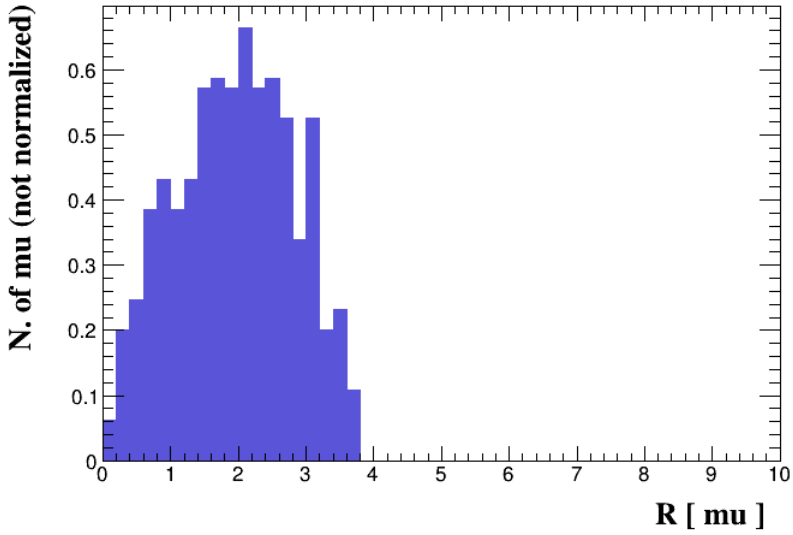

Figure 57.

3.58 Histogram 58

*** Plot: Y (mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	0.0359047	1.13	0.0	0.0

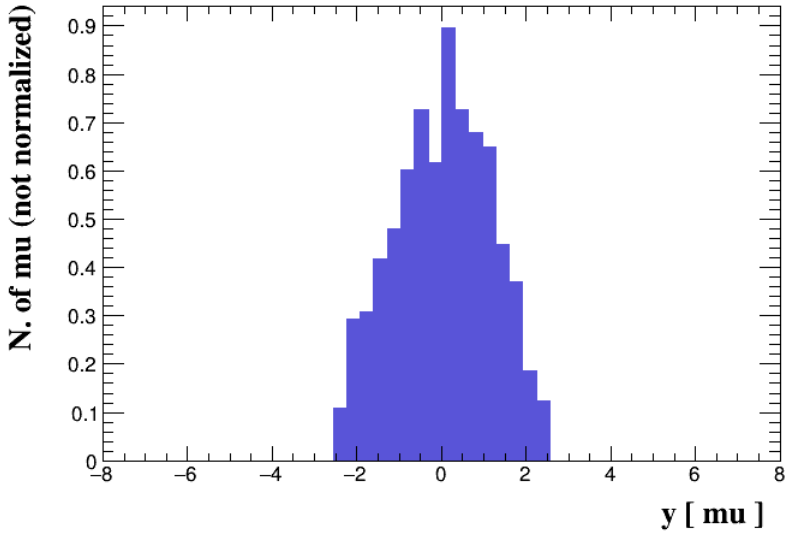

Figure 58.

3.59 Histogram 59

*** Plot: BETA (mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	1.0	0.0	0.0	0.0

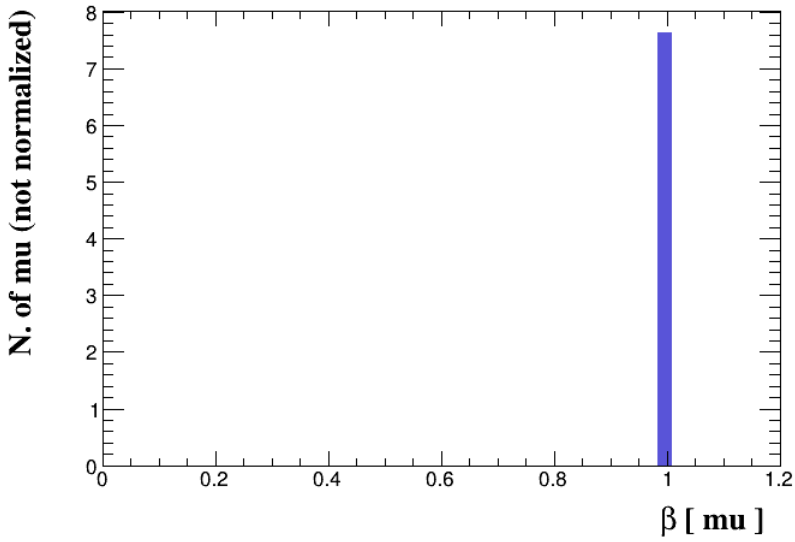

Figure 59.

3.60 Histogram 60

*** Plot: GAMMA (mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	nan	nan	0.0	52.43

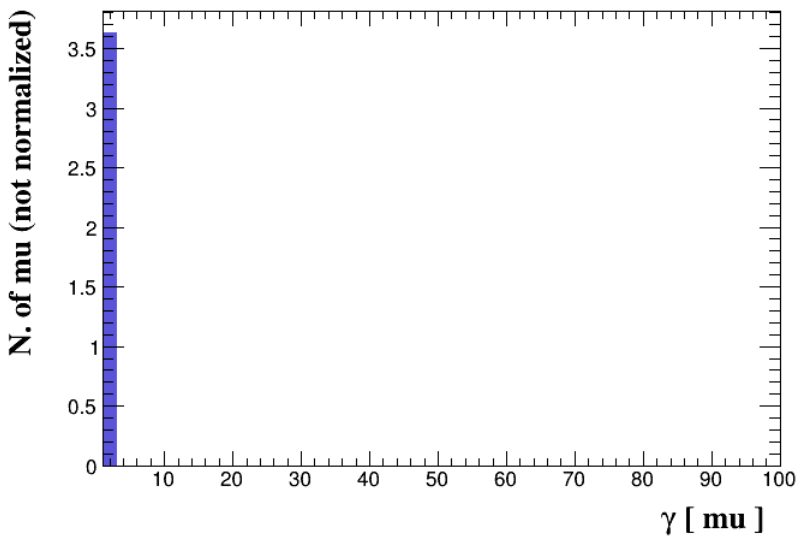

Figure 60.

3.61 Histogram 61

*** Plot: PT (mu[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	48.5205	33.1	0.0	0.0

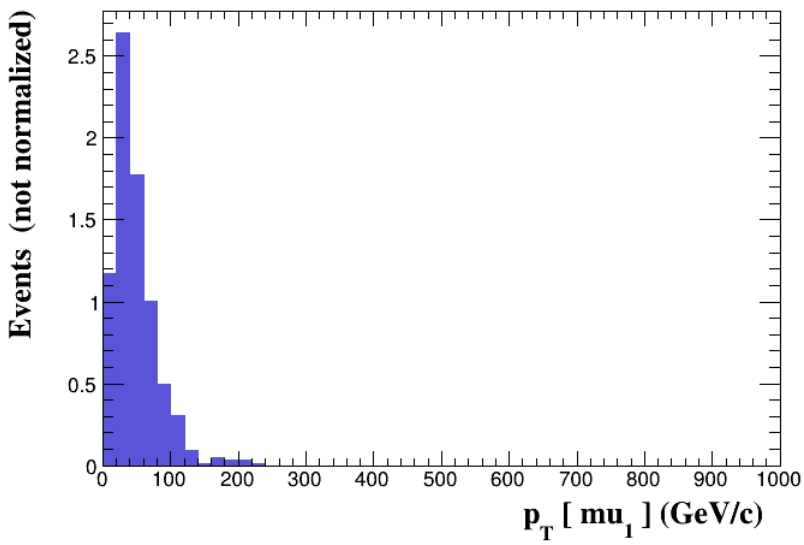

Figure 61.

3.62 Histogram 62

*** Plot: ETA (mu[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	0.0359047	1.13	0.0	0.0

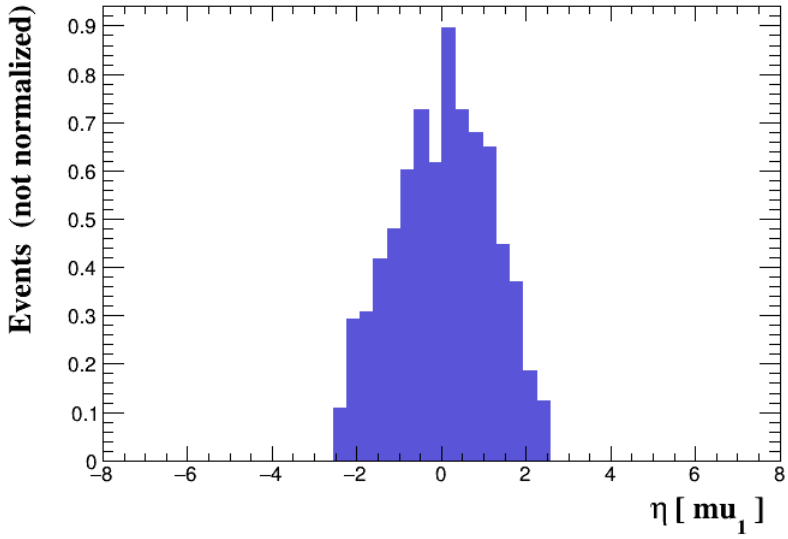

Figure 62.

3.63 Histogram 63

*** Plot: PHI (mu[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	-0.0970943	1.804	0.0	0.0

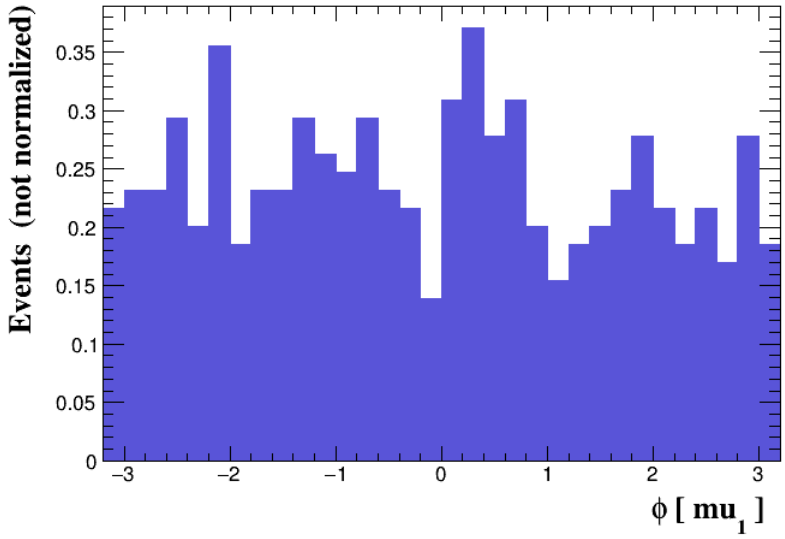

Figure 63.

3.64 Histogram 64

*** Plot: E (mu[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	83.6679	68.85	0.0	0.0

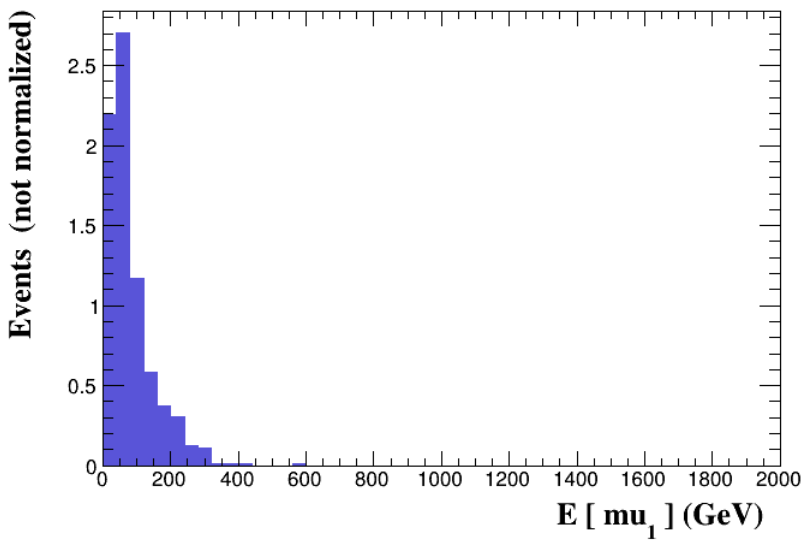

Figure 64.

3.65 Histogram 65

*** Plot: MT_MET (mu[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	59.2324	21.72	0.0	0.0

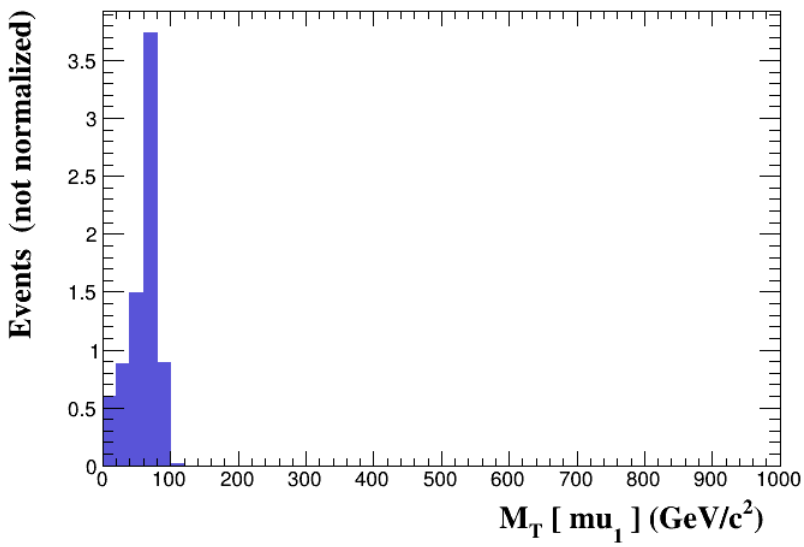

Figure 65.

3.66 Histogram 66

*** Plot: PT (mu[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

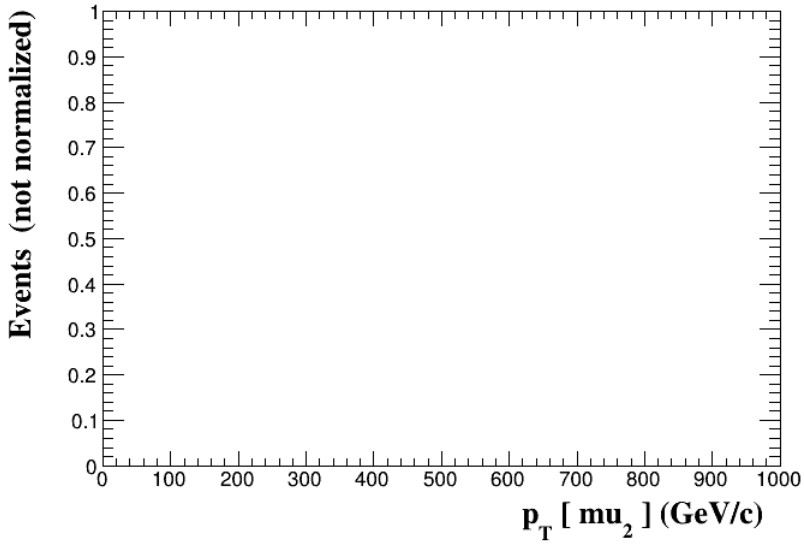

Figure 66.

3.67 Histogram 67

*** Plot: ETA (mu[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

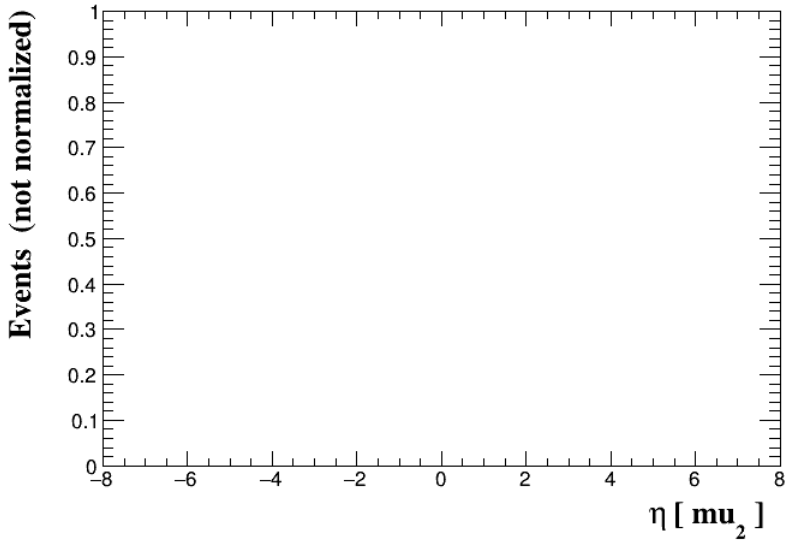

Figure 67.

3.68 Histogram 68

*** Plot: PHI (mu[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

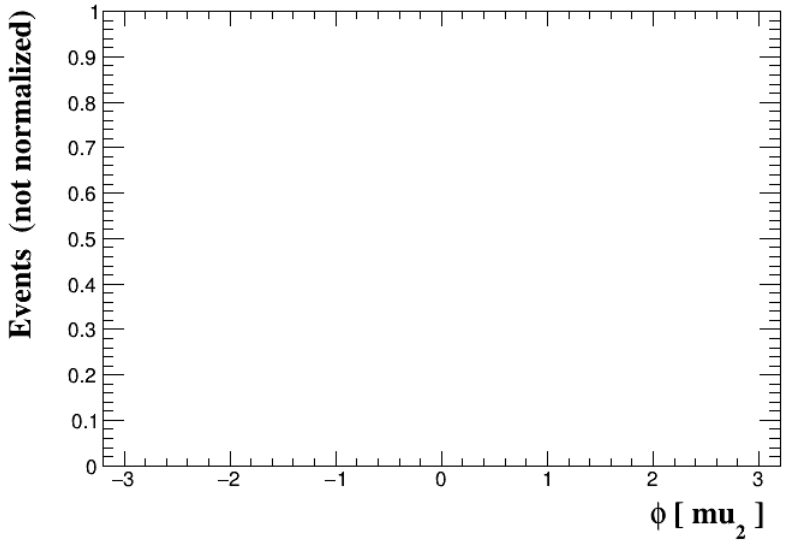

Figure 68.

3.69 Histogram 69

*** Plot: E (mu[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

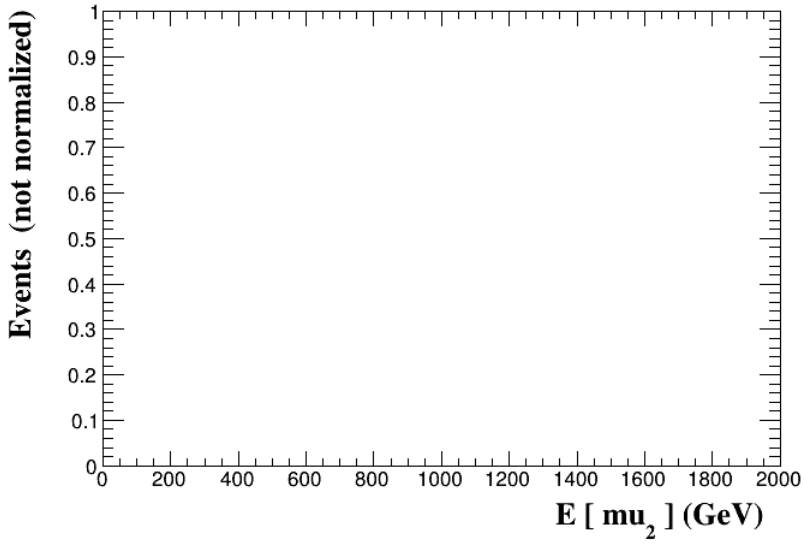

Figure 69.

3.70 Histogram 70

*** Plot: MT_MET (mu[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

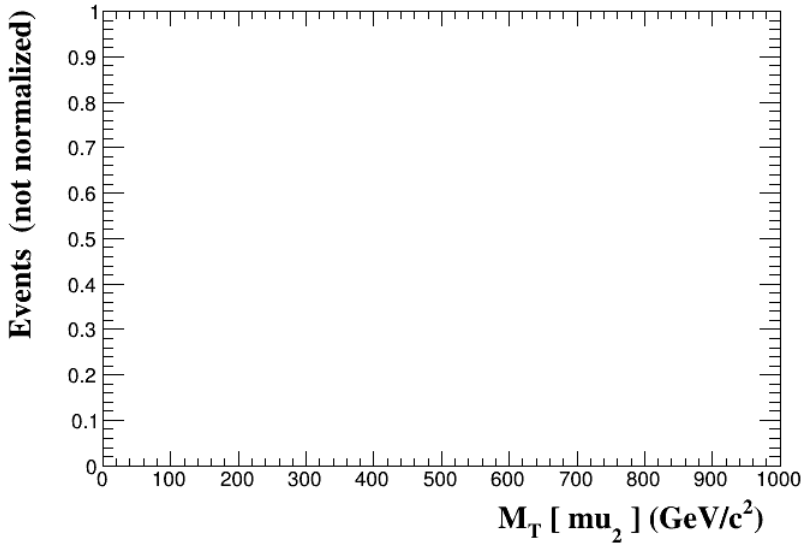

Figure 70.

3.71 Histogram 71

*** Plot: M (mu[1] mu[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

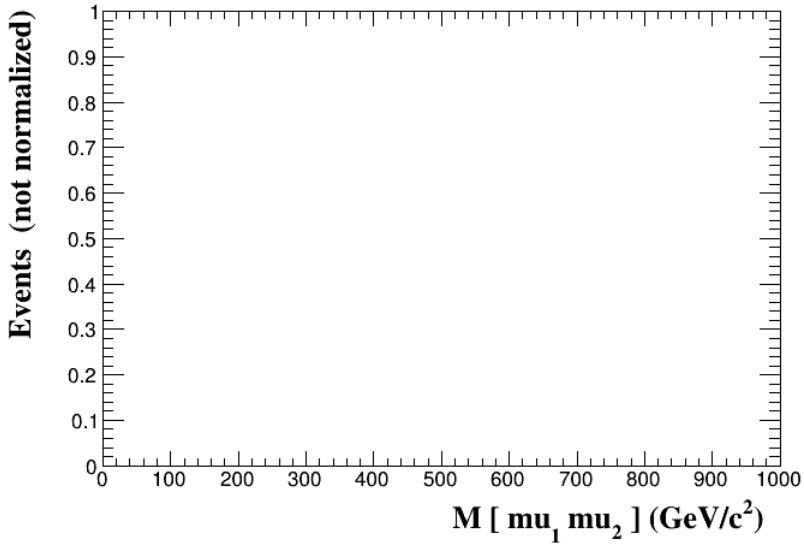

Figure 71.

3.72 Histogram 72

*** Plot: PT (mu[1] mu[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

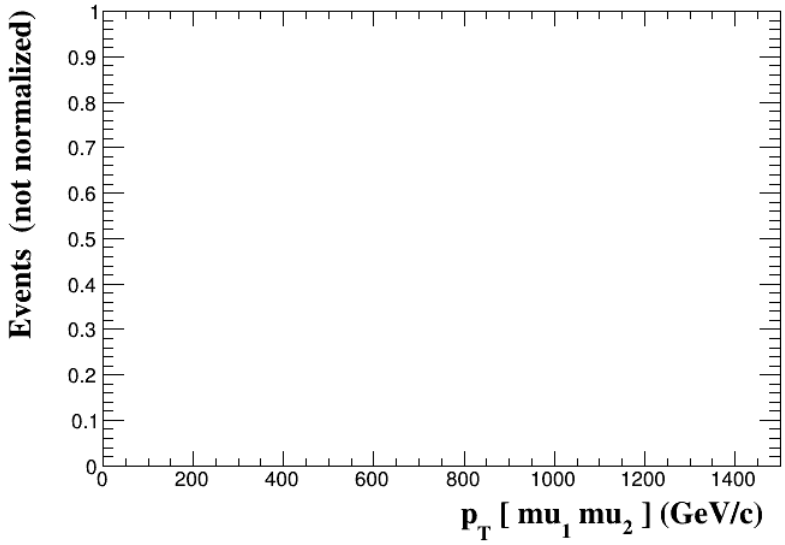

Figure 72.

3.73 Histogram 73

*** Plot: DELTAR (mu[1] , mu[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

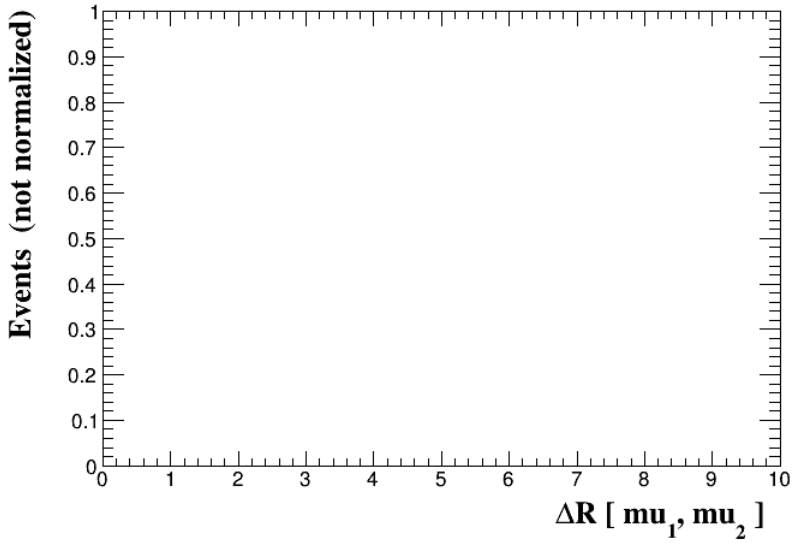

Figure 73.

3.74 Histogram 74

*** Plot: DPHI_0_PI (mu[1] , mu[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

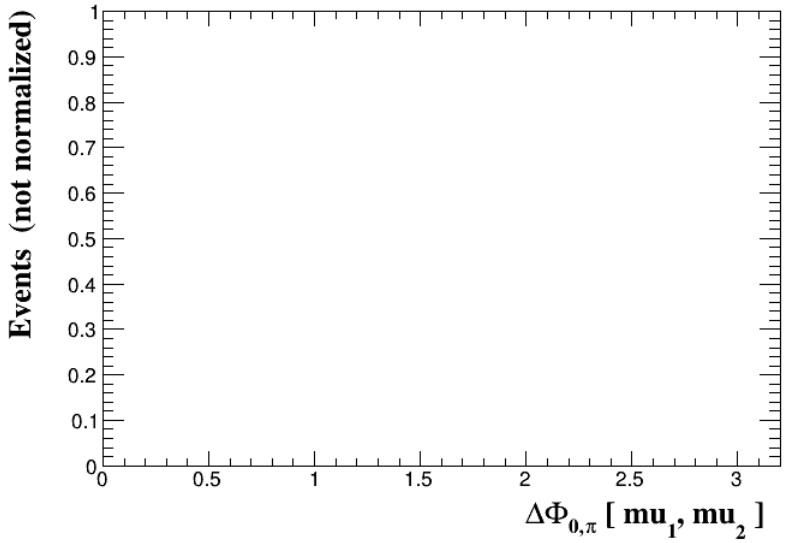

Figure 74.

3.75 Histogram 75

*** Plot: DPHI_0_2PI (mu[1] , mu[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

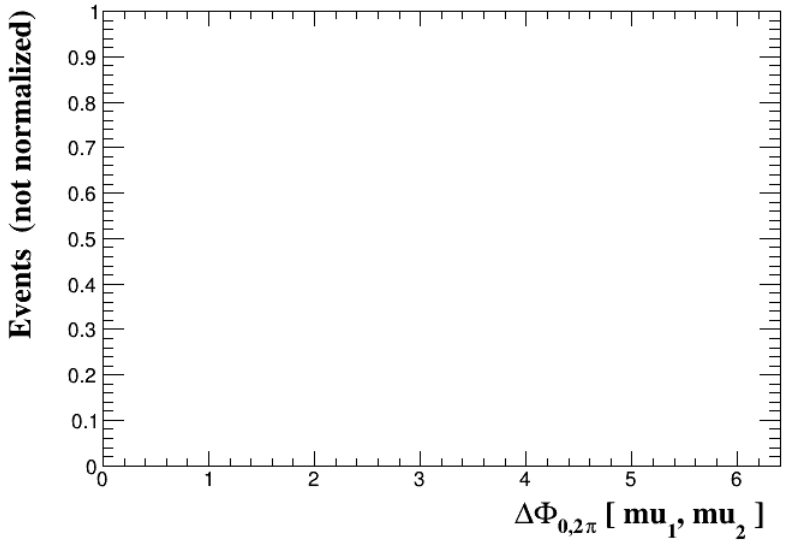

Figure 75.

3.76 Histogram 76

*** Plot: PT (b)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	72.2733	38.02	0.0	0.0

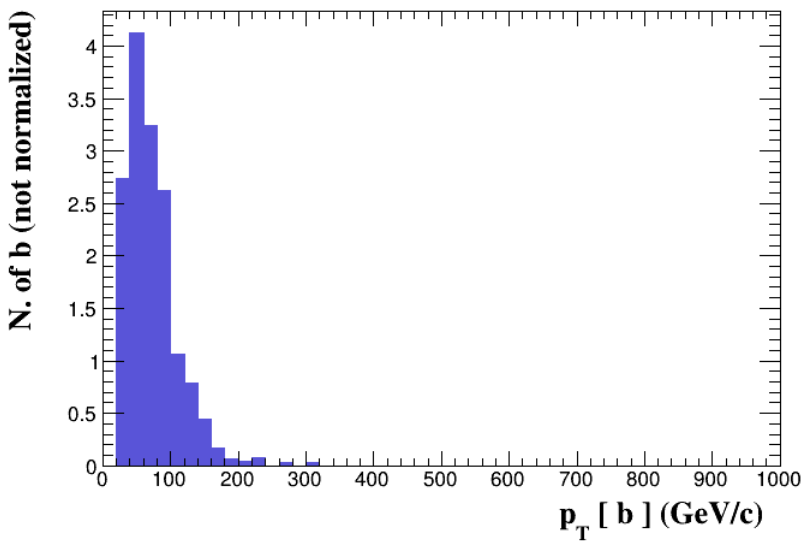

Figure 76.

3.77 Histogram 77

*** Plot: ETA (b)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	0.048521	1.264	0.0	0.0

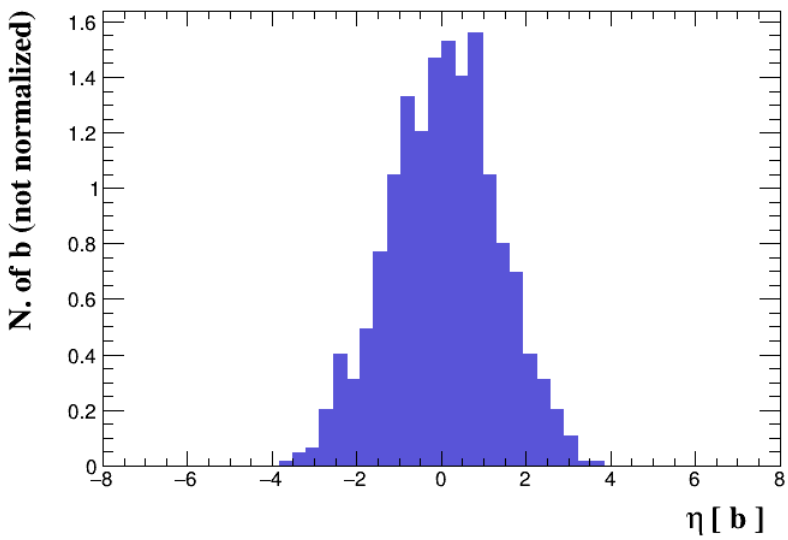

Figure 77.

3.78 Histogram 78

*** Plot: ABSETA (b)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	1.02001	0.7485	0.0	0.0

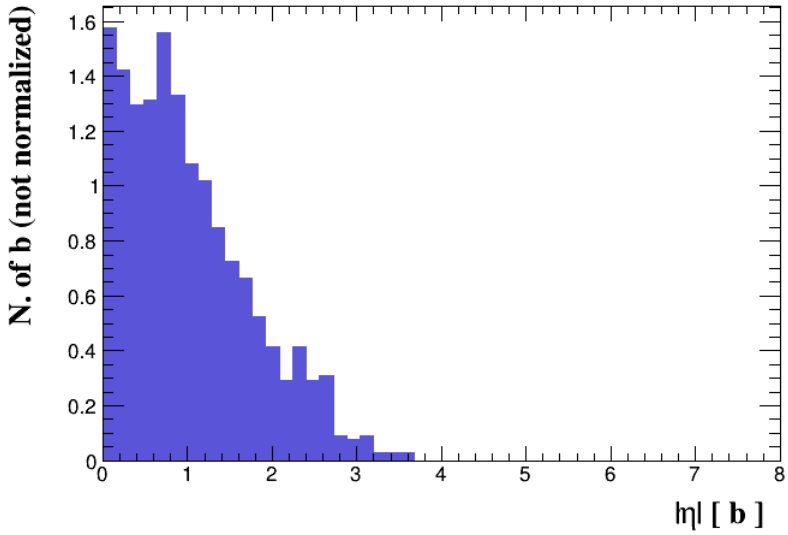

Figure 78.

3.79 Histogram 79

*** Plot: PHI (b)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	0.0257839	1.82	0.0	0.0

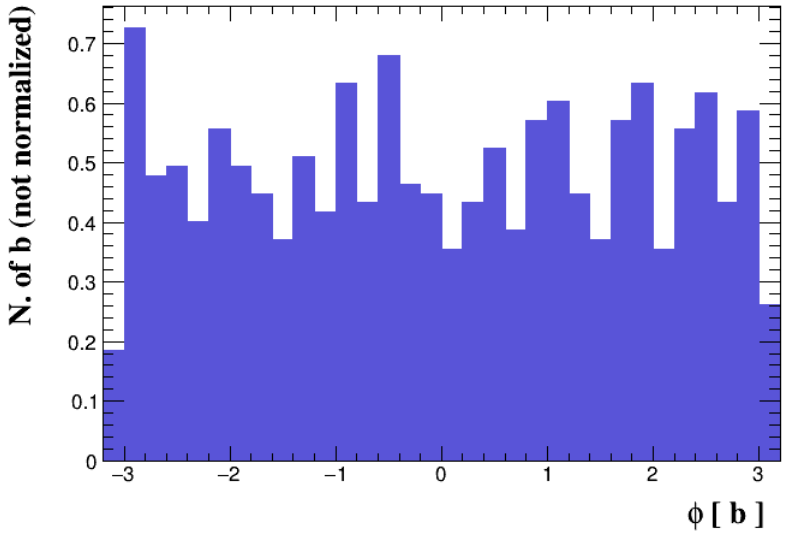

Figure 79.

3.80 Histogram 80

*** Plot: E (b)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	139.461	106.8	0.0	0.0

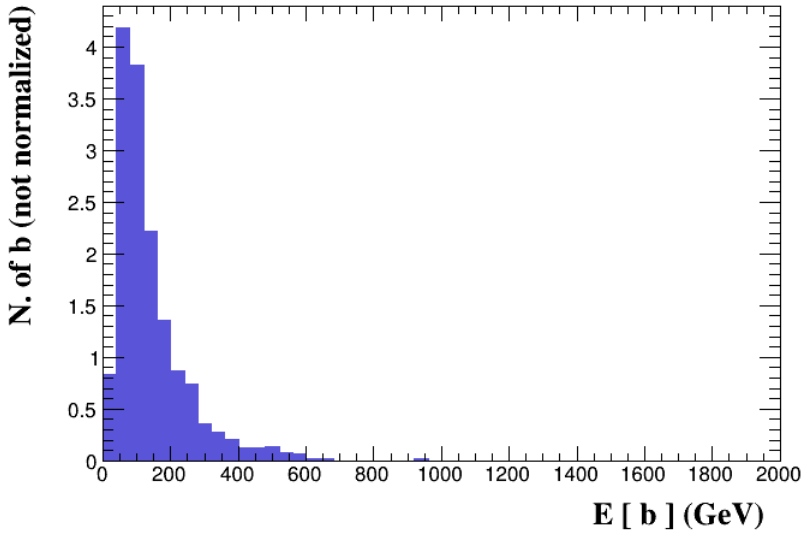

Figure 80.

3.81 Histogram 81

*** Plot: ET (b)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	72.3671	37.99	0.0	0.0

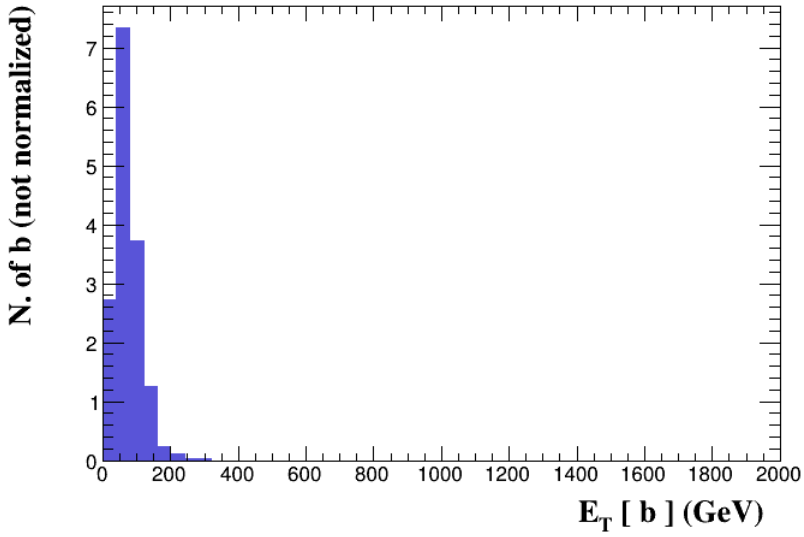

Figure 81.

3.82 Histogram 82

*** Plot: M (b)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	4.7	0.001892	0.0	0.0

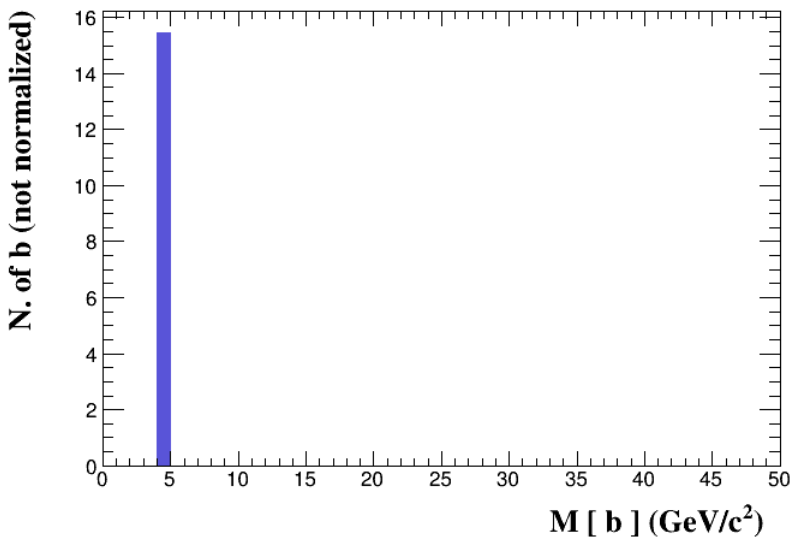

Figure 82.

3.83 Histogram 83

*** Plot: MT (b)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	3.10162	1.297	0.0	0.0

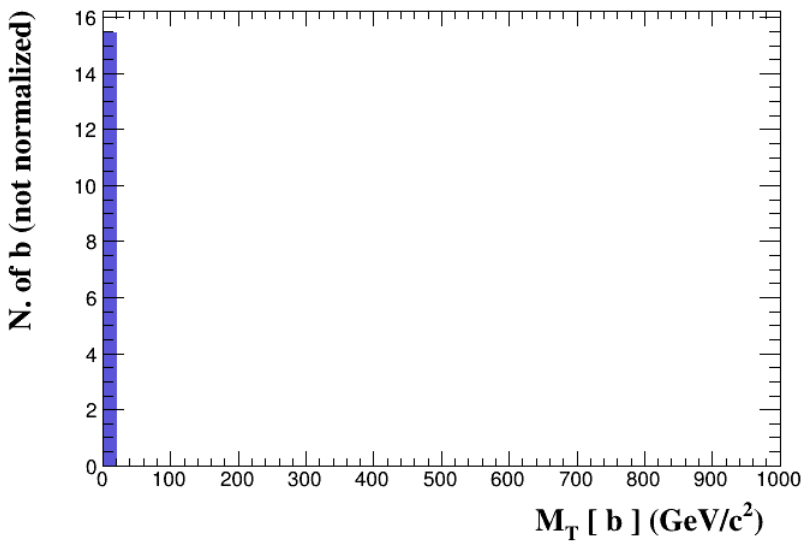

Figure 83.

3.84 Histogram 84

*** Plot: MT_MET (b)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	87.6682	54.39	0.0	0.0

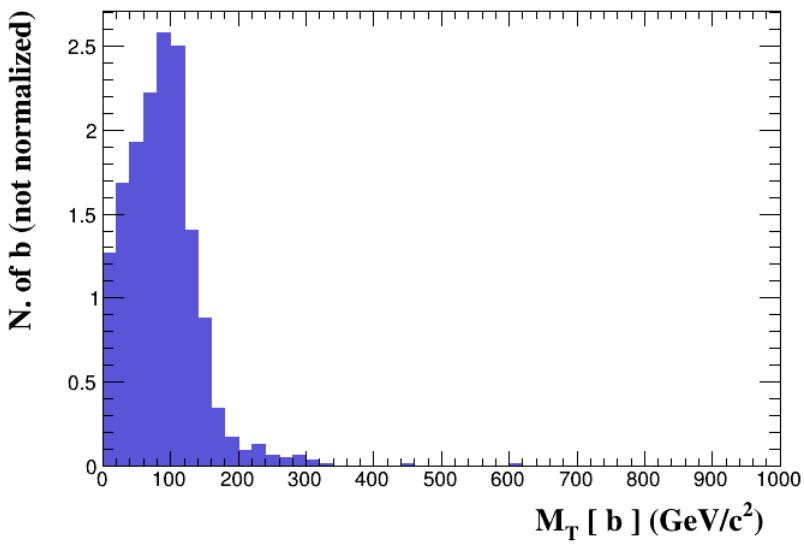

Figure 84.

3.85 Histogram 85

*** Plot: P (b)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	139.338	106.9	0.0	0.0

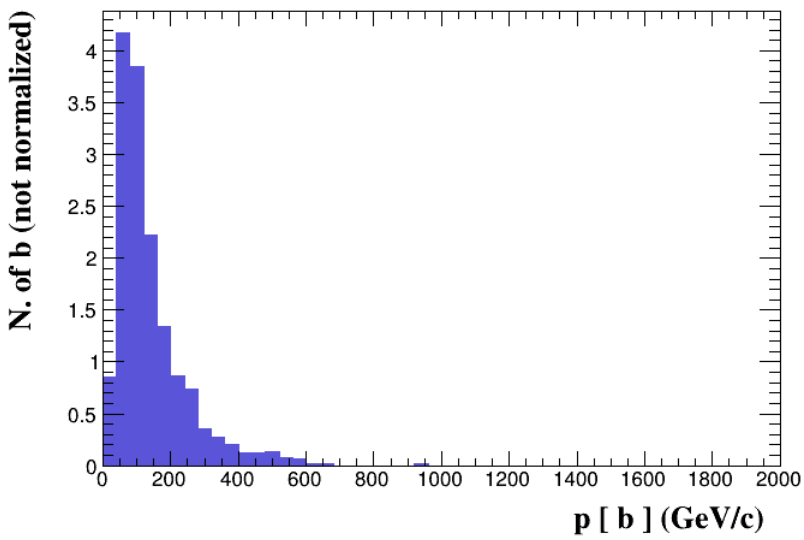

Figure 85.

3.86 Histogram 86

*** Plot: PX (b)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	-1.79812	56.57	0.0	0.0

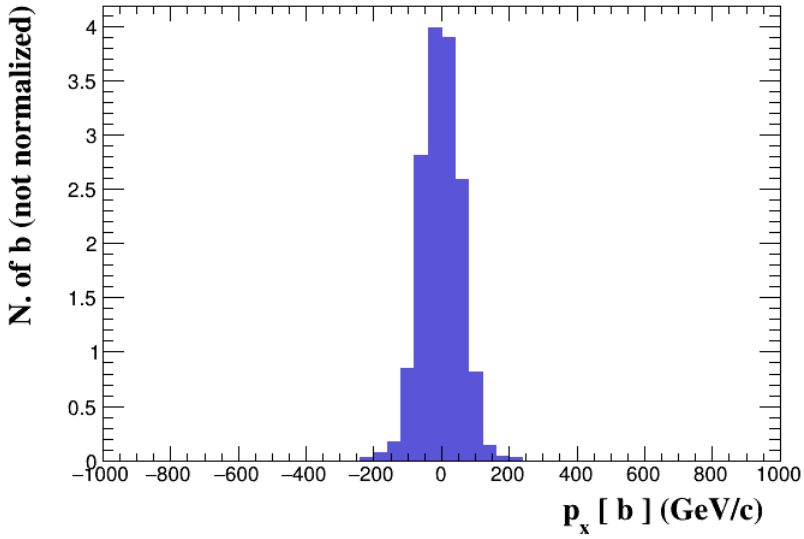

Figure 86.

3.87 Histogram 87

*** Plot: PY (b)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	1.05422	58.87	0.0	0.0

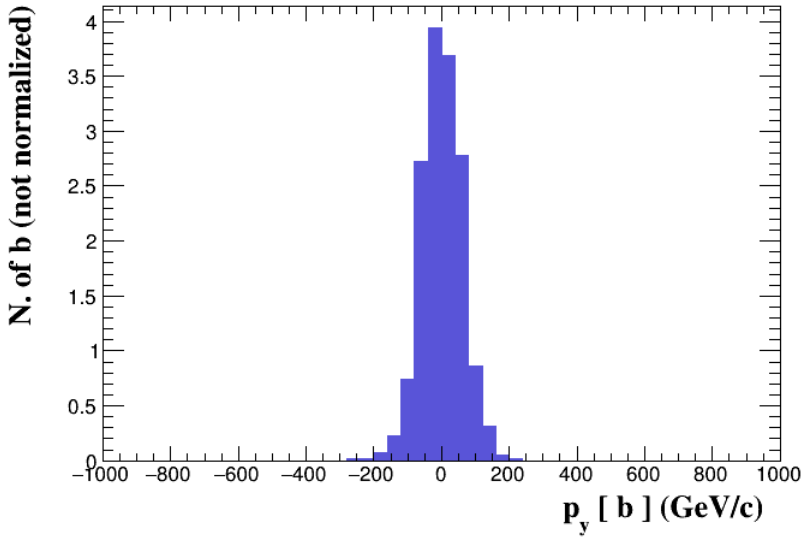

Figure 87.

3.88 Histogram 88

*** Plot: PZ (b)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	2.7842	155.5	0.0	0.0

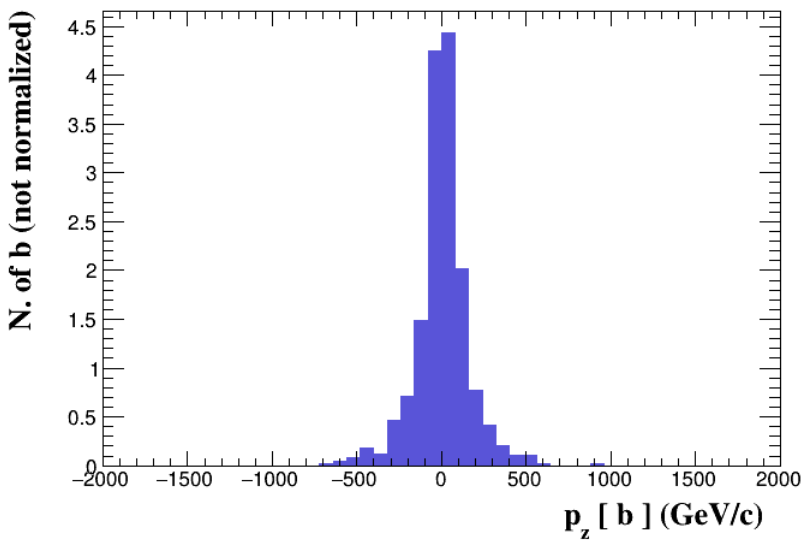

Figure 88.

3.89 Histogram 89

*** Plot: R (b)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	2.03579	0.878	0.0	0.0

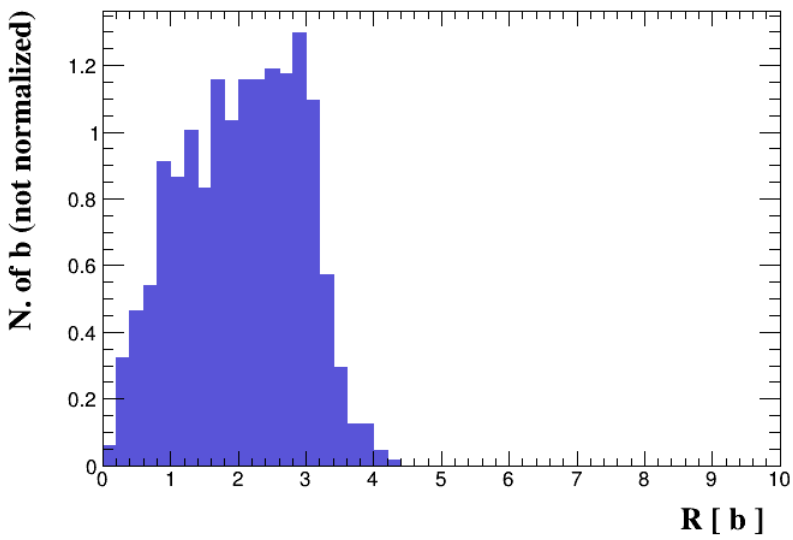

Figure 89.

3.90 Histogram 90

*** Plot: Y (b)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	0.0482562	1.261	0.0	0.0

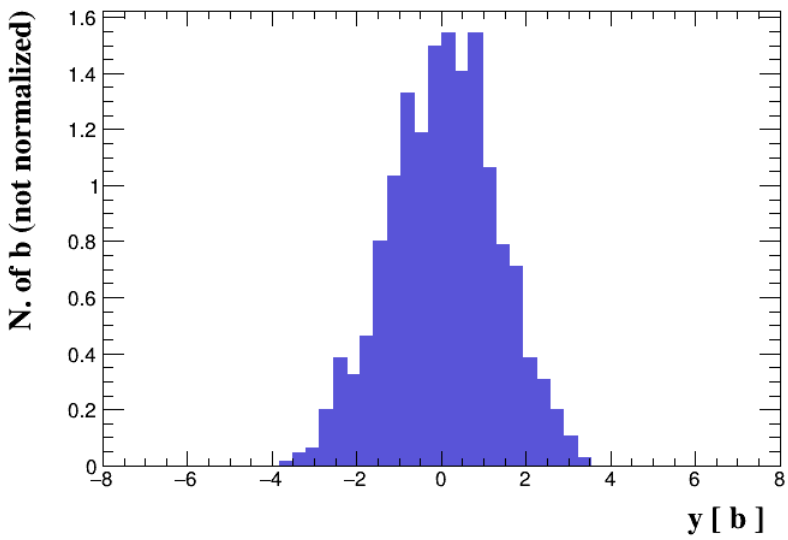

Figure 90.

3.91 Histogram 91

*** Plot: BETA (b)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	0.998042	0.002796	0.0	0.0

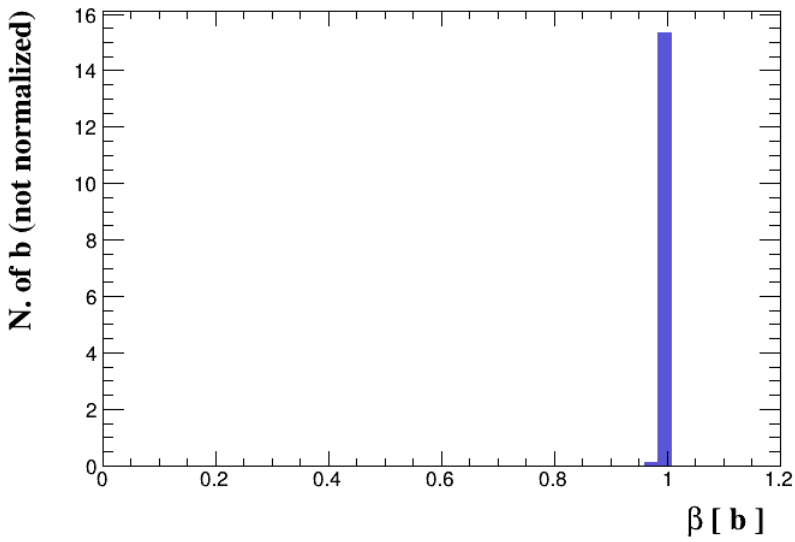

Figure 91.

3.92 Histogram 92

*** Plot: GAMMA (b)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	29.6726	22.73	0.0	2.3

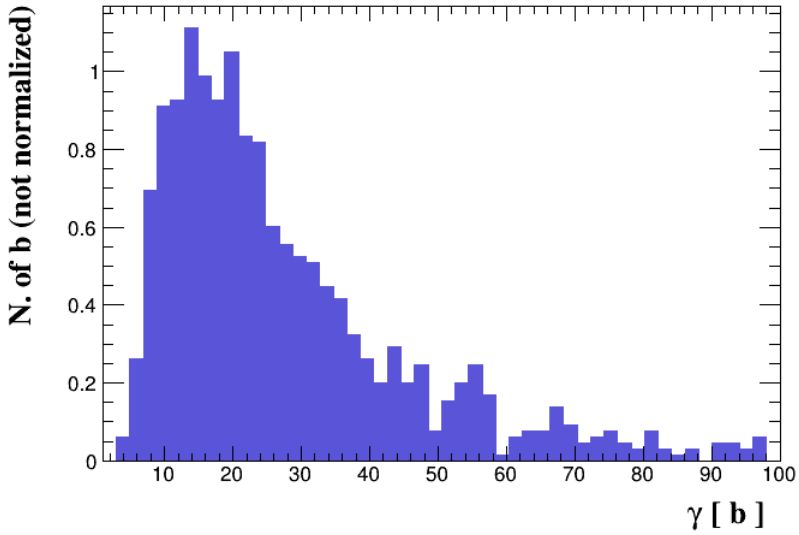

Figure 92.

3.93 Histogram 93

*** Plot: PT (b[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	72.2733	38.02	0.0	0.0

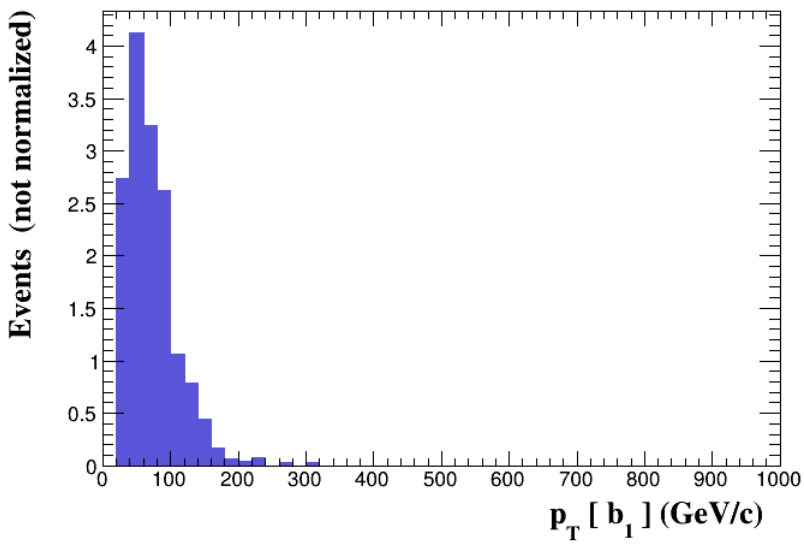

Figure 93.

3.94 Histogram 94

*** Plot: ETA (b[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	0.048521	1.264	0.0	0.0

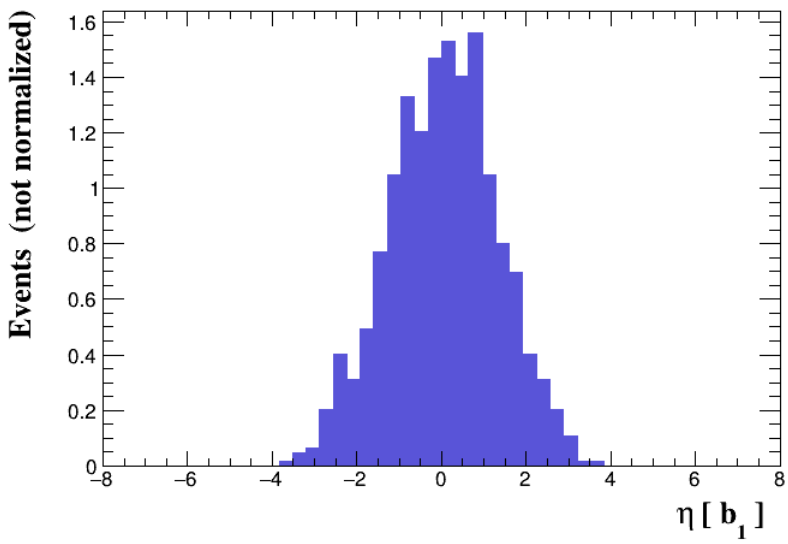

Figure 94.

3.95 Histogram 95

*** Plot: PHI (b[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	0.0257839	1.82	0.0	0.0

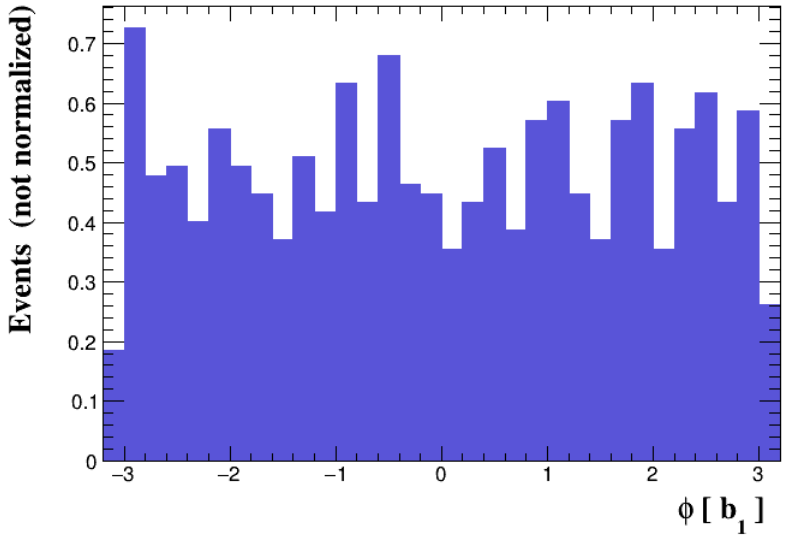

Figure 95.

3.96 Histogram 96

*** Plot: E (b[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	139.461	106.8	0.0	0.0

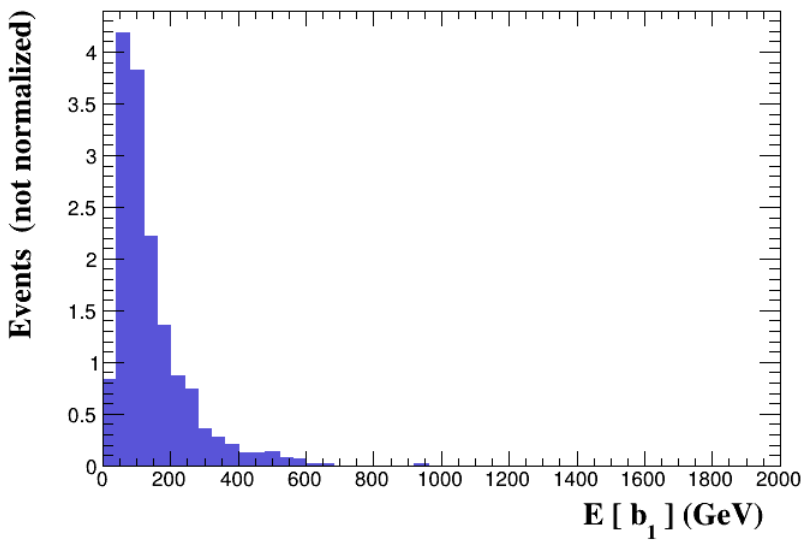

Figure 96.

3.97 Histogram 97

*** Plot: MT_MET (b[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	87.6682	54.39	0.0	0.0

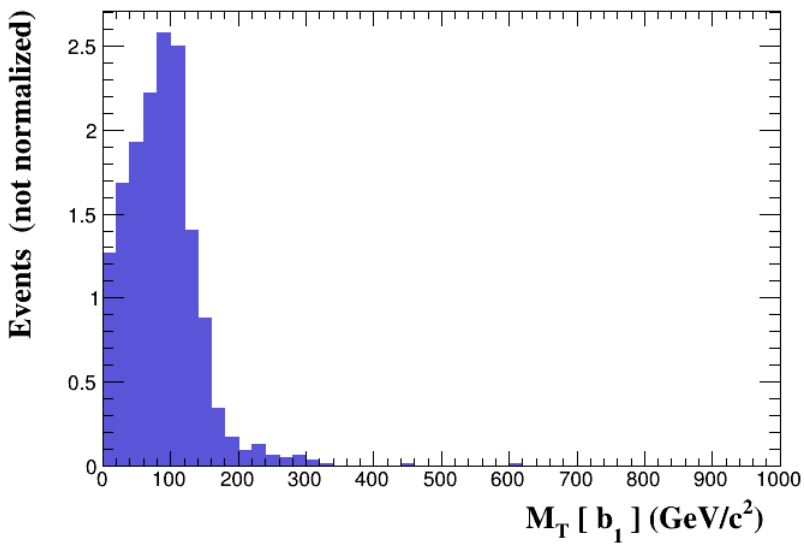

Figure 97.

3.98 Histogram 98

*** Plot: PT (b[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

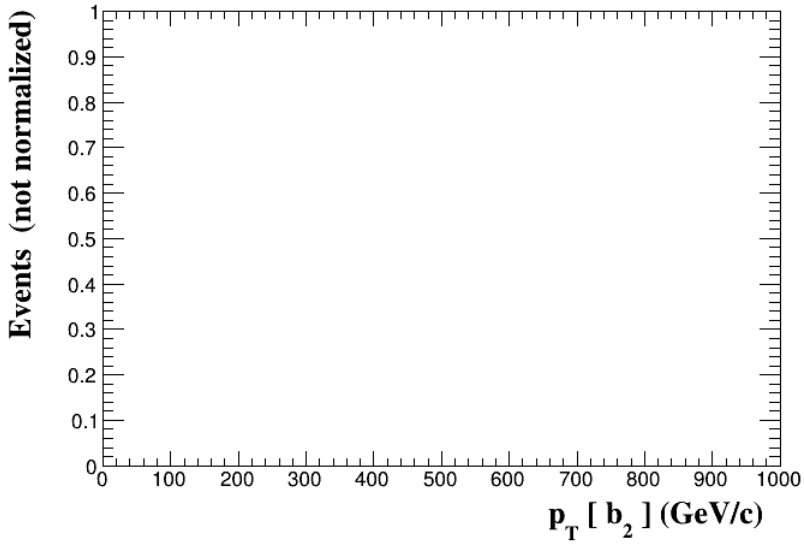

Figure 98.

3.99 Histogram 99

*** Plot: ETA (b[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

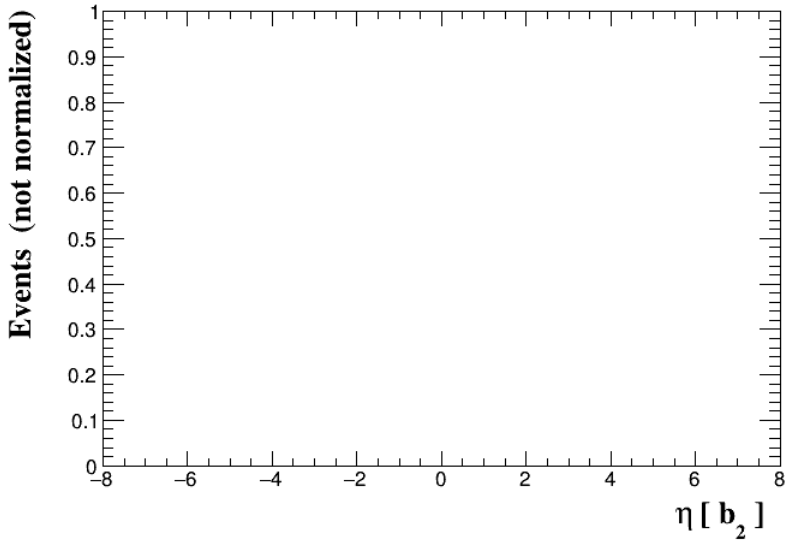

Figure 99.

3.100 Histogram 100

*** Plot: PHI (b[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

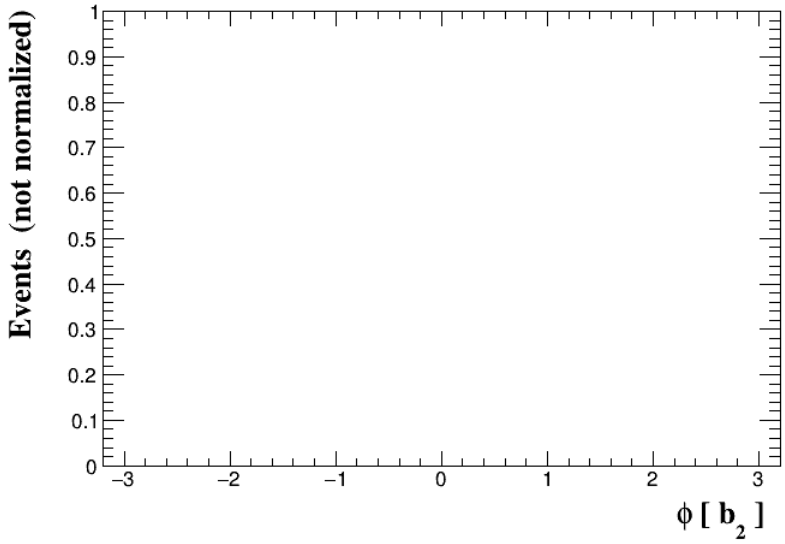

Figure 100.

3.101 Histogram 101

*** Plot: E (b[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

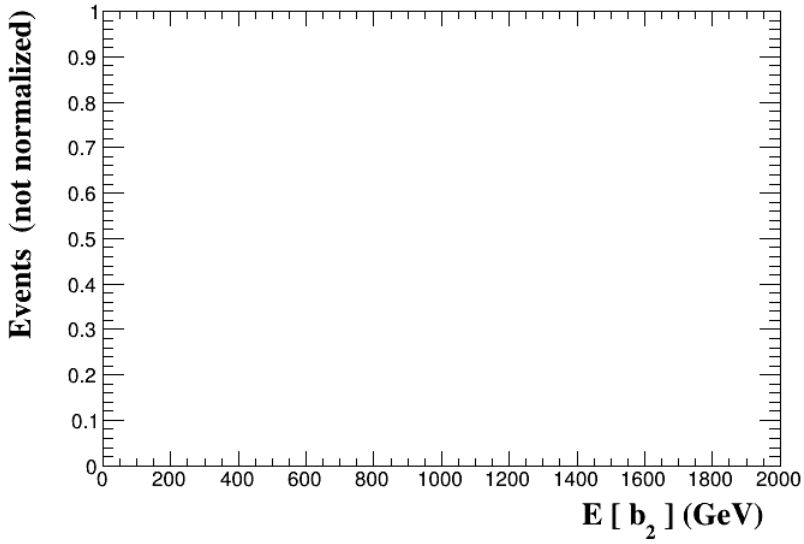

Figure 101.

3.102 Histogram 102

*** Plot: MT_MET (b[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

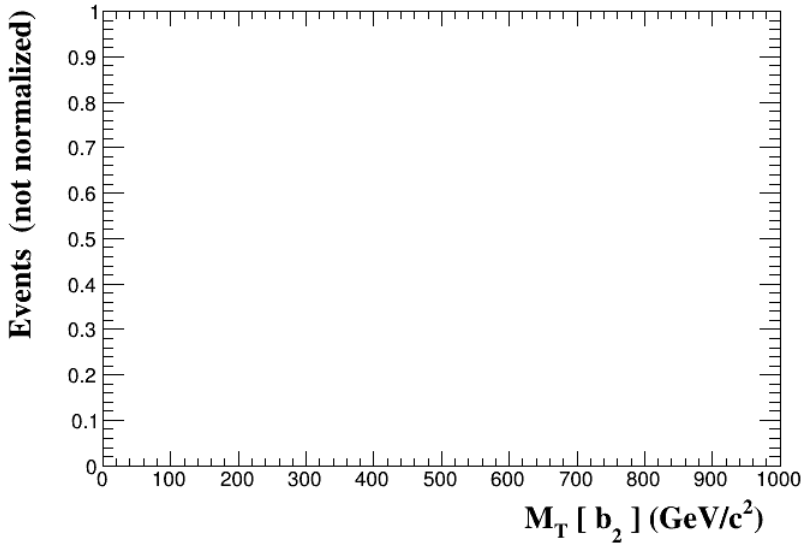

Figure 102.

3.103 Histogram 103

*** Plot: M (b[1] b[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

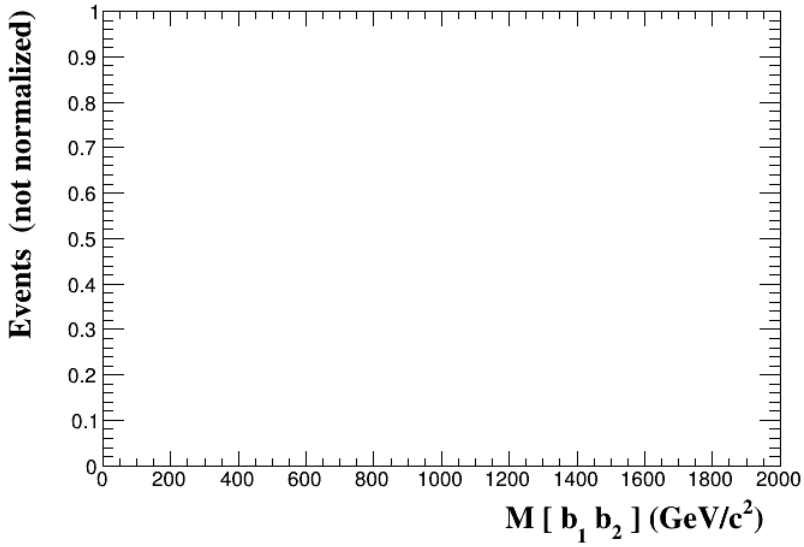

Figure 103.

3.104 Histogram 104

*** Plot: PT (b[1] b[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

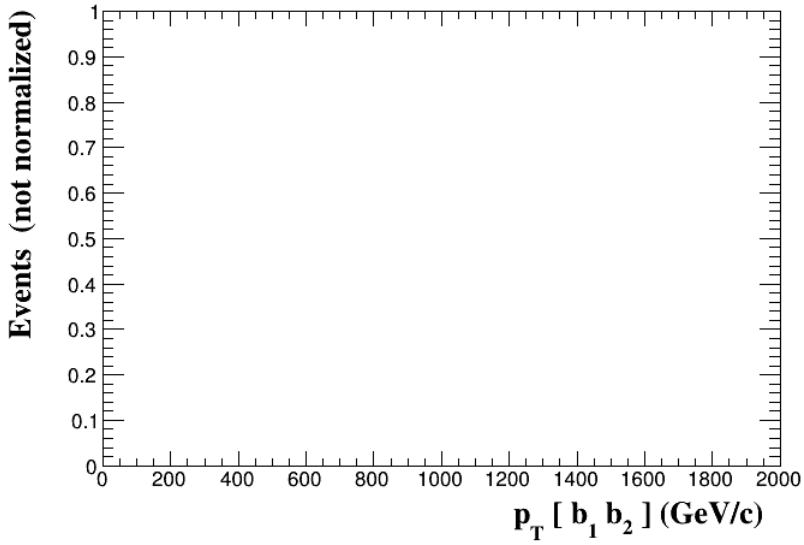

Figure 104.

3.105 Histogram 105

*** Plot: DELTAR (b[1] , b[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

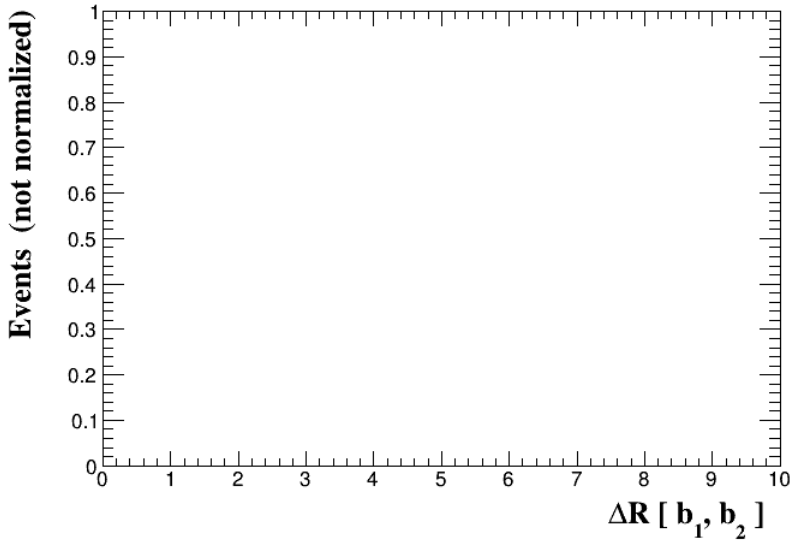

Figure 105.

3.106 Histogram 106

*** Plot: DPHI_0_PI (b[1] , b[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

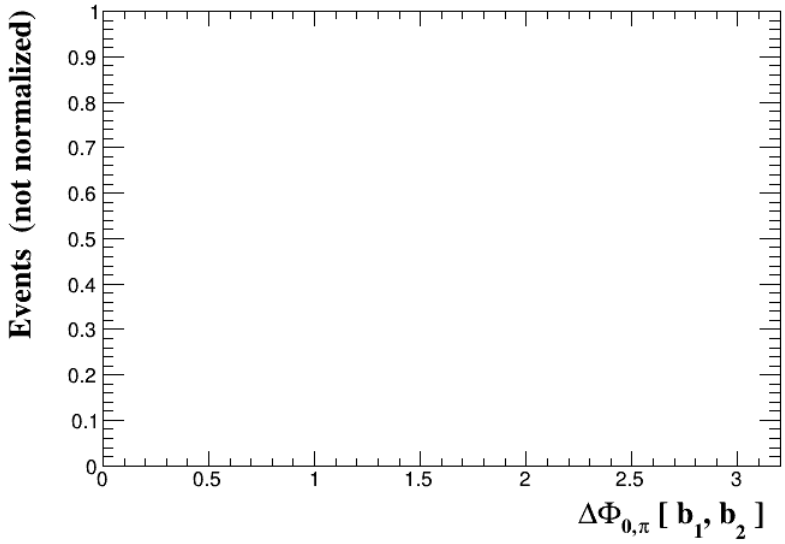

Figure 106.

3.107 Histogram 107

*** Plot: DPHI_0_2PI (b[1] , b[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

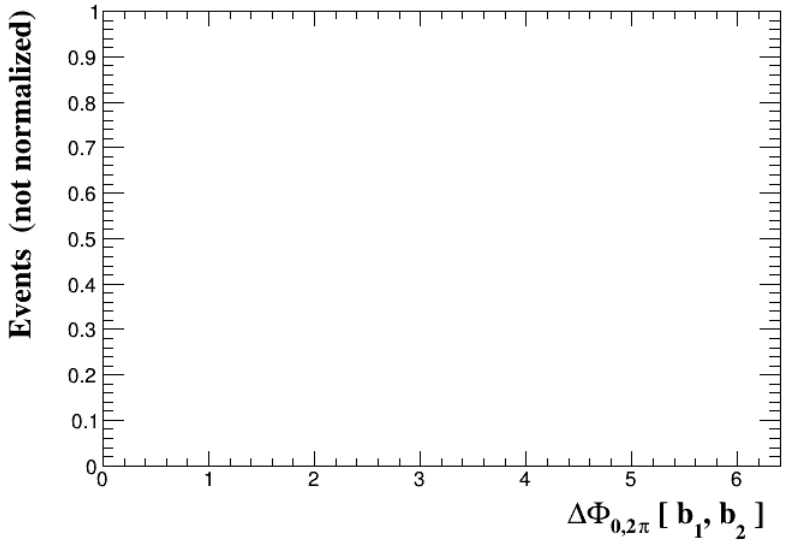

Figure 107.

3.108 Histogram 108

*** Plot: PT (j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	56.3608	32.57	0.0	0.0

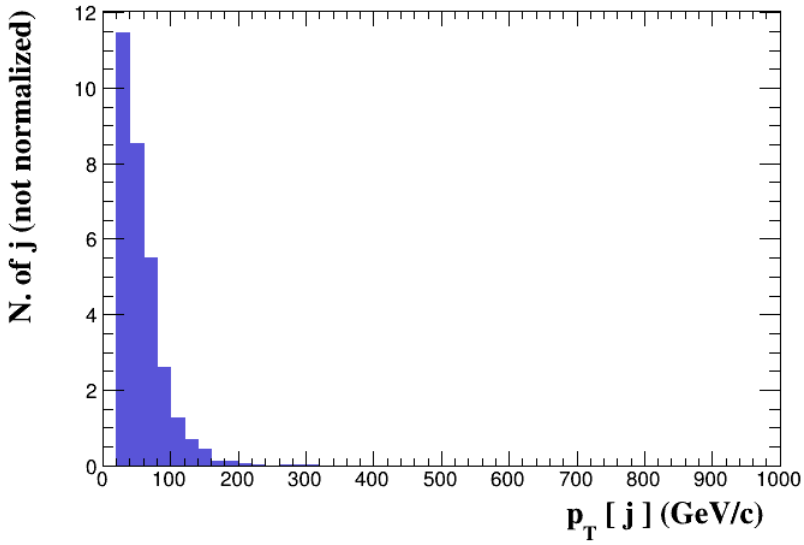

Figure 108.

3.109 Histogram 109

*** Plot: ETA (j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	0.0219264	1.2	0.0	0.0

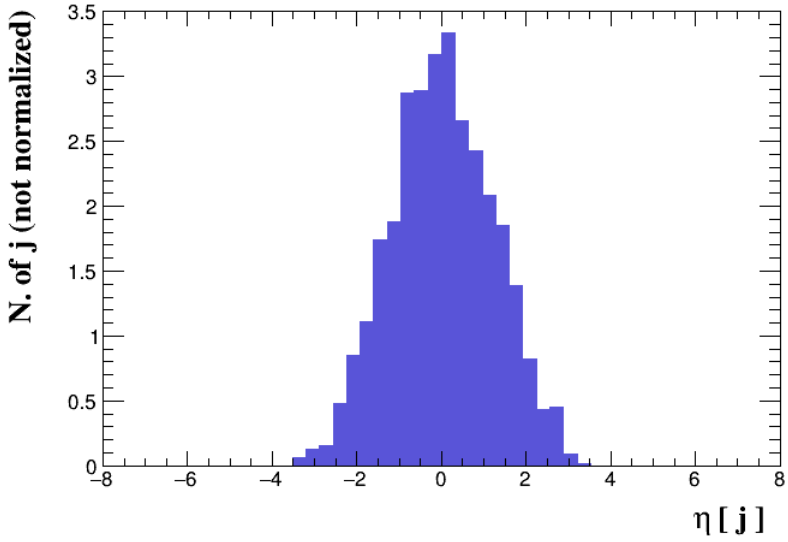

Figure 109.

3.110 Histogram 110

*** Plot: ABSETA (j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	0.971377	0.7051	0.0	0.0

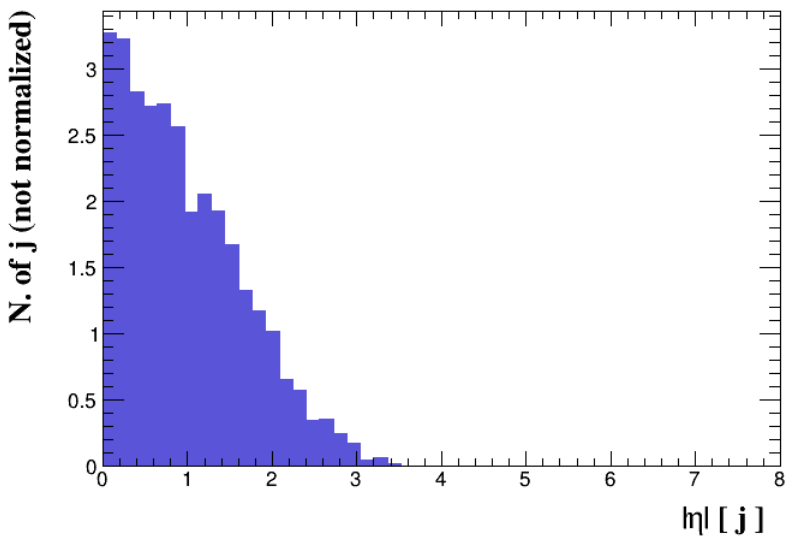

Figure 110.

3.111 Histogram 111

*** Plot: PHI (j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	0.0781512	1.851	0.0	0.0

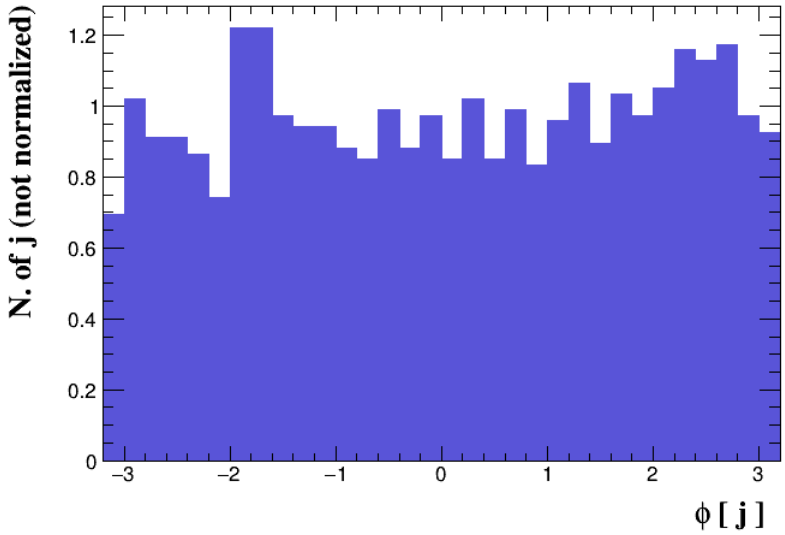

Figure 111.

3.112 Histogram 112

*** Plot: E (j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	104.57	85.46	0.0	0.0

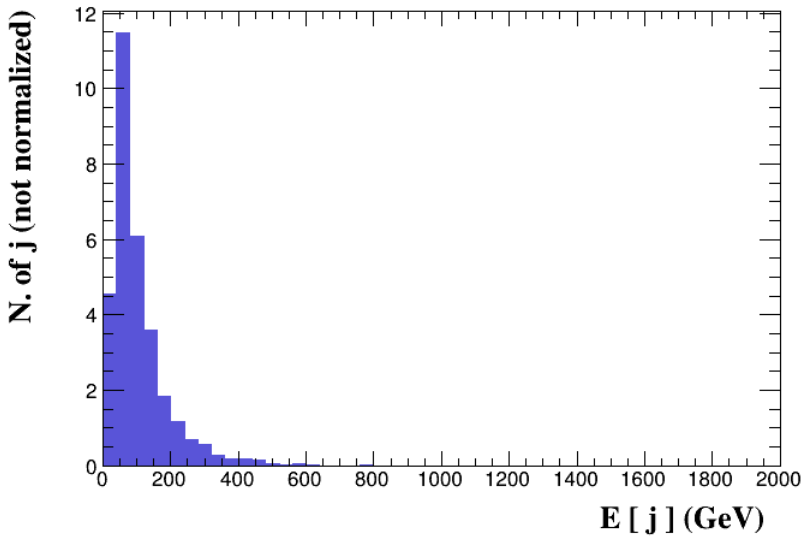

Figure 112.

3.113 Histogram 113

*** Plot: ET (j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	56.3608	32.57	0.0	0.0

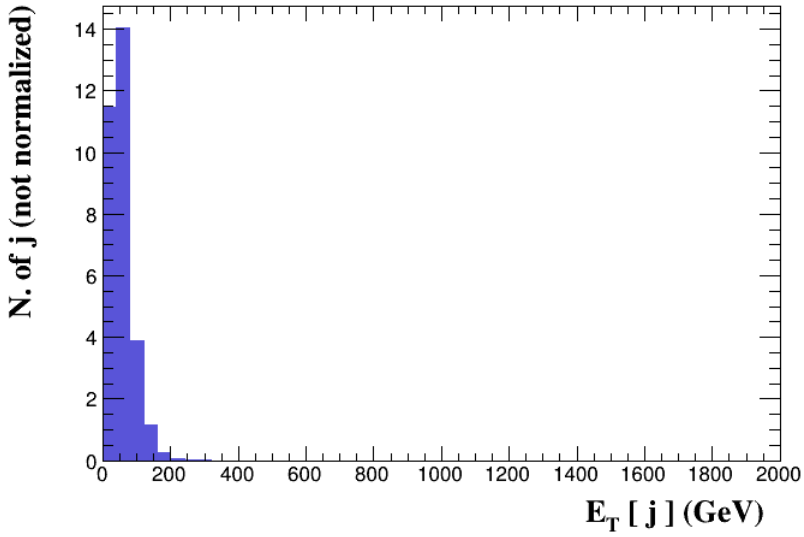

Figure 113.

3.114 Histogram 114

*** Plot: M (j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	-1.3327e-05	0.0006733	50.9	0.0

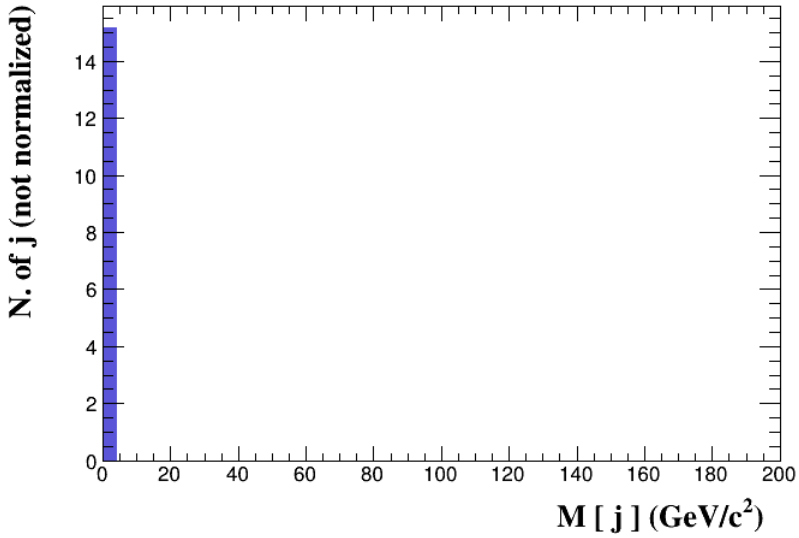

Figure 114.

3.115 Histogram 115

*** Plot: MT (j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	0.000121728	0.0002075	0.0	0.0

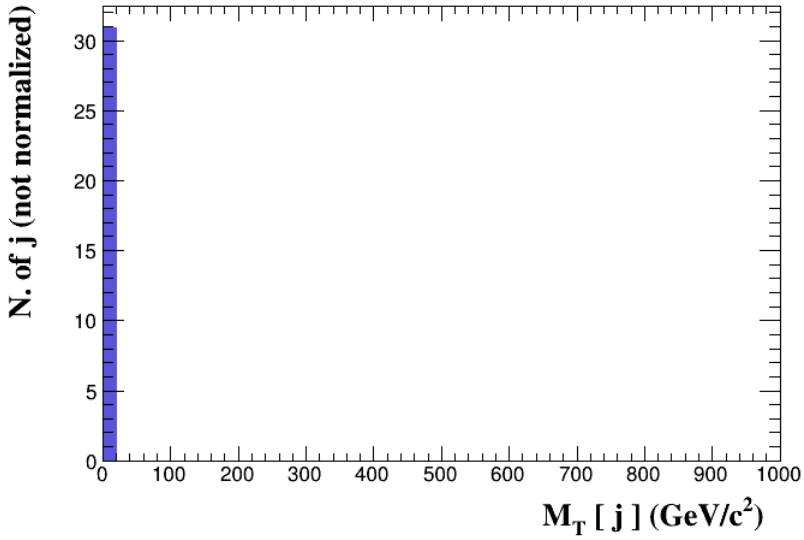

Figure 115.

3.116 Histogram 116

*** Plot: MT_MET (j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	82.308	60.37	0.0	0.0

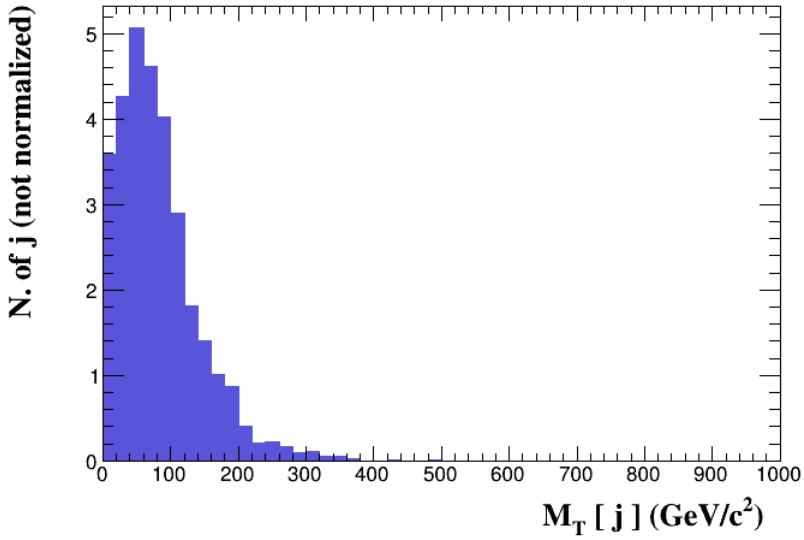

Figure 116.

3.117 Histogram 117

*** Plot: P (j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	104.57	85.46	0.0	0.0

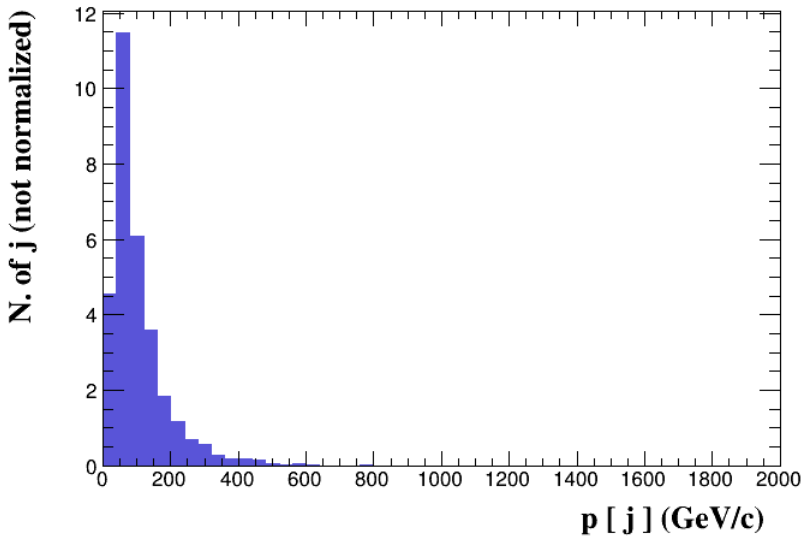

Figure 117.

3.118 Histogram 118

*** Plot: PX (j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	-2.36396	45.86	0.0	0.0

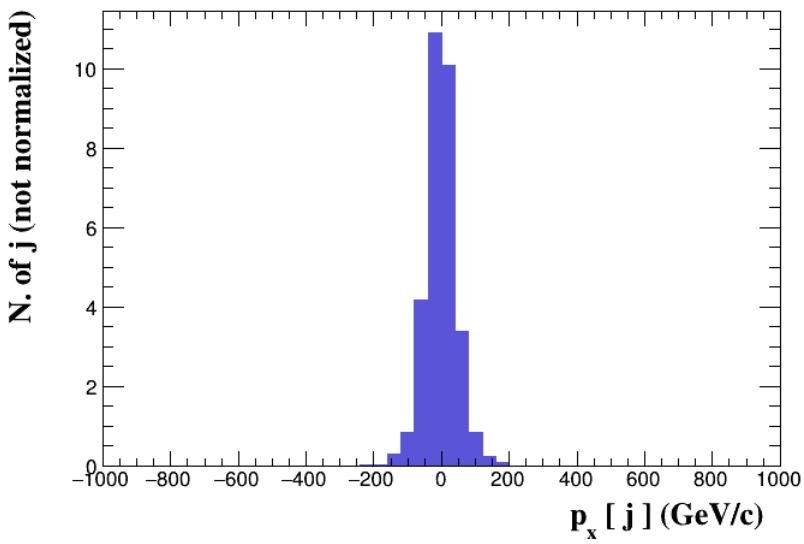

Figure 118.

3.119 Histogram 119

*** Plot: PY (j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	1.06079	46.13	0.0	0.0

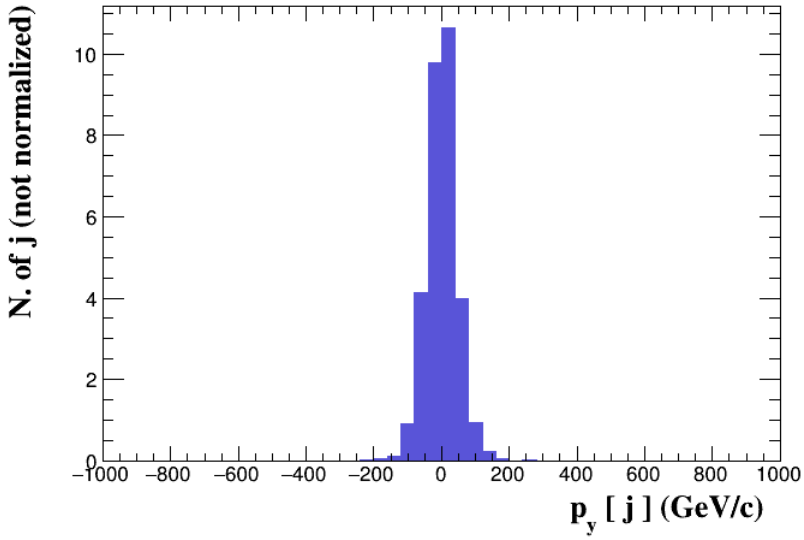

Figure 119.

3.120 Histogram 120

*** Plot: PZ (j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	4.14477	118.3	0.0	0.0

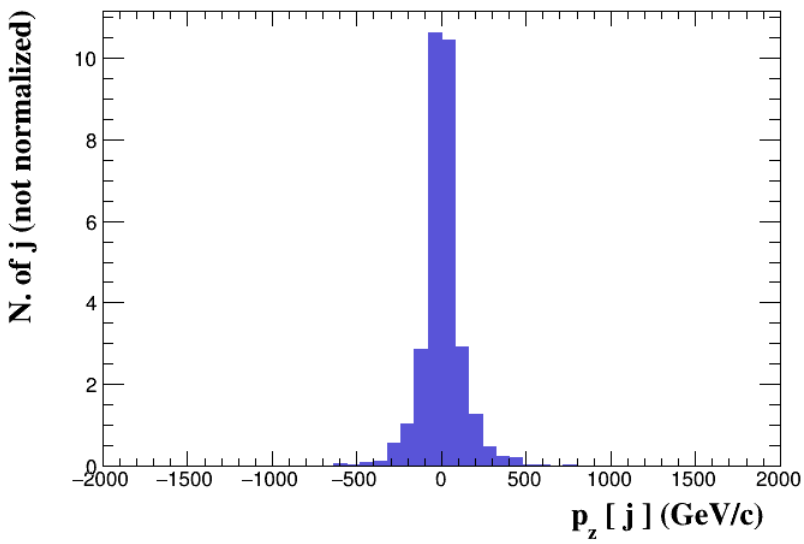

Figure 120.

3.121 Histogram 121

*** Plot: R (j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	2.036	0.8521	0.0	0.0

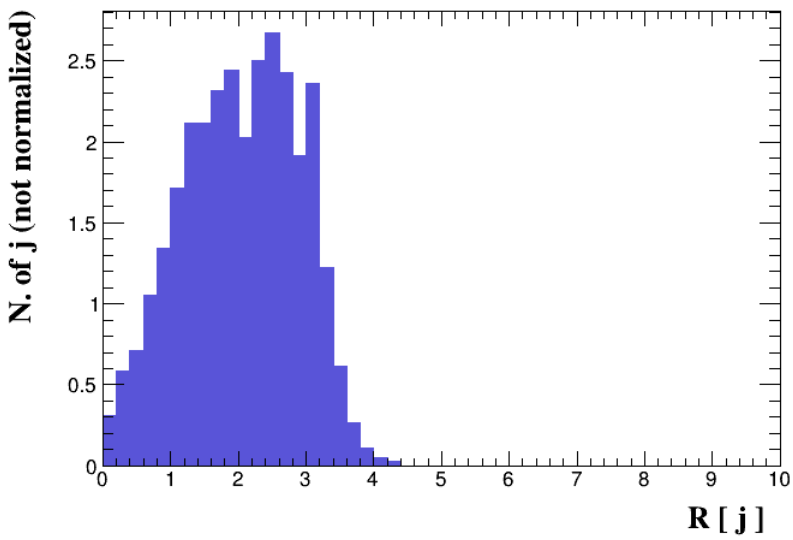

Figure 121.

3.122 Histogram 122

*** Plot: Y (j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	0.0219264	1.2	0.0	0.0

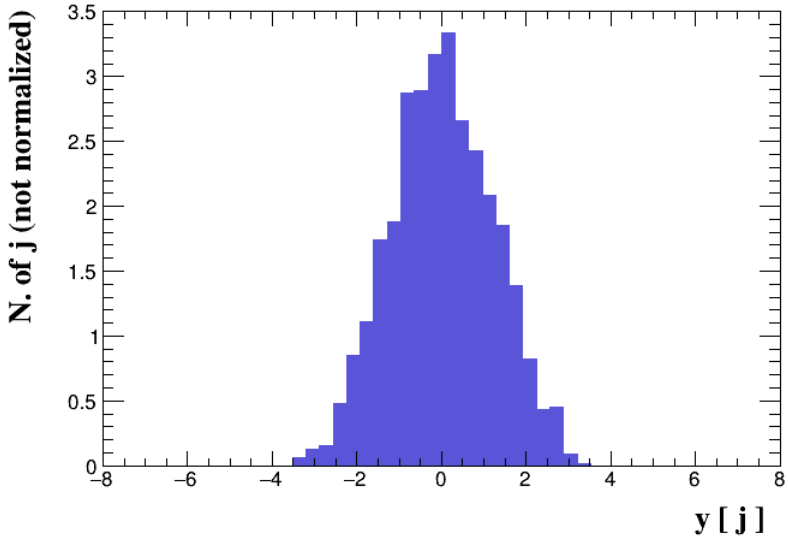

Figure 122.

3.123 Histogram 123

*** Plot: BETA (j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	1.0	0.0	0.0	0.0

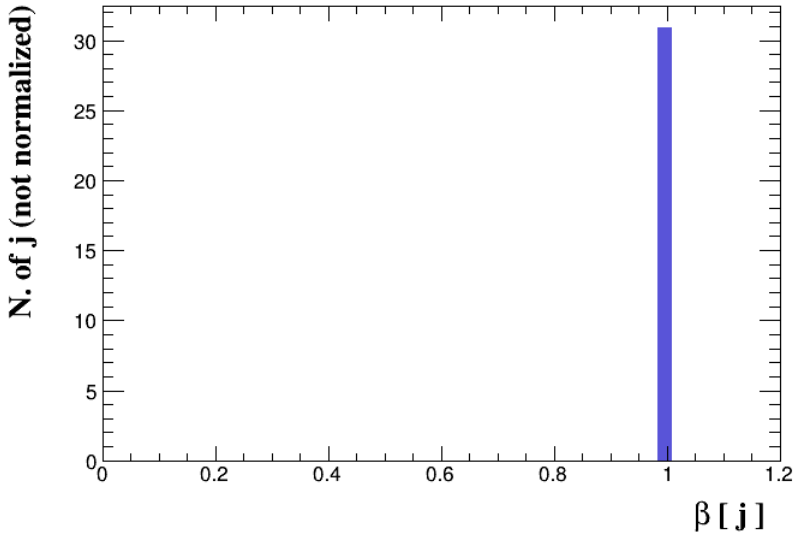

Figure 123.

3.124 Histogram 124

*** Plot: GAMMA (j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	nan	nan	0.0	49.1

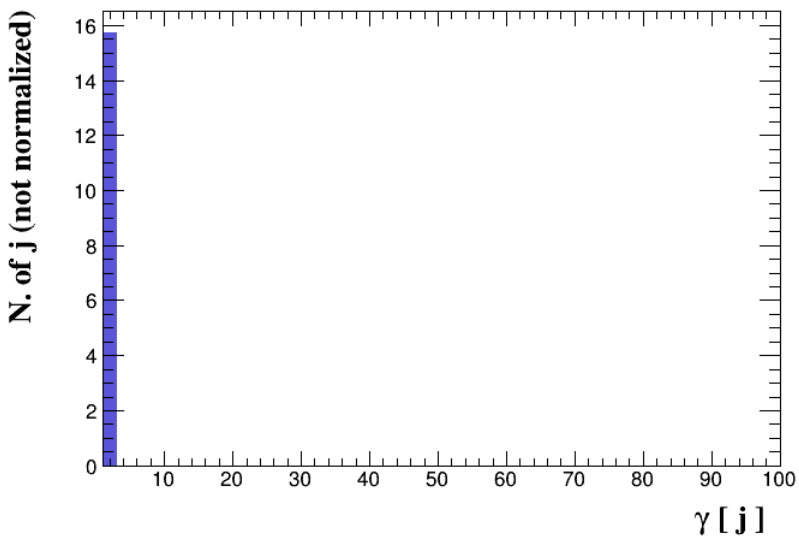

Figure 124.

3.125 Histogram 125

*** Plot: PT (j[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	74.0942	34.32	0.0	0.0

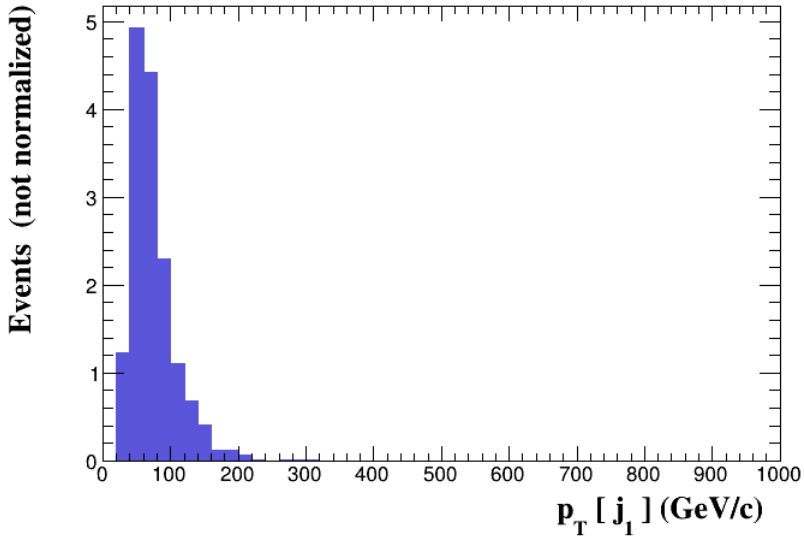

Figure 125.

3.126 Histogram 126

*** Plot: ETA (j[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	0.0200712	1.117	0.0	0.0

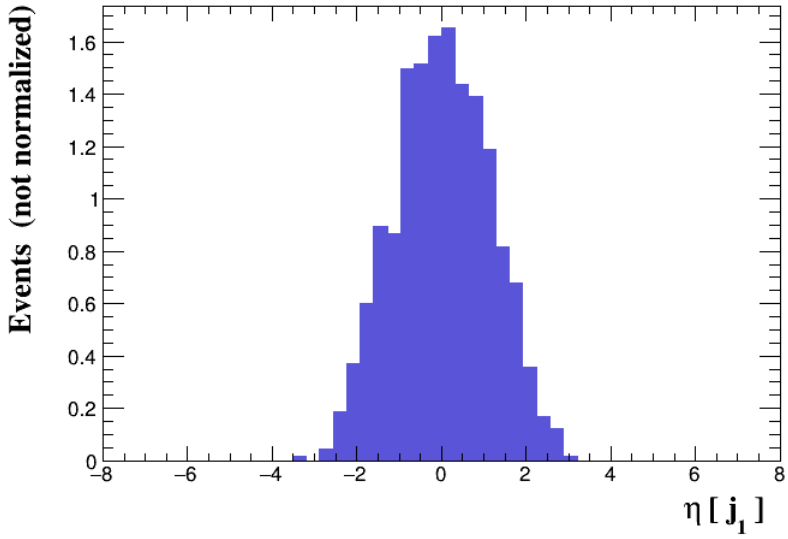

Figure 126.

3.127 Histogram 127

*** Plot: PHI (j[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	0.0625323	1.856	0.0	0.0

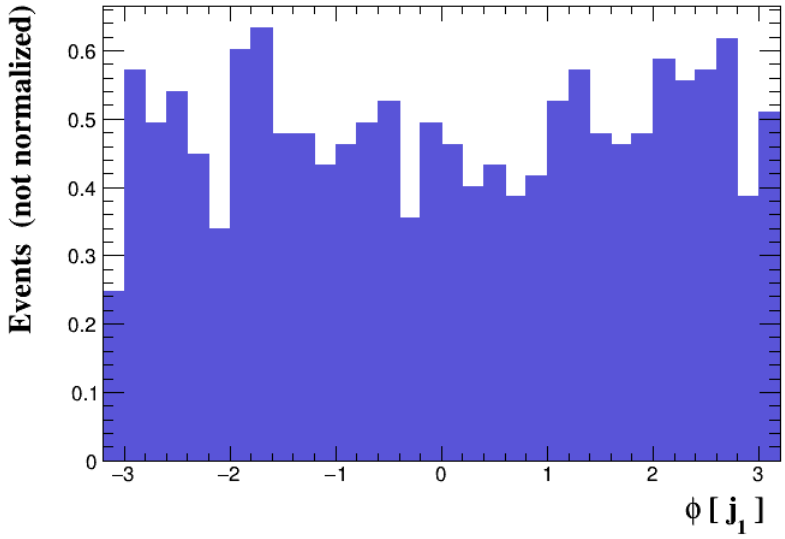

Figure 127.

3.128 Histogram 128

*** Plot: E (j[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	129.519	93.36	0.0	0.0

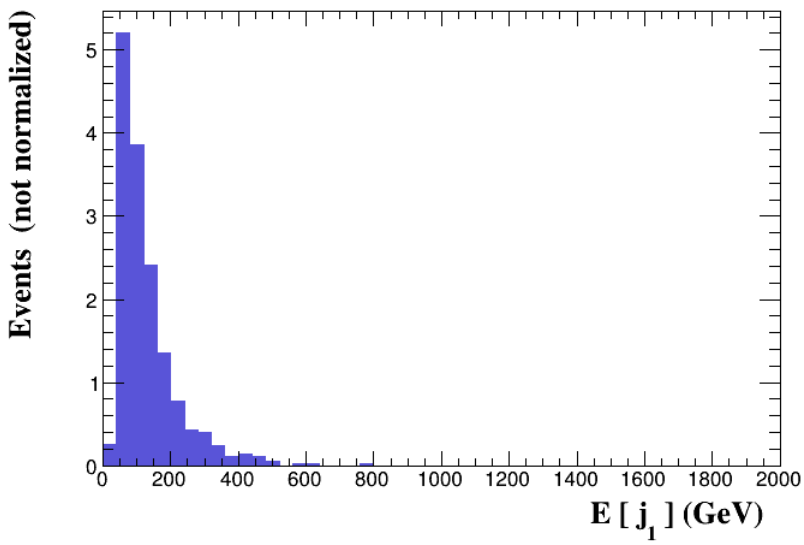

Figure 128.

3.129 Histogram 129

*** Plot: MT_MET (j[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	98.7583	67.56	0.0	0.0

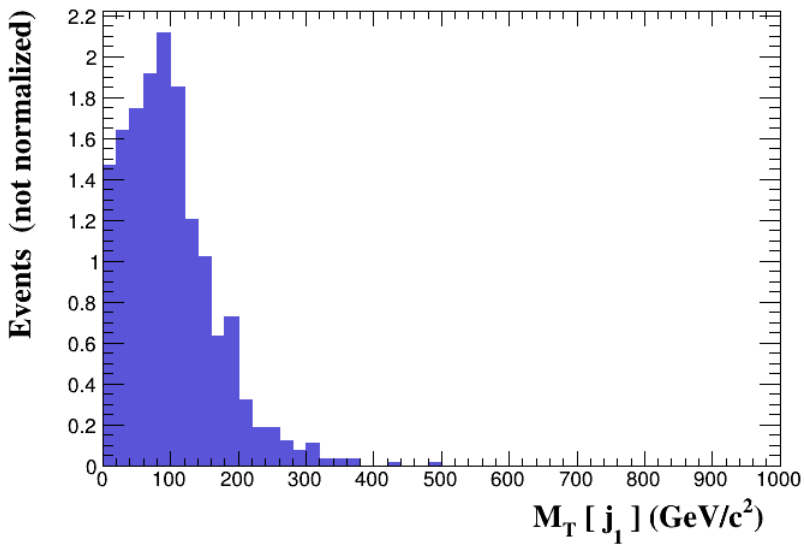

Figure 129.

3.130 Histogram 130

*** Plot: PT (j[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	38.6273	17.75	0.0	0.0

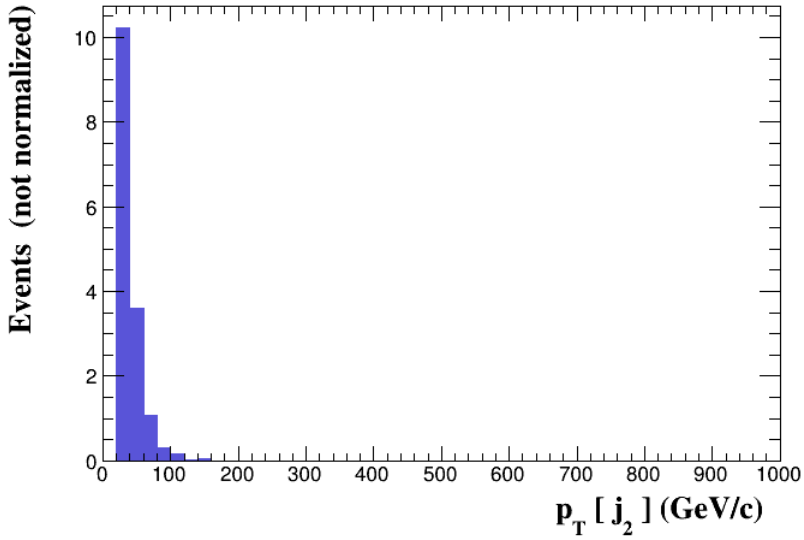

Figure 130.

3.131 Histogram 131

*** Plot: ETA (j[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	0.0237817	1.278	0.0	0.0

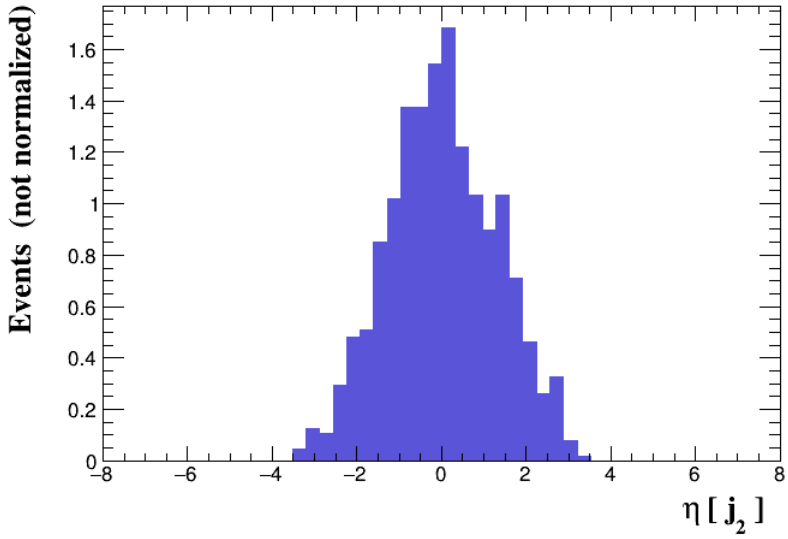

Figure 131.

3.132 Histogram 132

*** Plot: PHI (j[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	0.0937701	1.845	0.0	0.0

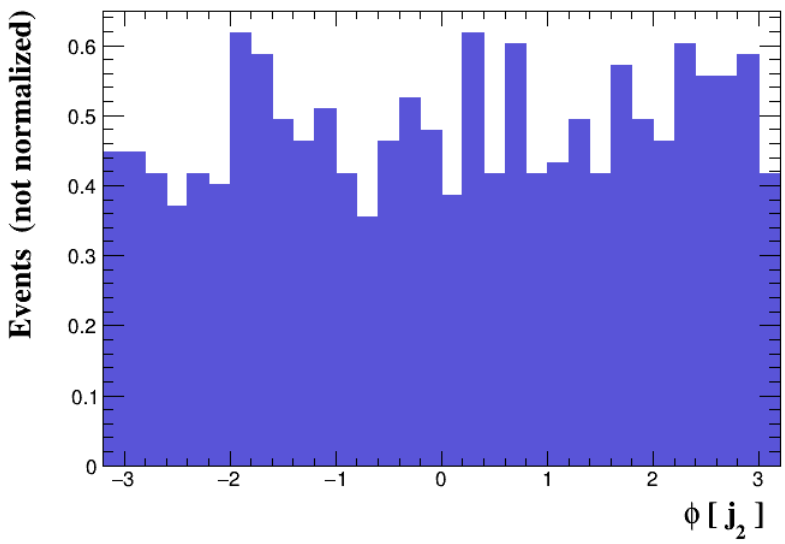

Figure 132.

3.133 Histogram 133

*** Plot: E (j[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	79.6223	68.15	0.0	0.0

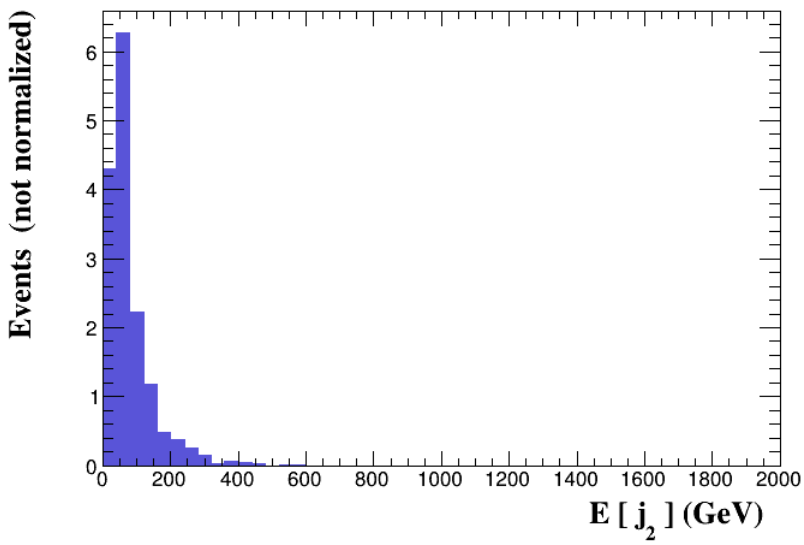

Figure 133.

3.134 Histogram 134

*** Plot: MT_MET (j[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	65.8576	46.73	0.0	0.0

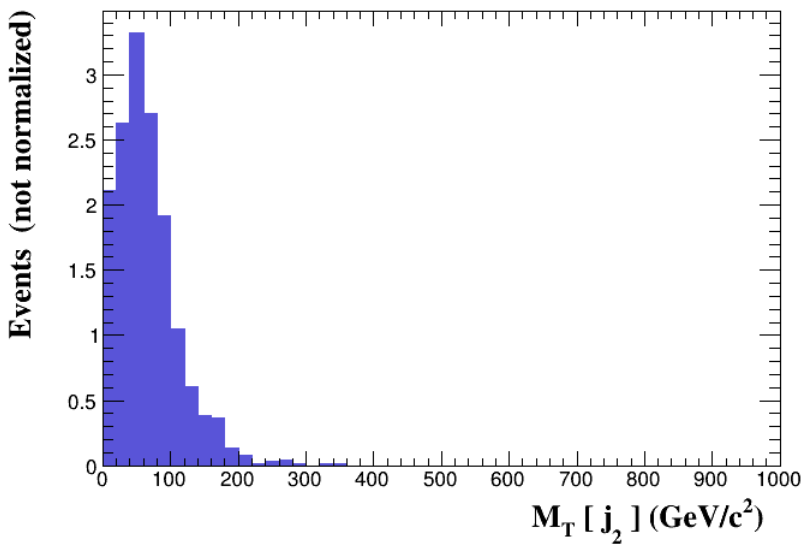

Figure 134.

3.135 Histogram 135

*** Plot: M (j[1] j[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	80.3339	4.323	0.0	0.0

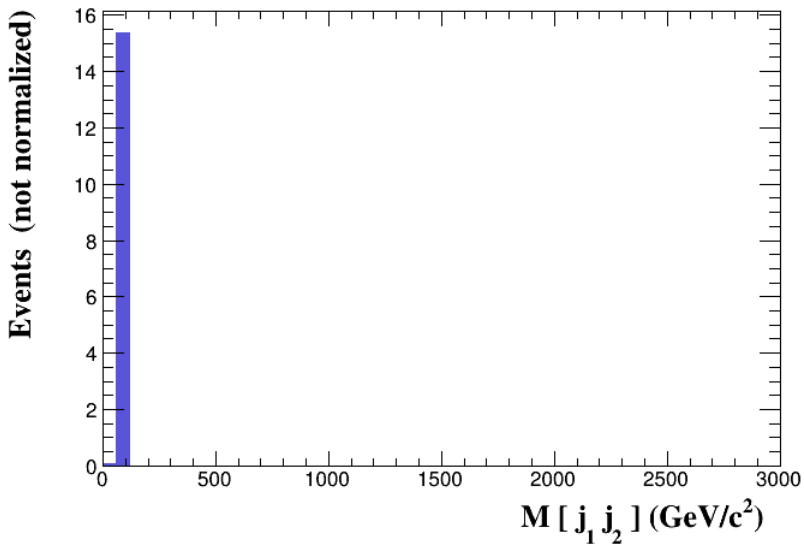

Figure 135.

3.136 Histogram 136

*** Plot: PT (j[1] j[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	86.6463	54.96	0.0	0.0

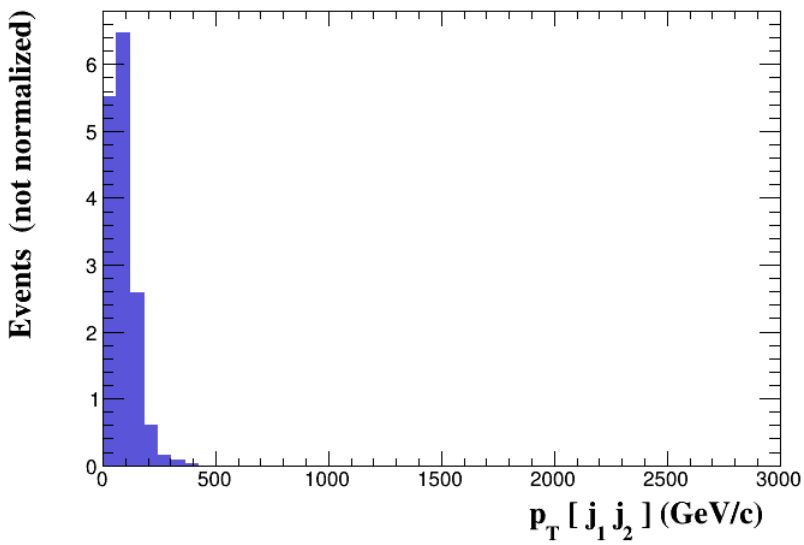

Figure 136.

3.137 Histogram 137

*** Plot: DELTAR (j[1] , j[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	1.90737	0.6749	0.0	0.0

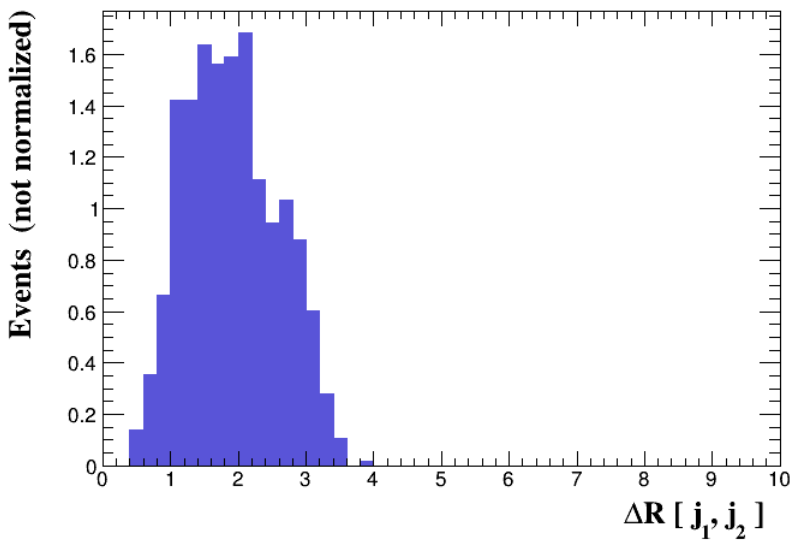

Figure 137.

3.138 Histogram 138

*** Plot: DPHI_0_PI (j[1] , j[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	1.55671	0.809	0.0	0.0

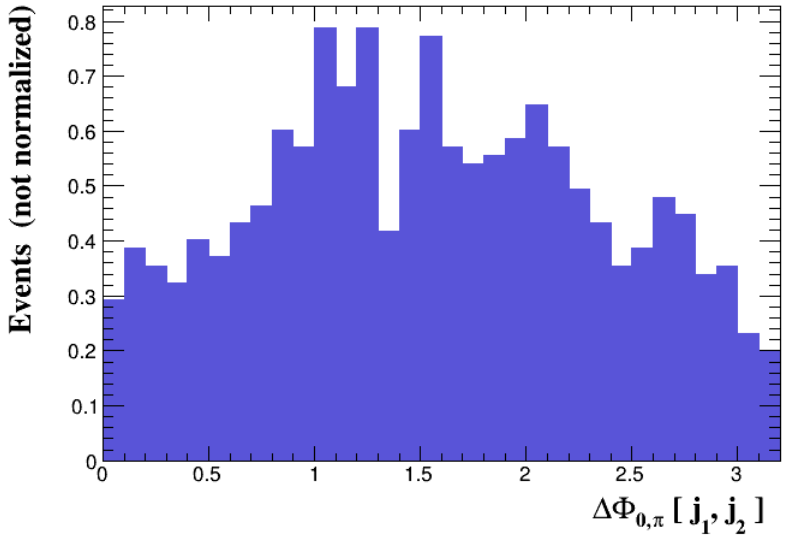

Figure 138.

3.139 Histogram 139

*** Plot: DPHI_0_2PI (j[1] , j[2])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	3.02239	1.775	0.0	0.0

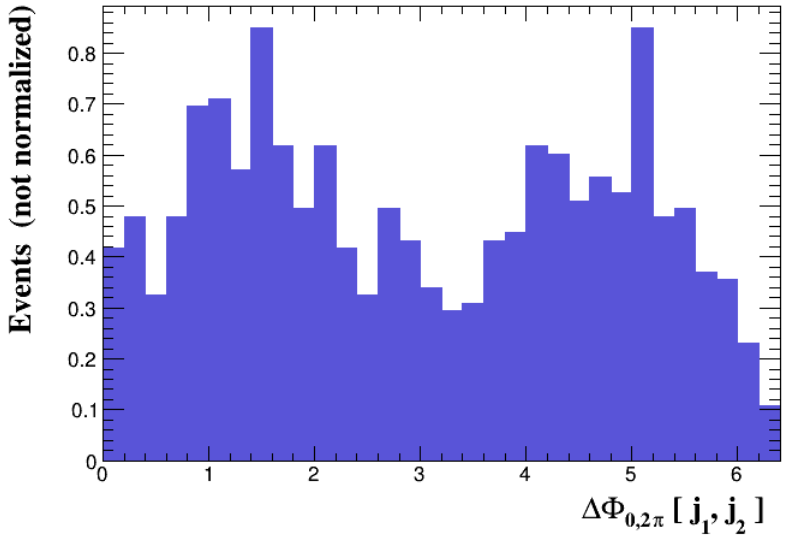

Figure 139.

3.140 Histogram 140

*** Plot: DELTAR (e , e)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

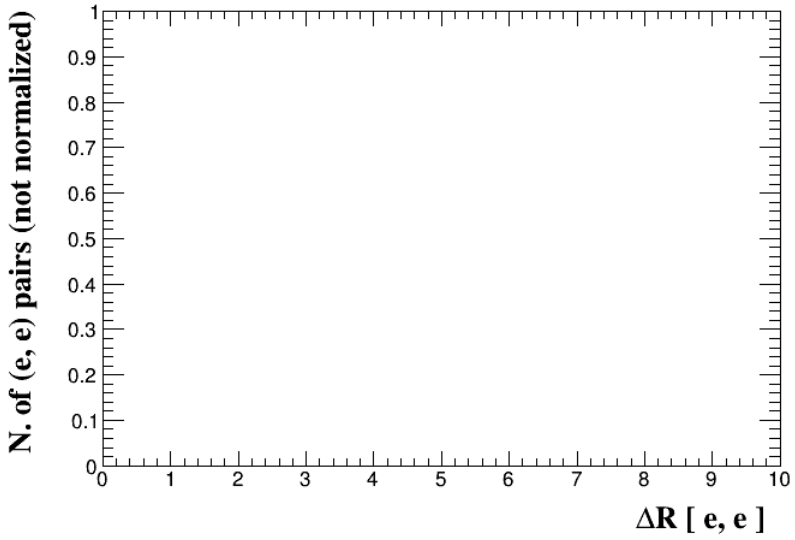

Figure 140.

3.141 Histogram 141

*** Plot: DPHI_0_PI (e , e)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

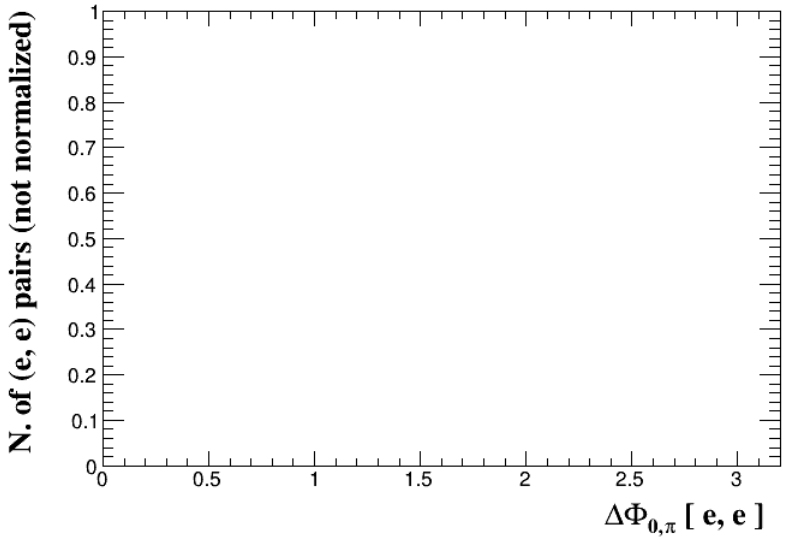

Figure 141.

3.142 Histogram 142

*** Plot: DPHI_0_2PI (e , e)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

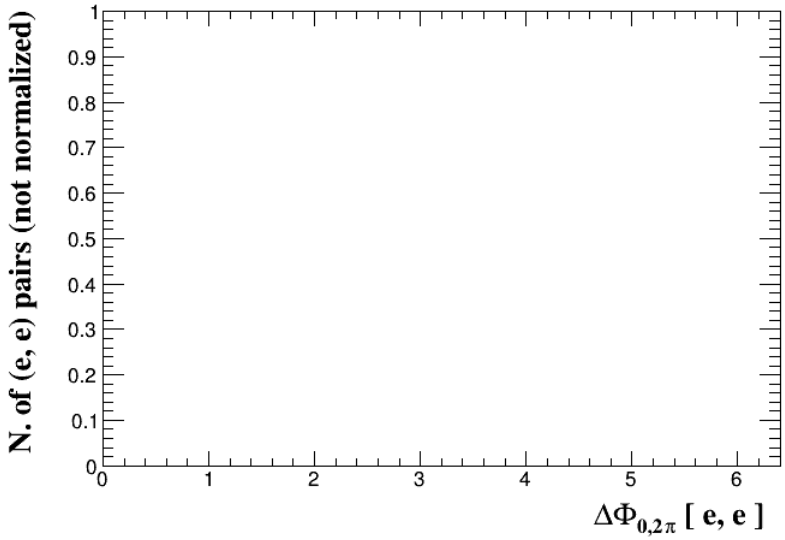

Figure 142.

3.143 Histogram 143

*** Plot: M (e e)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

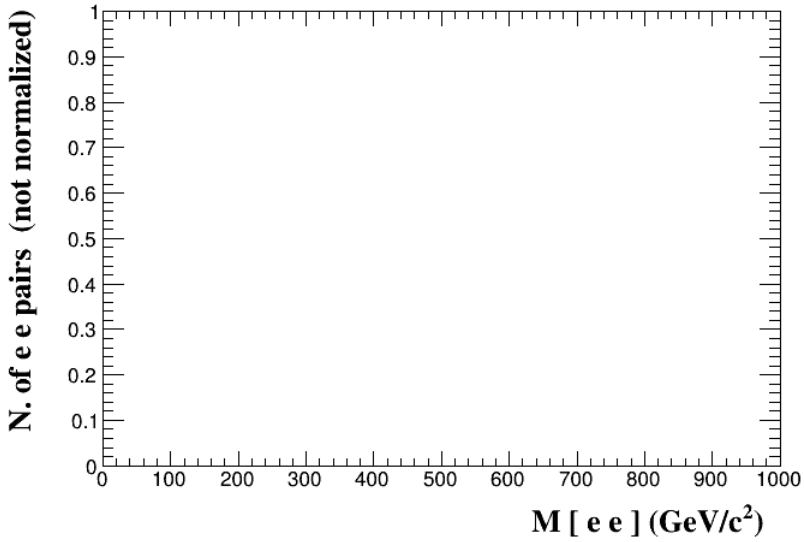

Figure 143.

3.144 Histogram 144

*** Plot: PT (e e)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

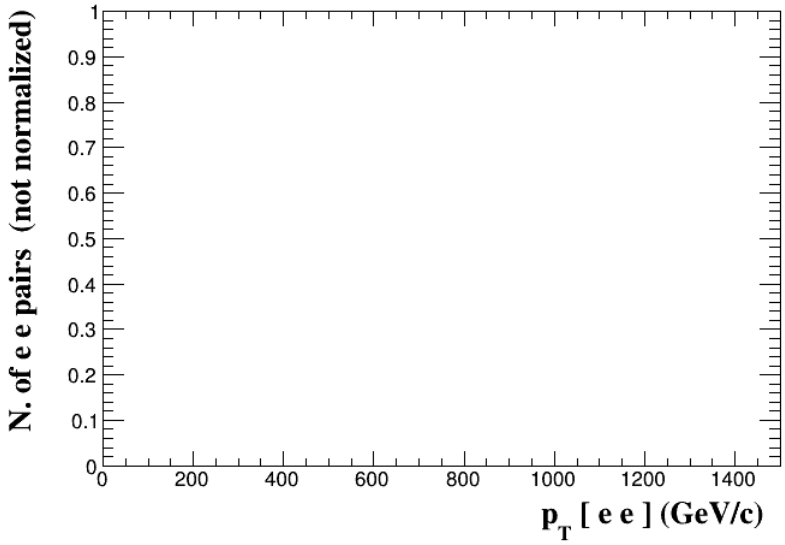

Figure 144.

3.145 Histogram 145

*** Plot: DELTAR (mu , mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

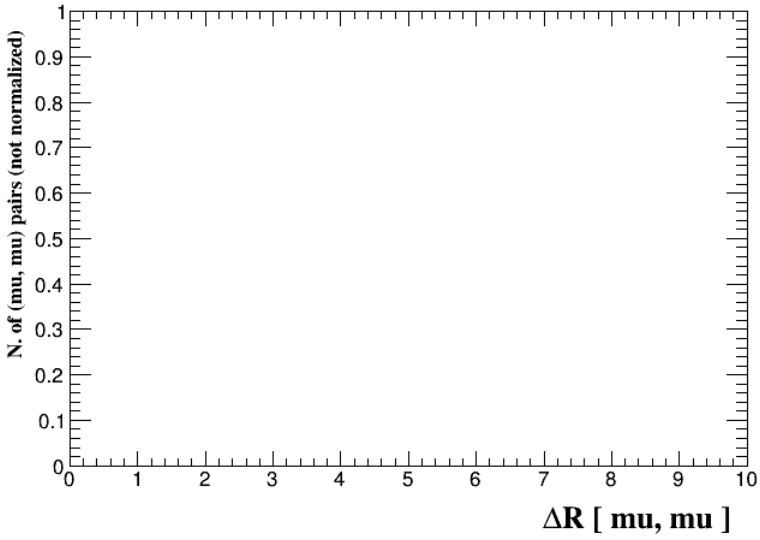

Figure 145.

3.146 Histogram 146

*** Plot: DPHI_0_PI (mu , mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

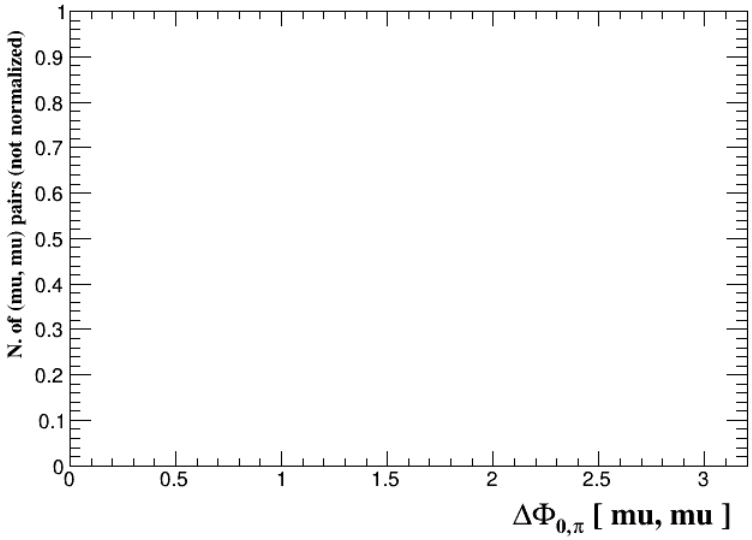

Figure 146.

3.147 Histogram 147

*** Plot: DPHI_0_2PI (mu , mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

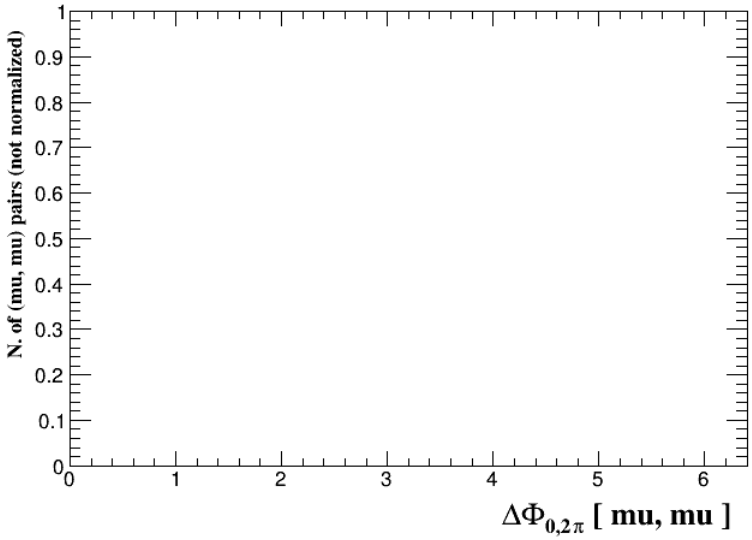

Figure 147.

3.148 Histogram 148

*** Plot: M (mu mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

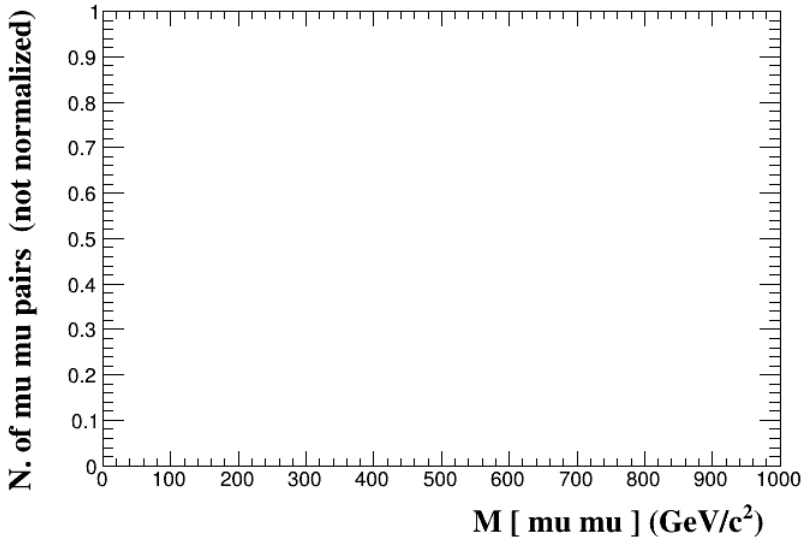

Figure 148.

3.149 Histogram 149

*** Plot: PT (mu mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

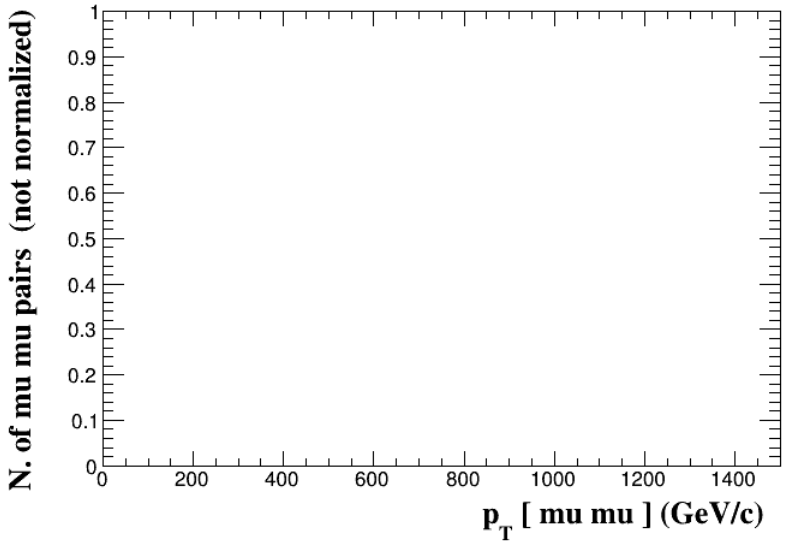

Figure 149.

3.150 Histogram 150

*** Plot: DELTAR (b , b)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

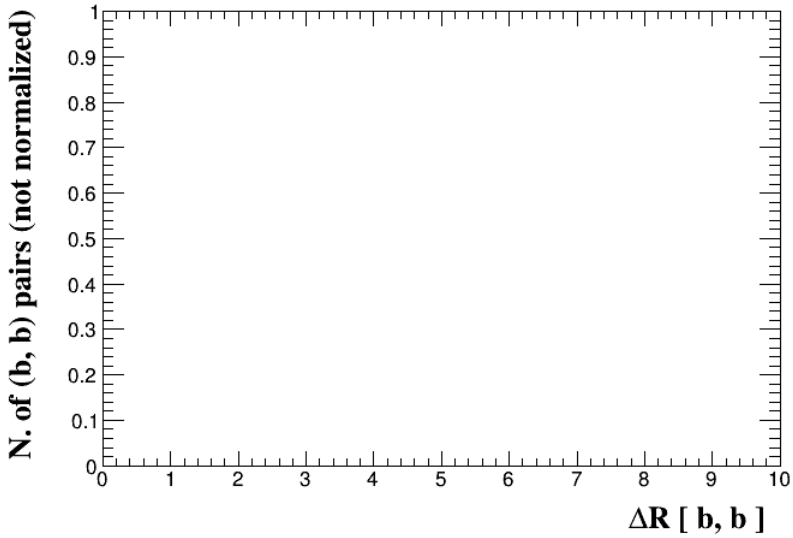

Figure 150.

3.151 Histogram 151

*** Plot: DPHI_0_PI (b , b)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

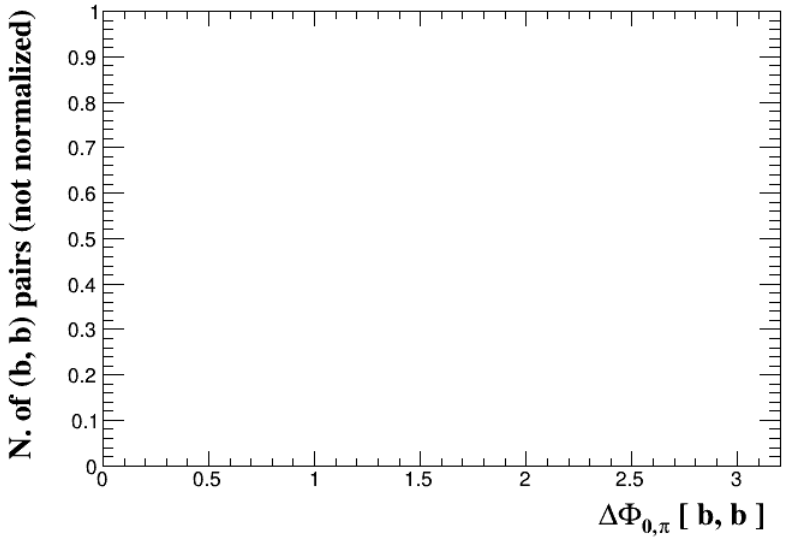

Figure 151.

3.152 Histogram 152

*** Plot: DPHI_0_2PI (b , b)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

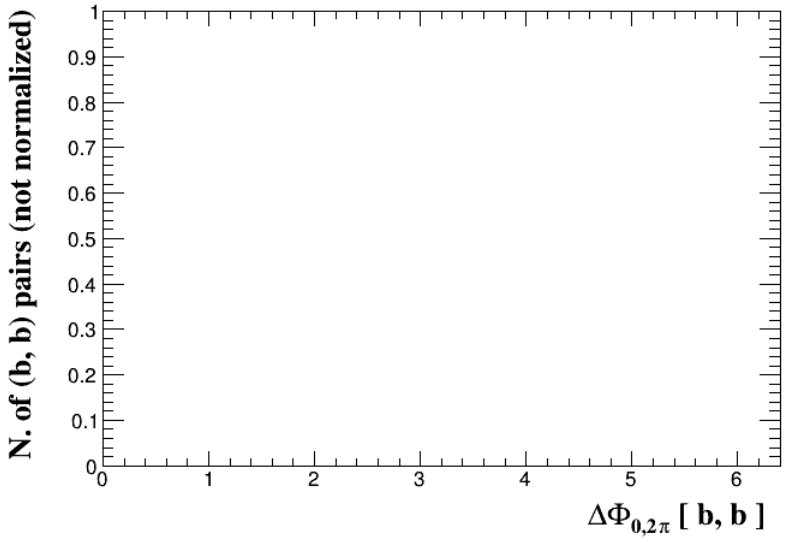

Figure 152.

3.153 Histogram 153

*** Plot: M (b b)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

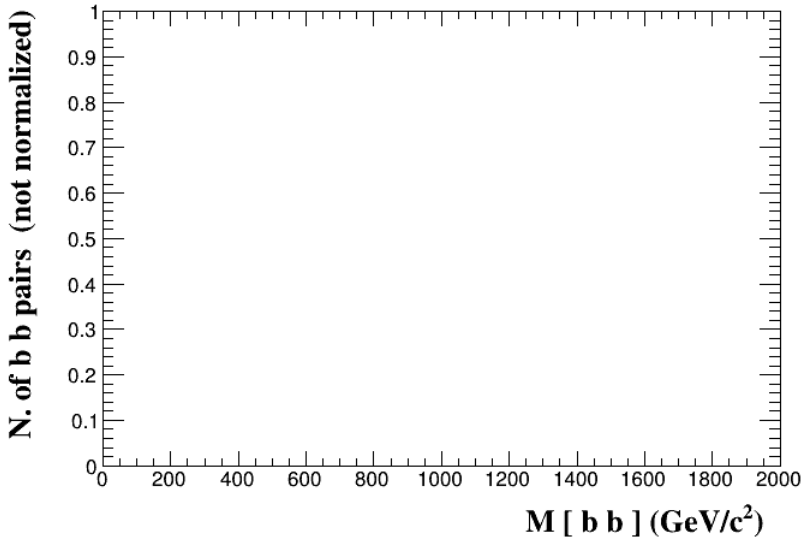

Figure 153.

3.154 Histogram 154

*** Plot: PT (b b)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

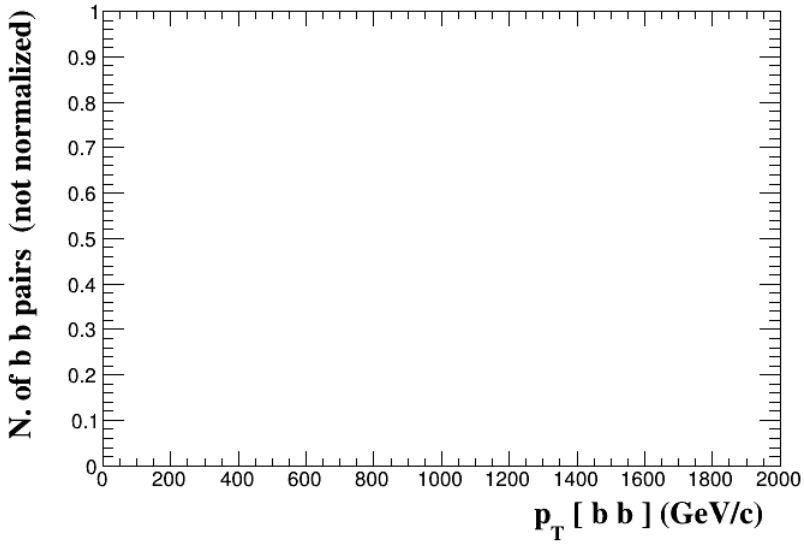

Figure 154.

3.155 Histogram 155

*** Plot: DELTAR (j , j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	1.90737	0.6749	0.0	0.0

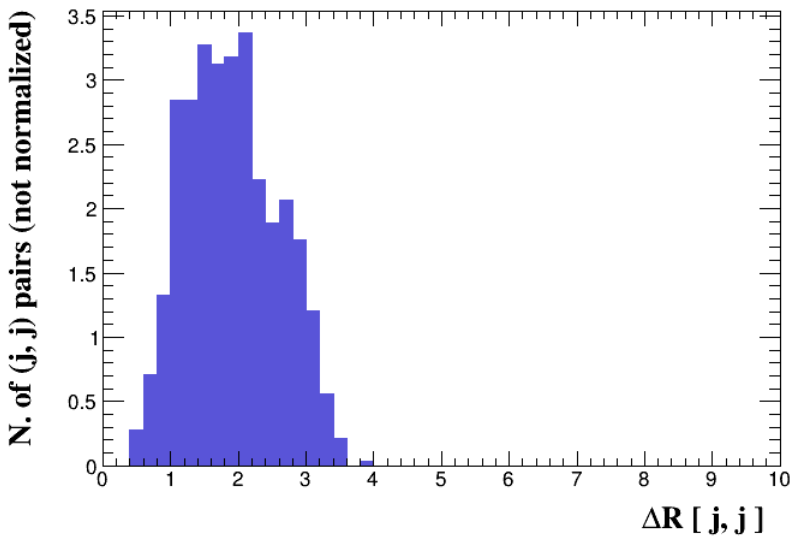

Figure 155.

3.156 Histogram 156

*** Plot: DPHI_0_PI (j , j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	1.55671	0.809	0.0	0.0

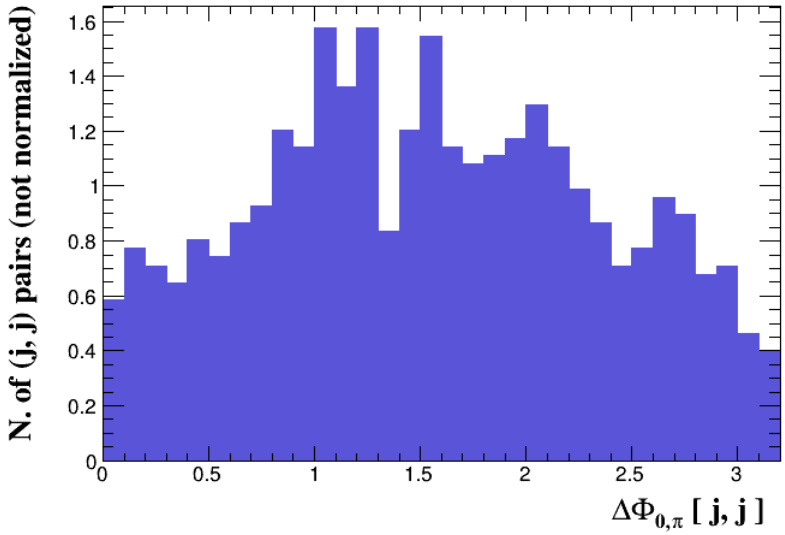

Figure 156.

3.157 Histogram 157

*** Plot: DPHI_0_2PI (j , j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	3.14159	1.779	0.0	0.0

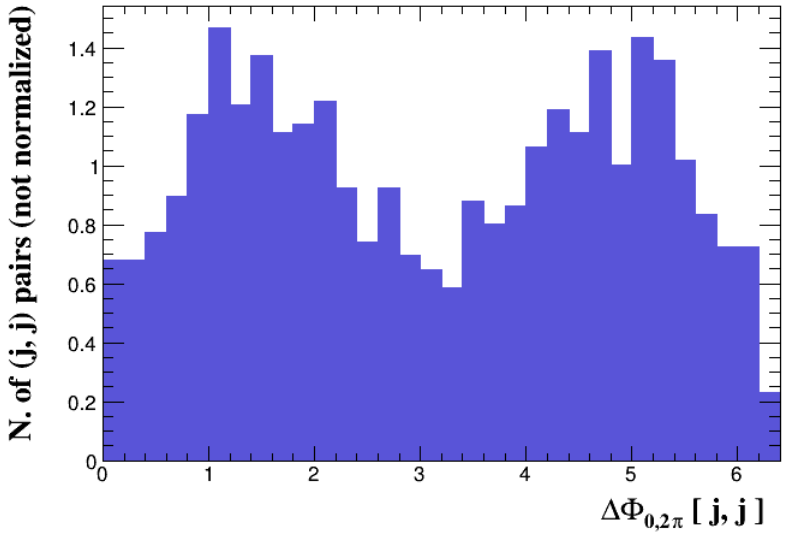

Figure 157.

3.158 Histogram 158

*** Plot: M (j j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	80.3339	4.323	0.0	0.0

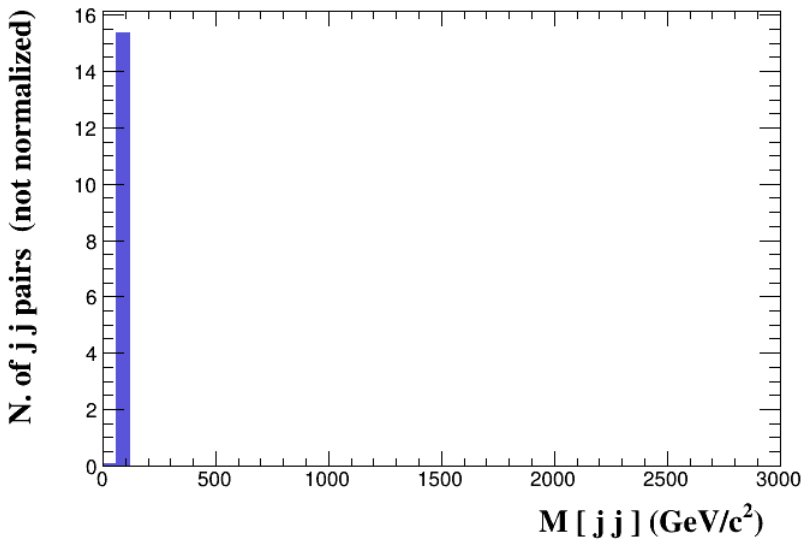

Figure 158.

3.159 Histogram 159

*** Plot: PT (j j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	86.6463	54.96	0.0	0.0

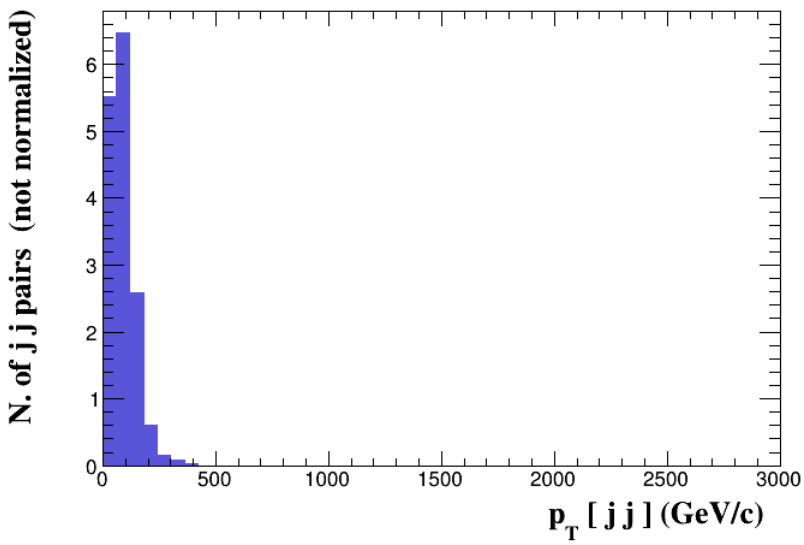

Figure 159.

3.160 Histogram 160

*** Plot: DELTAR (e , mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

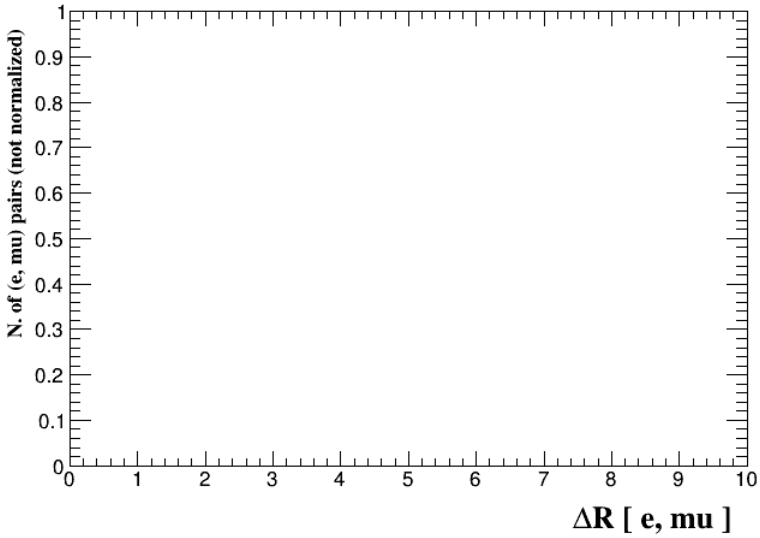

Figure 160.

3.161 Histogram 161

*** Plot: DELTAR (e , j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.6	2.0	2.3081	0.8766	0.0	0.0

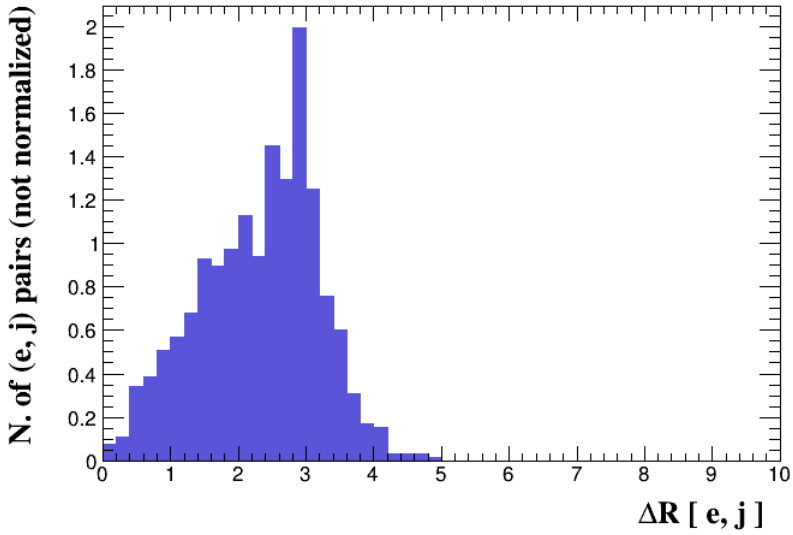

Figure 161.

3.162 Histogram 162

*** Plot: DELTAR (mu , j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.3	2.0	2.25996	0.8857	0.0	0.0

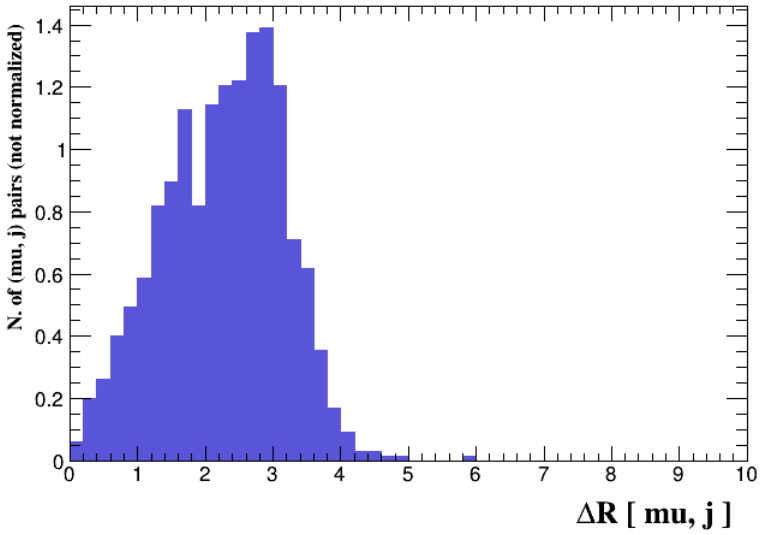

Figure 162.

3.163 Histogram 163

*** Plot: DELTAR (b , j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	2.25698	0.8185	0.0	0.0

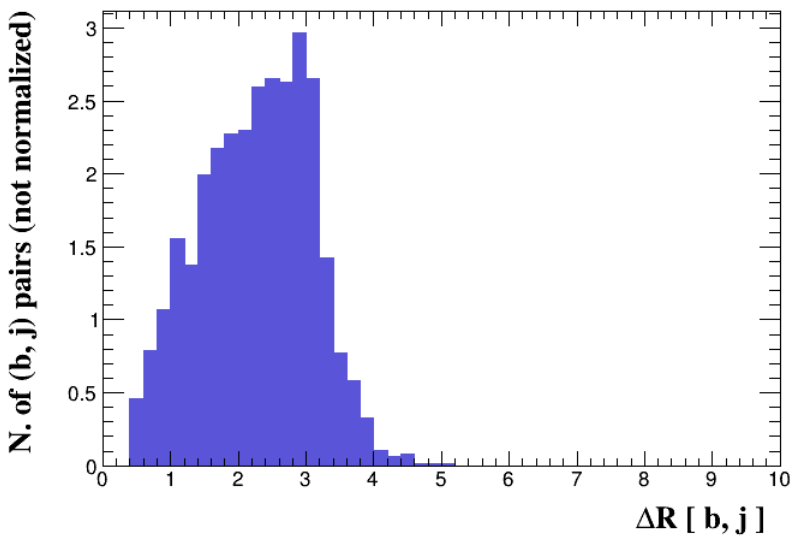

Figure 163.

3.164 Histogram 164

*** Plot: M (e mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

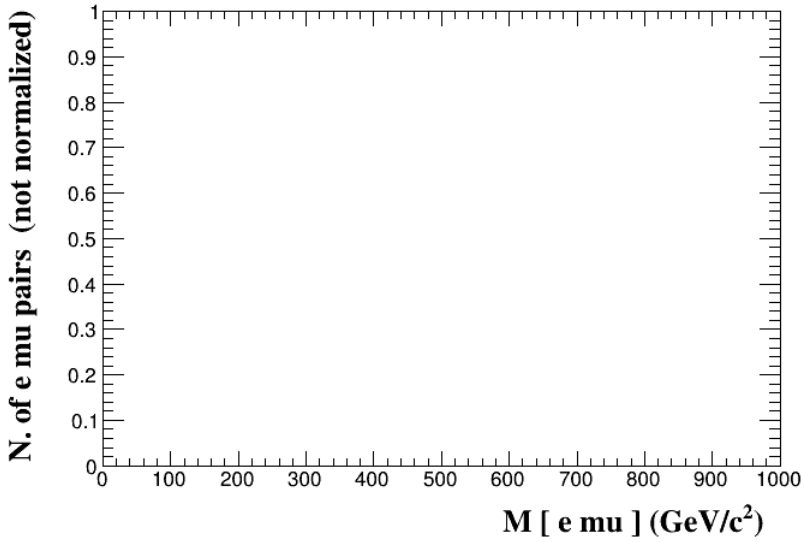

Figure 164.

3.165 Histogram 165

*** Plot: M (e j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.6	2.0	106.004	73.86	0.0	0.0

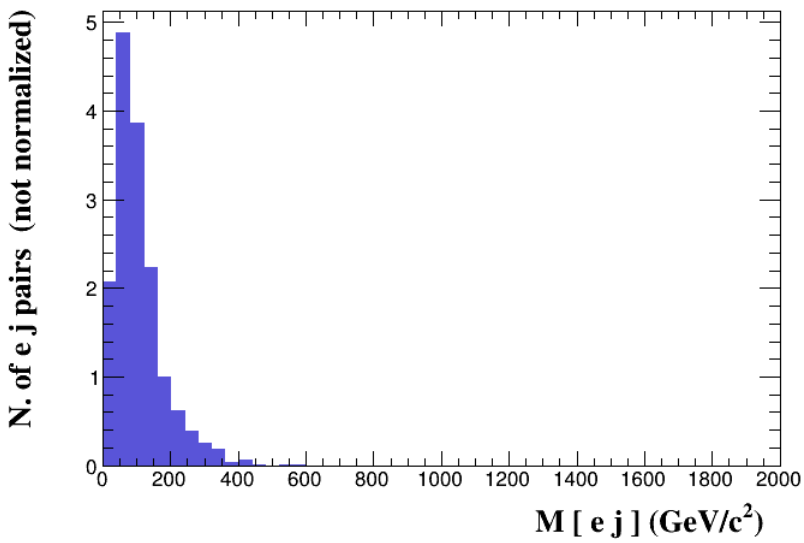

Figure 165.

3.166 Histogram 166

*** Plot: M (j mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.3	2.0	101.989	66.26	0.0	0.0

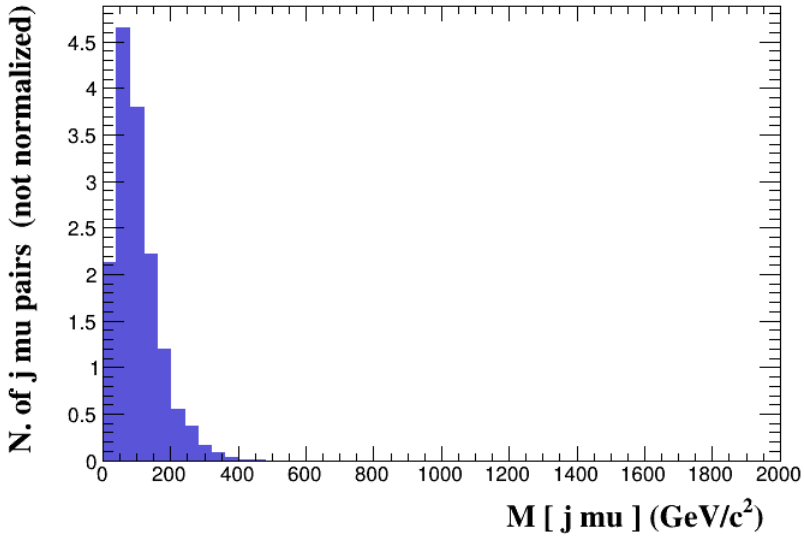

Figure 166.

3.167 Histogram 167

*** Plot: M (b j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	120.153	62.93	0.0	0.0

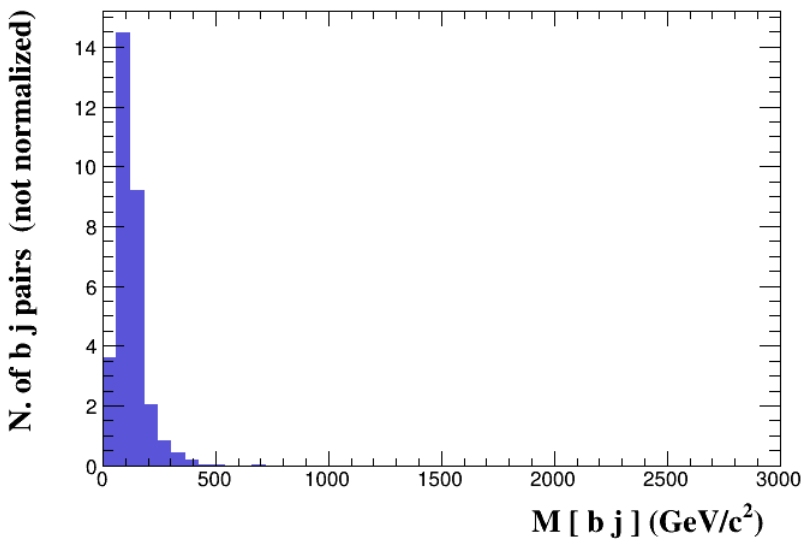

Figure 167.

3.168 Histogram 168

*** Plot: PT (e mu)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

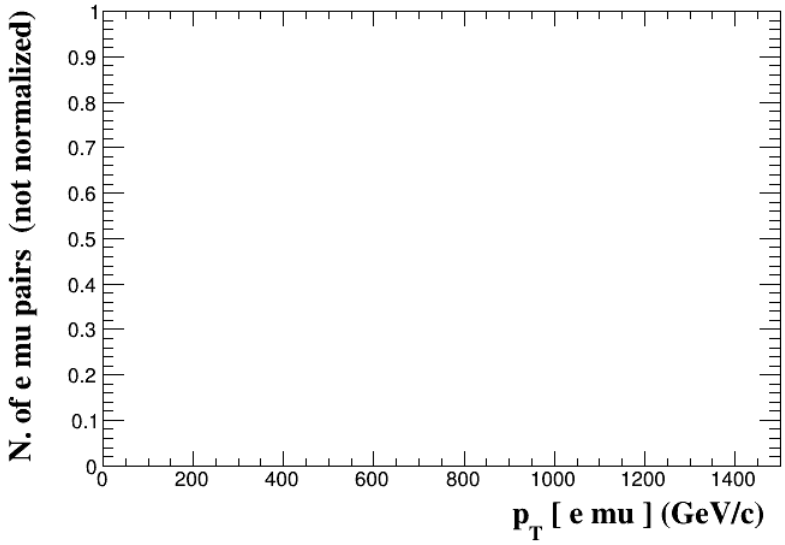

Figure 168.

3.169 Histogram 169

*** Plot: PT (b j)**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	30.9	2.0	80.8933	46.34	0.0	0.0

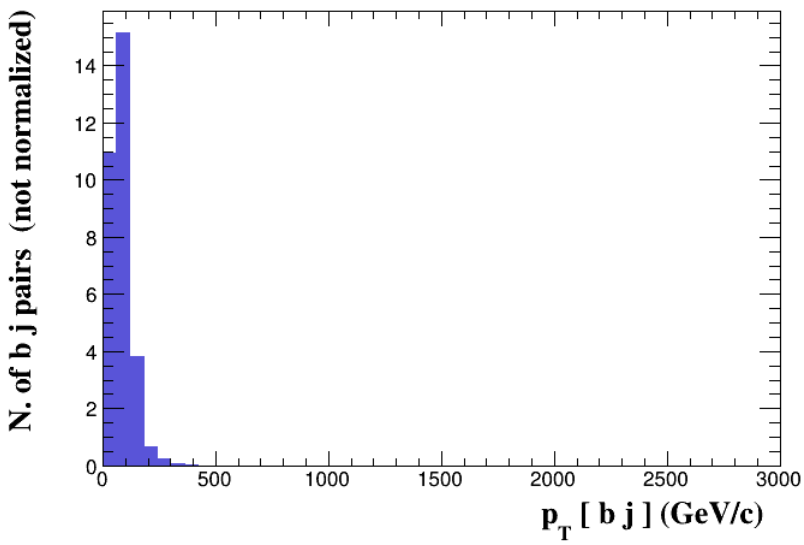

Figure 169.

3.170 Histogram 170

*** Plot: DELTAR (e[1] , mu[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

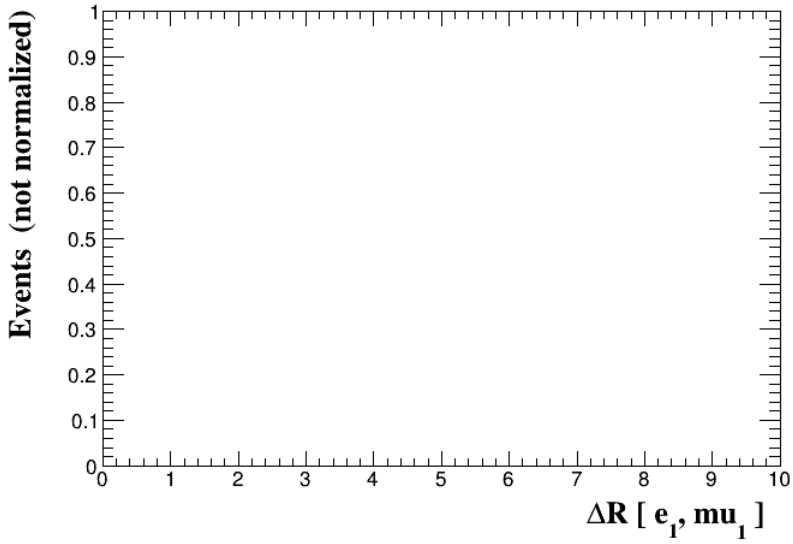

Figure 170.

3.171 Histogram 171

*** Plot: DELTAR (e[1] , j[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	2.34591	0.8529	0.0	0.0

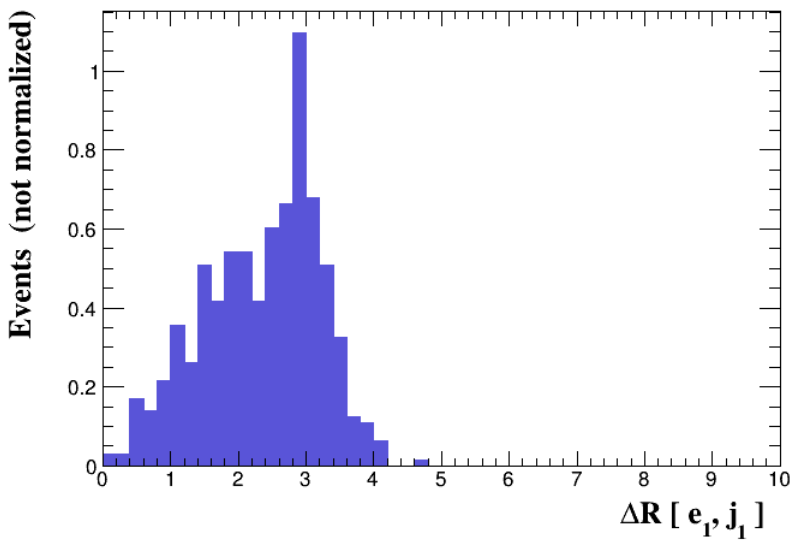

Figure 171.

3.172 Histogram 172

*** Plot: DELTAR (mu[1] , j[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	2.32408	0.8288	0.0	0.0

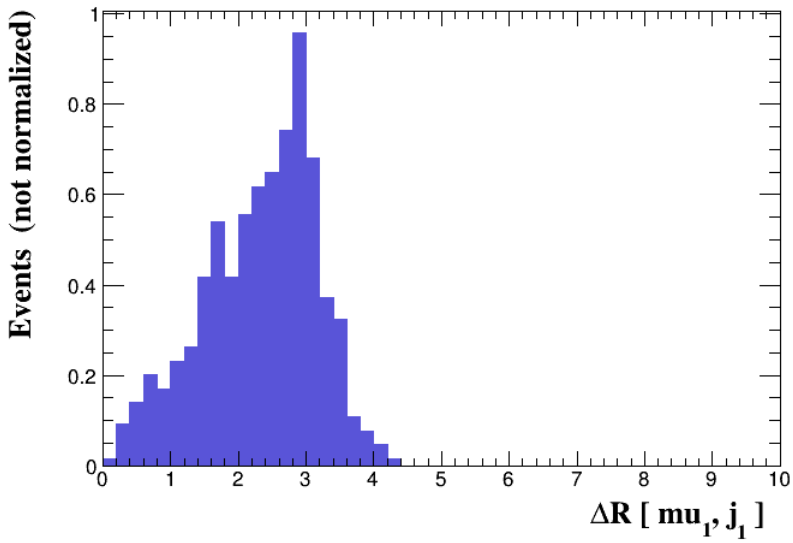

Figure 172.

3.173 Histogram 173

*** Plot: DELTAR (b[1] , j[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	2.26008	0.798	0.0	0.0

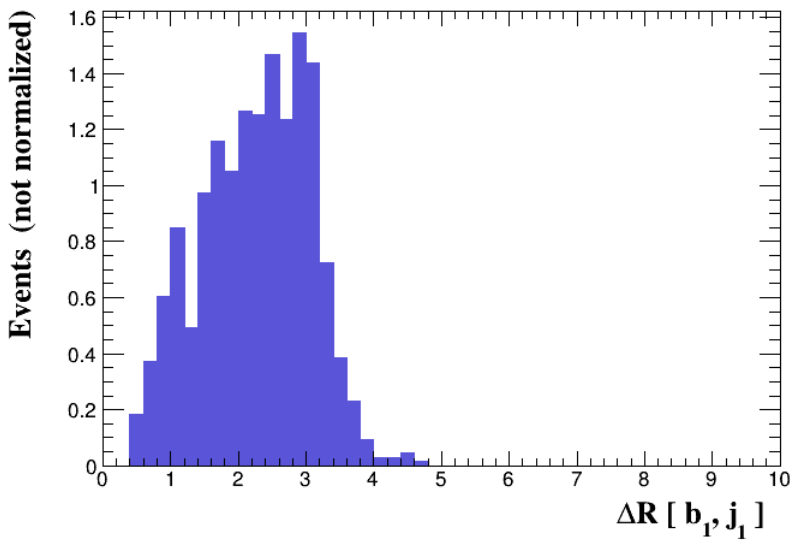

Figure 173.

3.174 Histogram 174

*** Plot: M (e[1] mu[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

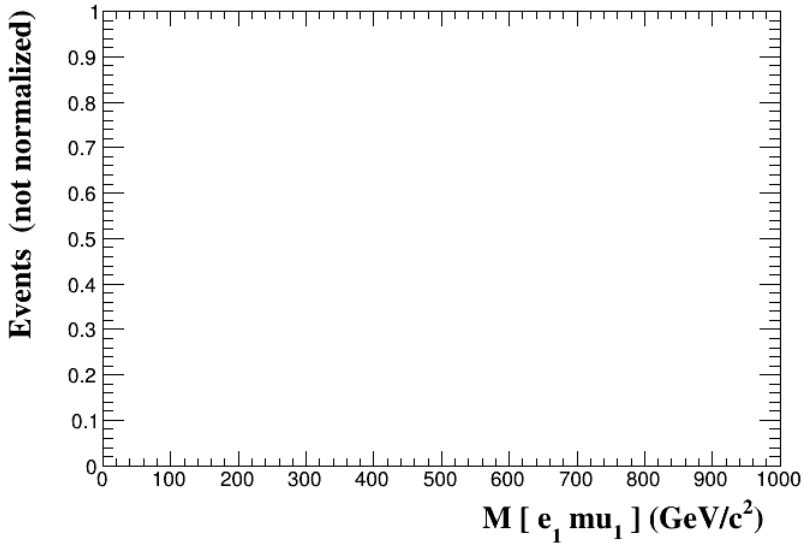

Figure 174.

3.175 Histogram 175

*** Plot: M (e[1] j[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.82	1.0	123.783	81.94	0.0	0.0

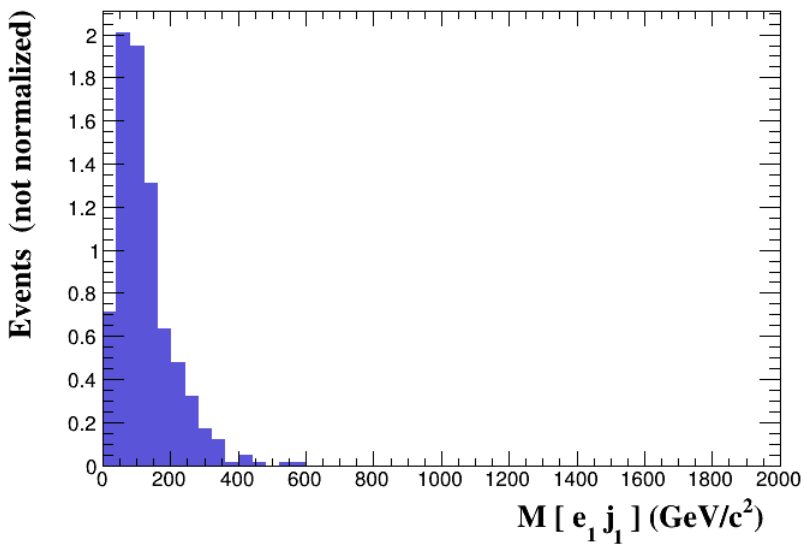

Figure 175.

3.176 Histogram 176

*** Plot: M (j[1] mu[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	7.63	1.0	117.989	70.7	0.0	0.0

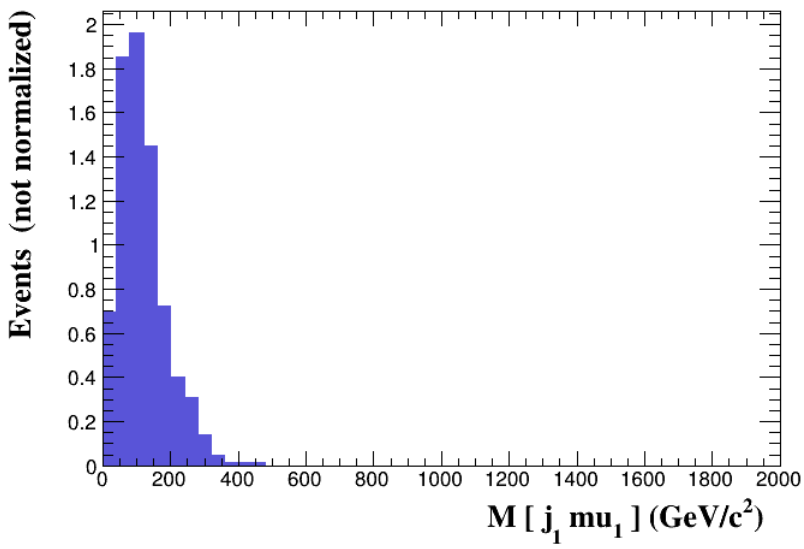

Figure 176.

3.177 Histogram 177

*** Plot: M (b[1] j[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	137.213	66.95	0.0	0.0

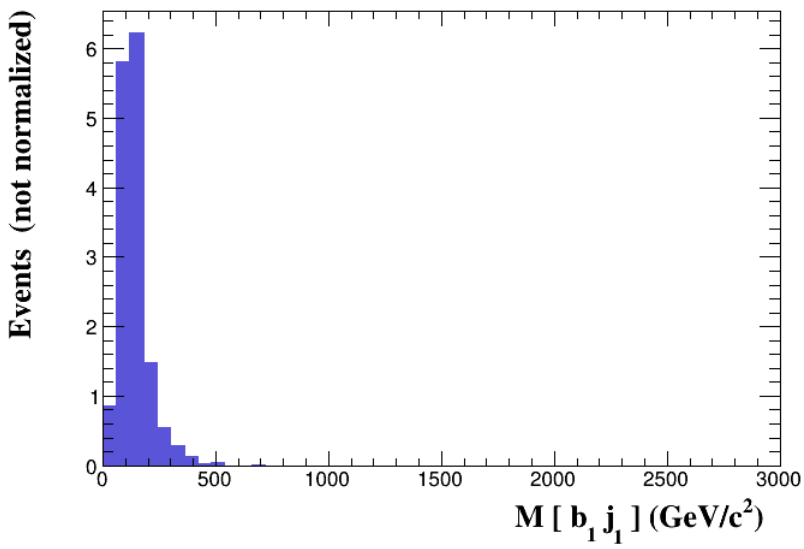

Figure 177.

3.178 Histogram 178

*** Plot: PT (e[1] mu[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	0.0 +/- 0.0	0.	0.0	0.0	0.0	0.0

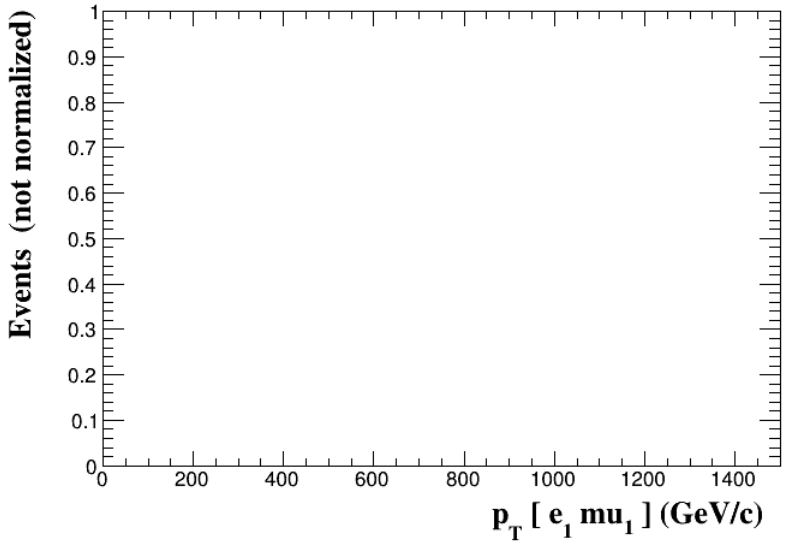

Figure 178.

3.179 Histogram 179

*** Plot: PT (b[1] j[1])**

Dataset	Integral	Entries per event	Mean	RMS	% underflow	% overflow
testset	15.4	1.0	87.1047	50.81	0.0	0.0

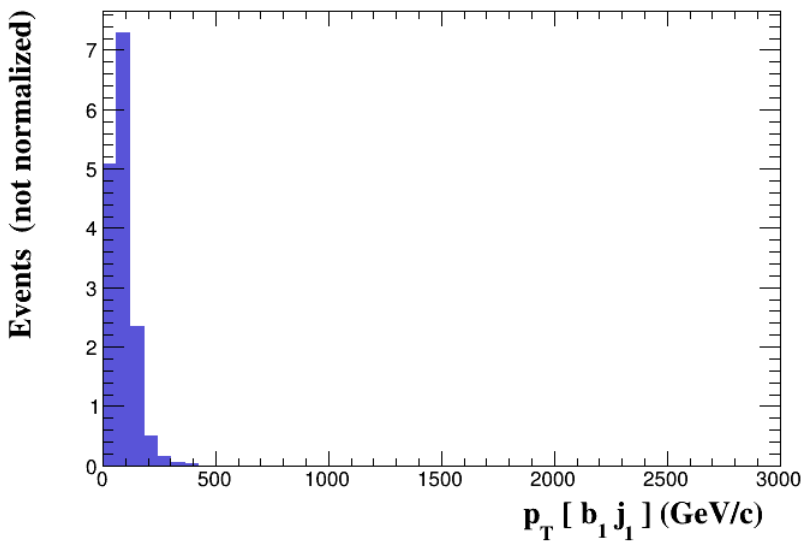

Figure 179.